L09 — study notes

Contents

Class 9 — Multivariate time series · Map of the notes **4**
The layers . . . 4
Reading order . . . 5
Part I — the VAR and what it can say . . . 5
Part II — too many parameters . . . 5
Part III — letting things move . . . 5
Support . . . 5
The one result to hold . . . 6
Numbers and cases to have memorized . . . 6
Figures . . . 6
Interactive companions . . . 7
Scope and ceiling . . . 7
Course-wide position . . . 7
Bibliography actually used in these notes . . . 7

Slides 3–9 · A VAR(p) is a multivariate regression: OLS is the MLE, the companion matrix decides stationarity, and forecasts are powers of it **9**
1 · U — why a VAR . . . 9
2 · C — the model (slide 4; T eq. 8.13, p. 403) . . . 10
3 · C — the multivariate regression form (slides 5–7) . . . 10
4 · C — OLS is the conditional MLE, and also the SUR/GLS estimator . . . 11
4.1 · The likelihood . . . 11
4.2 · Completing the square in B . . . 11
4.3 · Why SUR/GLS adds nothing here . . . 12
4.4 · Worked — the site's VAR(3) for UK/US/CA GDP growth (VAR-Example.html §2) 12
5 · C — companion form and stationarity (slides 8–9; T eq. 8.15, pp. 404–405) . . . 12
5.1 · Worked — the slide-13 simulated VAR(2) . . . 13
5.2 · Worked — the UK/US/CA VAR(3) . . . 14
6 · C — forecasting (slide 8; T p. 409) . . . 15
6.1 · Worked — Tsay's VAR(1) for IBM and S&P 500 (pp. 414–416) . . . 15
7 · C — choosing p (T §8.2.4, pp. 405–407) . . . 16
7.1 · Worked — `VARorder(rates)` on the GDP data (VAR-Example.html §3.1) . . . 16
8 · C — the curse of dimensionality (slides 27, 45) . . . 16
9 · U — the site's R and its Python twin . . . 17
Links . . . 17

Slides 10–12 · Impulse responses and variance decompositions: the Ψ_s recursion, the Cholesky factor, and what the ordering decides **18**

1 · C — the VMA(∞) representation (slide 10; T eq. 8.21, p. 412) . . . 18
2 · C — the impulse-response function (slide 12; T p. 412) . . . 19
2.1 · Worked — Ψ_2 of the UK/US/CA VAR(3) . . . 19
3 · C — orthogonalised responses (slide 11; T eq. 8.22, p. 413) . . . 19
3.1 · Worked — Tsay's VAR(1) for IBM and the S&P 500 (pp. 414–417) . . . 20
3.2 · Worked — the UK/US/CA VAR(3) . . . 20
4 · C — forecast-error variance decomposition (slide 11) . . . 21
4.1 · Worked — reproducing the site's `FEVdec` table (VAR-Example.html §3.2) . . . 22
5 · U — uncertainty bands, two ways . . . 23
Links . . . 23

Slides 31–37 · Structural VARs: the reduced form cannot tell orthogonal rotations apart, so $q(q-1)/2$ restrictions must come from economics **24**
1 · U — the question . . . 24
2 · C — from structure to reduced form (slide 33) . . . 24
3 · C — why the reduced form cannot identify A_0 . . . 25
4 · C — recursive identification is the Cholesky factor . . . 26
4.1 · Worked — Tsay Example 8.3 (pp. 400–401), both orderings . . . 26
5 · C — non-recursive identification on the slides . . . 27
6 · U — Python . . . 28
Links . . . 28

Slides 22–30 and Part 2 slides 3–13, 56–60 · The Minnesota prior: shrink every equation toward a random walk, tighter for distant lags and other variables, and the posterior is ridge regression in closed form **29**
1 · U — why shrink . . . 29
2 · C — the Minnesota prior (slides 23–24) . . . 30
3 · C — the closed-form posterior . . . 30
3.1 · Worked — the slide-13 VAR(2), 58 observations . . . 31
4 · C — other priors on (B, Σ) (slides 25–26) . . . 33
5 · C — variable selection: the ON/OFF switch and SSVS (Part 2, slides 5–13) . . . 34
6 · C — large BVARs work (Part 2, slides 50, 56–60) . . . 35
7 · U — Python . . . 35
Links . . . 35

Part 2 slides 14–37, 51–55 · Factor models: k common factors carry all the cross-correlation, $\Omega = \beta\beta' + \Sigma$; a FAVAR puts them in a VAR **36**
1 · C — the basic model (slide 15; T eq. 9.2, p. 469) . . . 36
2 · C — rotation, identification and the bound on k (slides 16–19) . . . 37
3 · C — priors and full conditionals (slides 20–21) . . . 38
3.1 · Worked — slide 22's simulation exercise . . . 39
3.2 · Worked — exchange rates (slides 26–28; Lopes and West 2004) . . . 39
4 · C — forecasting with many predictors: the FAVAR (slides 51–55) . . . 40
5 · G — named only . . . 40
Links . . . 41

Slides 38–44 · Letting the VAR move: GARCH and SV in the errors, random-walk coefficients, and the Kalman filter that makes them estimable **42**
1 · C — VAR-GARCH (slides 38–39) . . . 42
2 · C/G — VAR-SV (slide 40; Uhlig 1997) . . . 43

3 · C — TVP-VAR-SV (slides 41–44; Primiceri 2005) . . . 43
4 · C — state space and the Kalman filter (Tsay §11.1, pp. 558–563) . . . 44
4.1 · The steady state . . . 45
4.2 · Worked — Tsay Example 11.1: Alcoa realised volatility (pp. 559–560) . . . 45
5 · C — time-varying regression coefficients . . . 46
6 · C — large TVP-VARs without MCMC (Part 2, slide 61) . . . 47
Links . . . 47

Part 2 slides 38–43 and the two site pages · Time-varying covariances: DCC models the correlations directly, factor SV gets them from the factor's volatility 48
1 · C — the problem (T p. 506) . . . 48
2 · C — the exponentially weighted benchmark (T §10.1, eq. 10.2, p. 506) . . . 48
3 · C — correlations separately: $\Sigma_t = D_t R_t D_t$ (T eq. 10.7, p. 516) . . . 49
4 · C — dynamic conditional correlation (T §10.4.3, pp. 531–532) . . . 49
4.1 · Worked — W&T Offshore vs the S&P 500 (wti-sp500-timevaryingcovariance.html) 50
5 · C — factor stochastic volatility (Part 2, slide 38; Pitt and Shephard 1999) . . . 51
6 · C — the site's two FSV pages . . . 52
7 · C — DCC vs FSV, side by side . . . 53
Links . . . 53

Class 9 · Formulas — the exam specification, every result, every number 54
0 · The exam specification, mapped . . . 54
0.1 · The site's description of Class 9 (`Aula/COURSE.md`) . . . 54
0.2 · The class plan's goals (`Map/class-plan.md`, Class 9) . . . 54
1 · Definitions and results . . . 55
1.1 · VAR(p) (L09-01) . . . 55
1.2 · Impulse responses and FEVD (L09-02) . . . 55
1.3 · Structural VAR (L09-03) . . . 56
1.4 · Bayesian VAR (L09-04) . . . 56
1.5 · Factor models and FAVAR (L09-05) . . . 57
1.6 · Time-varying VARs and the Kalman filter (L09-06) . . . 57
1.7 · Multivariate volatility (L09-07) . . . 58
2 · Every number . . . 58

Class 9 · Pitfalls — the confusions, each with its minimal counterexample, and an original True/False self-test 61
Part I — the confusions this material produces . . . 61
1 · A coefficient above one vs a non-stationary VAR . . . 61
2 · Reduced-form vs orthogonalised impulse responses . . . 61
3 · The ordering as a detail vs an identifying assumption . . . 61
4 · The order condition vs identification . . . 61
5 · OLS equation by equation vs "ignoring the correlation between equations" . . . 62
6 · AIC vs BIC lag choice . . . 62
7 · Shrinkage as bias vs shrinkage as a better estimator . . . 62
8 · λ as "prior strength" vs a variance scale . . . 62
9 · Normal-Wishart vs Minnesota . . . 62
10 · Factors identified vs loadings identified . . . 62
11 · More factors always fit better vs the identification bound . . . 63
12 · The Kalman gain as a tuning knob vs a derived quantity . . . 63
13 · Time-varying parameters as "more realistic, hence better" . . . 63

14 · DCC's parameters as pair-specific . . . 63
15 · Correlations rising in crises as a separate phenomenon . . . 63
16 · GARCH persistence below one by assumption . . . 63
17 · Ticker vs asset . . . 63
18 · Labels in the site scripts . . . 64
Part II — True/False self-test (original) . . . 64
Block A — VAR, IRF, FEVD . . . 64
Block B — structural identification . . . 64
Block C — Bayesian VAR . . . 65
Block D — factors, TVP, covariances . . . 65
Part III — the distinctions that cost most points . . . 65

Class 9 · Computation in Python — every routine, with the assert that must pass 67
0 · Setup . . . 67
§1 · VAR(p) as a multivariate regression: OLS = MLE = SUR/GLS (L09-01 (11)-(13)) . . . 68
§2 · Companion form, stationarity, means, forecasts (L09-01 (2), (14)-(18)) . . . 69
§3 · VMA weights, orthogonalised IRFs, FEVD, recursive SVAR (L09-02 (3), (6), (8); L09-03 (9)-(11)) . . . 70
§4 · Minnesota posterior, limits in lambda, SSVS inclusion probability (L09-04 (2), (7)-(9), (15)) 71
§5 · Factor model algebra (L09-05 (2)-(9)) . . . 72
§6 · Kalman filter, steady state, ARIMA link, discounting (L09-06 (8)-(14), (18); (3)) . . . 73
§7 · EWMA, DCC and GARCH persistence, FSV implied correlation (L09-07 (1)-(2), (6), (11)-(12)) . . . 74
Links . . . 75

Class 9 — Multivariate time series · Map of the notes

One claim. A VAR is a multivariate regression: OLS equation by equation is its MLE, the companion matrix decides stationarity, and impulse responses and forecasts are powers of it. **Its shocks are correlated, so every structural reading needs restrictions** — $q(q-1)/2$ of them — and the familiar Cholesky ordering is one such set, not a neutral device. **Its parameters grow like** q^2p, and the two answers of the lecture are shrinkage (the Minnesota prior, whose posterior is ridge regression toward a random walk) and factors (a few common components carrying the cross-section, in a FAVAR or in the covariance). **When coefficients and volatilities move**, each is a state and the Kalman filter updates it; **when only covariances move**, DCC models correlations directly while factor SV derives them from the factor's volatility.

Status: taught (slides on disk). **PDF:** L09-notes.pdf.

The layers

Layer	In this unit
U — intuition	why VARs (Sims), why shrink, what a factor "is", what the gain means, why correlations rise in crises

Layer	In this unit
C — course, examinable	VAR(p) algebra, OLS = MLE, companion form, forecasts, order selection; Ψ_s, Cholesky IRF, FEVD; SVAR and the order condition; Minnesota prior and posterior, other priors, SSVS; factor model, identification, full conditionals, FAVAR; VAR-SV, TVP-VAR-SV, Kalman filter; EWMA, DCC, FSV
G — graduate, not examinable unless asked	Uhlig's scalar martingale condition, West–Harrison discounting formula, rank deficiency in factor models, parsimonious BFA, spatial DFM

Reading order

Part I — the VAR and what it can say

- L09-01-var-p-estimation-forecasting — slides 3–9, 13, 27, 45; T §8.2.3–8.2.4 (pp. 403–409); G §9.1. **Settles:** OLS = conditional MLE = SUR/GLS when regressors are common; $\text{Cov}(\hat{\beta}) = \Sigma \otimes (X'X)^{-1}$; companion eigenvalues (slide VAR(2): 0.9099); forecasts and their MSE (Tsay's limits 7.0436/5.5454); VARorder (AIC 2, BIC 1).
- L09-02-vma-irf-fevd — slides 10–12, 17, 20–21; T §8.2.5 (pp. 412–417). **Settles:** the Ψ_s recursion; Cholesky-orthogonalised responses (Tsay's Θ_1); the FEVD, reproducing the site's table (0.0758 at $h = 1$); two label errata in the site script.
- L09-03-structural-var — slides 31–37; T Ex. 8.3 (pp. 400–401). **Settles:** $\Sigma = (A_0' A_0)^{-1}$; the rotation problem and $q(q-1)/2$; recursive = Cholesky; Example 8.3's two structural equations (13.6% vs 48.5% FEVD); counting zeros in the slides' illustrations.

Part II — too many parameters

- L09-04-bvar-minnesota-ssvs — slides 22–30; part 2 slides 3–13, 50, 56–60. **Settles:** the Minnesota prior and its closed-form posterior; λ from random walk to OLS; why Normal-Wishart forces $\theta = 1$; SSVS inclusion threshold 0.30 se; large BVARs beat OLS (7.56 vs 0.46).
- L09-05-factor-models-favar — part 2 slides 14–37, 51–55, 59; T §9.1–9.2. **Settles:** $\Omega = \beta\beta' + \Sigma$; rotation and the bound $k \leq (2m + 1 - \sqrt{8m+1})/2$; factor-score posterior; exchange-rate example; FAVAR and RMSFE.

Part III — letting things move

- L09-06-tvp-var-sv-state-space — slides 38–44; part 2 slide 61; T §11.1, §11.3.1; G §10.3. **Settles:** VAR-GARCH, VAR-SV, Primiceri's TVP-VAR-SV as a drifting recursive SVAR; the Kalman filter derived; steady-state gain $K^* = 0.1418 = 1 - \hat{\theta}$ for Tsay's Alcoa volatility; discounting.
- L09-07-fsv-dcc-covariances — part 2 slides 38–43; T §10.1, §10.3.1, §10.4.3, §10.6; the two site pages. **Settles:** EWMA, DCC (Engle, Tse–Tsui) and their half-lives (7.2 vs 2.4 days); S&P GARCH persistence 1.0029; FSV's implied correlation rising with factor volatility.

Support

- L09-08-formulas — exam specification mapped; every result; every number.
- L09-09-pitfalls — 18 confusions with counterexamples; 24-item original True/False self-test.

- L09-10-python — reference code (`L09_python_checks.py`) with asserts against the printed numbers.

The one result to hold

> **Every structural or forecasting statement from a VAR is a function of (B, Σ) plus an assumption.** Ψ_s and forecasts need only (B, Σ); orthogonalised IRFs, FEVDs and "structural" equations need an ordering or other restrictions on A_0 ($\Sigma = (A_0' A_0)^{-1}$ leaves $q(q-1)/2$ degrees of freedom); and with many series, B itself needs a prior.

Numbers and cases to have memorized

Case	Number
Slide-13 VAR(2) companion max modulus	0.9099
UK/US/CA VAR(3): AIC / BIC lag; largest root	2 / 1; 0.6107
FEVD of US due to UK shock at $h = 1$	$\rho^2 = 0.0758$
T Ex. 8.3: share of r_1's 10-step variance due to r_2	13.6% vs 48.5% by ordering
Restrictions needed, $q = 3, 5, 6$	3, 10, 15
Parameter counts, VAR(1) with $q = 3, 20, 131$	15, 610, 25,807
Bańbura et al., EMPL $h = 1$	CEE OLS 7.56; LARGE BVAR 0.46
$k_{\max}$ for $m = 6, 12, 20, 50$	3, 7, 14, 40
Tsay Ex. 11.1 steady-state gain	$0.1418 = 1 - 0.858$
DCC half-lives (Engle, Tse–Tsui)	7.22, 2.43 days
FSV correlation, slide-22 pair, $h = 1$ / $h = 0.05$	0.939 / 0.489

Figures

Built by `figures/make_figures.py`. Run: `python Notas/L09/figures/make_figures.py`. Every figure asserts at least one plotted number.

Figure	Shows	Lives in
L09-companion-eigenvalues	companion eigenvalues of the slide-13 VAR(2) (0.9099) and the site's VAR(3) (0.6107)	L09-01-var-p-estimation-forecasting §5
L09-irf-var3-gdp	Ψ_h and Θ_h for the UK/US/CA VAR(3)	L09-02-vma-irf-fevd §3
L09-fevd-var3-gdp	FEVD of the refined VAR(3), asserted against FEVdec	L09-02-vma-irf-fevd §4
L09-cholesky-ordering	T Ex. 8.3 under both orderings; impact vs ρ	L09-03-structural-var §4
L09-minnesota-shrinkage	prior sd by lag; posterior means vs λ (seeded, $T = 58$)	L09-04-bvar-minnesota-ssvs §3
L09-factor-correlation	one-factor correlation vs factor log-variance; bound on k	L09-05-factor-models-favar §2
L09-kalman-gain-snr	K^* vs q; gain path for Tsay Ex. 11.1	L09-06-tvp-var-sv-state-space §4
L09-dcc-garch-persistence	DCC and GARCH reversion multipliers, site estimates	L09-07-fsv-dcc-covariances §4

Interactive companions

- The Ordering Dial (`companions/companion-var-irf-ordering.html`). Move the four coefficients of a bivariate VAR(1), the innovation correlation ρ and the Cholesky ordering. Watch the orthogonalised responses (solid: current ordering, dashed: the other), the eigenvalues on the unit circle, and the **FEVD share** of r_1's 10-step variance due to the r_2 shock (13.6% vs 48.5% at Tsay Ex. 8.3's defaults). Landmarks: $\rho = 0$, modulus 1. Linked from L09-02-vma-irf-fevd and L09-03-structural-var.
- The Factor-Volatility Dial (`companions/companion-fsv-correlation.html`). Move two loadings, two idiosyncratic variances and the factor log-variance λ_t. Watch ρ_t travel along the exact curve and the covariance ellipse change; **answer:** ρ_t (0.939 at slide 22's defaults). Landmarks: $\lambda = 0$ and the half-limit point. Linked from L09-05-factor-models-favar and L09-07-fsv-dcc-covariances.

Both pages are verified by `companions/check_companions.py` at three control vectors (both ends and the default) plus degenerate ones, against their own JavaScript run under `node`.

Scope and ceiling

- **Examinable (C):** everything the slides state as a model, prior, posterior or result, and the derivations in the notes built on them; the reading of the site pages' printed output.
- **Named only (G):** Uhlig's beta-Wishart dynamics beyond the scalar case; Waggoner–Zha and Rubio-Ramírez et al. algorithms; parsimonious BFA, British Cohort Study, spatial DFM; the ENC prior; sign restrictions (Ahmadi–Uhlig, in the references).
- **Out:** cointegration/ECM (T 8.7(d)–(e); listed in `Map/class-plan.md` as a syllabus item no class covers); the BEKK and diagonal-VEC estimation details of T §10.2; principal-component asymptotics (T §9.6).

Course-wide position

Class	Relation to Class 9
1 — L01 map	VAR(1) mean, Γ_0 by the vec formula, T Ex. 8.3 (L01-07 §9); OLS algebra (L01-06)
2 — L02 map	bias–variance: why shrinkage helps forecasts
3+4 (`Notas/L03/`)	conjugate priors, Gibbs sampling, ridge/shrinkage priors — the machinery of L09-04 and L09-05
5+6, 7	causal inference and hierarchical models; the SVAR's identification problem is the time-series twin of Class 5+6's
8 (`Notas/L08/`)	univariate AR/ARMA, unit roots, GARCH and SV — the univariate pieces of every model here

Bibliography actually used in these notes

- Lopes, H. F. — `BVAR-part11.pdf`, `BVAR-part2.pdf` (slides), and the three site pages.
- Tsay, R. S. (2010). *Analysis of Financial Time Series*, 3rd ed. §8.2, §9.1–9.2, §10.1, §10.3.1, §10.4.3, §10.6, §11.1, §11.3.1 (pages as in the front matter).
- Greenberg, E. (2007). *Introduction to Bayesian Econometrics*. §9.1, §10.3.

Related project material: quiz quiz-L09; book exercises `Listas/lista-L09-book-exercises.tex` with `Listas/L09-book-exercises-check.py`; NotebookLM audio prompt L09-audio; rules pack

rules/05; class plan class-plan.

Slides 3–9 · A VAR(p) is a multivariate regression: OLS is the MLE, the companion matrix decides stationarity, and forecasts are powers of it

> **Claim.** Stack the lags in x_t and a VAR(p) is $y_t = Bx_t + u_t$, a multivariate regression with the **same regressors in every equation**. That one fact makes equation-by-equation OLS the conditional MLE and the SUR/GLS estimator at once, and gives $\widehat{\mathrm{Cov}}(\hat{\beta}) = \hat{\Sigma} \otimes (X'X)^{-1}$. Writing the VAR(p) as a VAR(1) in qp dimensions (the **companion form**) settles both remaining questions: the process is stationary iff every eigenvalue of the companion matrix lies inside the unit circle, and the h-step forecast is its h-th power applied to the current state.

Prev: L01 §9 (VAR(1) review) · Next: L09-02-vma-irf-fevd · Map: L09-00-map

Notation.

Symbol	Meaning	Range
y_t	$q \times 1$ vector of series at time t (Tsay: r_t, dimension k)	$t = 1, \dots, n$
B_i	$q \times q$ autoregressive matrix at lag i (Tsay: Φ_i)	$i = 1, \dots, p$
$B = (B_1, \dots, B_p)$	$q \times qp$ coefficient matrix	
$x_t = (y'_{t-1}, \dots, y'_{t-p})'$	$qp \times 1$ stacked lags	
μ / ϕ_0	intercept (slide 13 calls it μ; Tsay ϕ_0)	$\mathbb{R}^q$
u_t	innovation, i.i.d. $N(0, \Sigma)$	
Σ	$q \times q$ innovation covariance, positive definite	
Y, X, U	$n \times q$, $n \times qp$, $n \times q$ data matrices with rows y'_t, x'_t, u'_t	
$\beta = \mathrm{vec}(B')$	$q^2 p \times 1$: equation 1's coefficients, then equation 2's, … (slide 22)	
A	$qp \times qp$ companion matrix	
Ψ_i	VMA weights (derived in L09-02-vma-irf-fevd)	$\Psi_0 = I_q$

---

1 · U — why a VAR

Slide 3 quotes Del Negro and Schorfheide (2011): VARs "appear to be straightforward multivariate generalizations of univariate autoregressive models. They turn out to be one of the key empirical tools in modern macroeconomics." Sims (1980) proposed them in place of large structural models whose exclusion restrictions he called "incredible". The VAR makes **no** exclusion: every variable depends on every lag of every variable. That buys flexibility and costs parameters — the tension the whole of Class 9 is about (§8 below, then L09-04-bvar-minnesota-ssvs).

---

2 · C — the model (slide 4; T eq. 8.13, p. 403)

$$y_t = \mu + B_1 y_{t-1} + \cdots + B_p y_{t-p} + u_t, \qquad u_t \overset{iid}{\sim} N(0, \Sigma). \tag{1}$$

The slide writes (1) without μ ("the (basic) VAR(p)"); the simulated example (slide 13) and Tsay keep it. **Stationary mean.** If y_t is weakly stationary with $E(y_t) = m$, take expectations of (1):

$$m = \mu + B_1 m + \cdots + B_p m$$

$$(I_q - B_1 - \cdots - B_p)\, m = \mu$$

$$m = (I_q - B_1 - \cdots - B_p)^{-1} \mu \tag{2}$$

(T p. 403: $\mu = [\Phi(1)]^{-1}\phi_0$), provided $I - \sum B_i$ is invertible.

---

3 · C — the multivariate regression form (slides 5–7)

Stack the lags: with $B = (B_1, \ldots, B_p)$ and $x_t = (y'_{t-1}, \ldots, y'_{t-p})'$, (1) without μ is

$$y_t = B x_t + u_t, \qquad u_t \sim N(0, \Sigma). \tag{3}$$

(A constant is one more column of x_t.) Transpose (3) and stack the n rows:

$$Y = XB' + U, \qquad U \sim N(0,\ I_n,\ \Sigma) \tag{4}$$

— the **matrix normal**: rows of U independent across t, each row with covariance Σ (slide 6 writes $Y = XB + U$ with B already transposed). Column i of (4) is equation i: $y^{(i)} = X\beta_i + u^{(i)}$, where β'_i is row i of B and $y^{(i)}$ the n observations of series i. Stacking the columns, with $\mathrm{vec}(XB') = (I_q \otimes X)\mathrm{vec}(B')$:

$$y = \mathrm{vec}(Y) = (I_q \otimes X)\beta + u, \qquad u \sim N(0,\ \Sigma \otimes I_n) \tag{5}$$

(slide 7). $\Sigma \otimes I_n$ says: errors of **different equations at the same date** are correlated (Σ_{ij}), errors at **different dates** are not. That is exactly the **seemingly unrelated regression** of Greenberg §9.1 (p. 134, eqs. 9.1–9.2: "the covariances permit nonzero correlations for disturbances with the same second subscript"), with the same regressor block x'_t in every equation.

---

4 · C — OLS is the conditional MLE, and also the SUR/GLS estimator

4.1 · The likelihood

Conditioning on the first p observations, (3) gives the log-likelihood

$$\ell(B,\Sigma) = -\frac{nq}{2}\log 2\pi - \frac{n}{2}\log|\Sigma| - \frac{1}{2}\sum_t (y_t - Bx_t)'\Sigma^{-1}(y_t - Bx_t). \tag{6}$$

A scalar is its own trace, and $\operatorname{tr}(ab') = b'a$, so

$$\sum_t (y_t - Bx_t)'\Sigma^{-1}(y_t - Bx_t) = \operatorname{tr}\Big(\Sigma^{-1}\sum_t (y_t - Bx_t)(y_t - Bx_t)'\Big) = \operatorname{tr}(\Sigma^{-1}S(B)), \tag{7}$$

with $S(B) = (Y - XB')'(Y - XB')$, a $q \times q$ matrix of residual cross-products.

4.2 · Completing the square in B

Define the OLS solution $\hat{B}' = (X'X)^{-1}X'Y$, so that the normal equations hold:

$$X'(Y - X\hat{B}') = 0. \tag{8}$$

Write $Y - XB' = (Y - X\hat{B}') + X(\hat{B} - B)'$ and expand $S(B)$:

$$S(B) = (Y - X\hat{B}')'(Y - X\hat{B}') + (Y - X\hat{B}')'X(\hat{B} - B)' + (\hat{B} - B)X'(Y - X\hat{B}') + (\hat{B} - B)X'X(\hat{B} - B)'.$$

By (8) the two middle terms are zero:

$$S(B) = S(\hat{B}) + (\hat{B} - B)X'X(\hat{B} - B)'. \tag{9}$$

Substituting (9) into (7):

$$\operatorname{tr}(\Sigma^{-1}S(B)) = \operatorname{tr}(\Sigma^{-1}S(\hat{B})) + \operatorname{tr}(\Sigma^{-1}(\hat{B} - B)X'X(\hat{B} - B)'). \tag{10}$$

The last trace equals $\operatorname{tr}(C'\Sigma^{-1}C) \geq 0$ with $C = (\hat{B} - B)(X'X)^{1/2}$, and is zero iff $B = \hat{B}$. So **for every** Σ, (6) is maximised in B at $\hat{B}$:

$$\boxed{\hat{B}' = (X'X)^{-1}X'Y \quad \Longleftrightarrow \quad \hat{\beta}_i = (X'X)^{-1}X'y^{(i)},\ i = 1, \dots, q} \tag{11}$$

— each equation by its own OLS (T p. 405: "Parameters … can be estimated by the ordinary least-squares (OLS) method"; p. 406: "the OLS estimates … are equivalent to the (conditional) ML estimates"). Then maximising $-\frac{n}{2}\log|\Sigma| - \frac{1}{2}\operatorname{tr}(\Sigma^{-1}S(\hat{B}))$ over Σ gives $\tilde{\Sigma} = S(\hat{B})/n$ (T eq. 8.19); the bias-adjusted version divides by $n - qp$ (one constant more if there is an intercept; T eq. 8.17 uses $T - 2i - 1$ for the univariate count).

4.3 · Why SUR/GLS adds nothing here

GLS on (5) is $\hat{\beta}_{GLS} = [(I_q \otimes X)'(\Sigma^{-1} \otimes I_n)(I_q \otimes X)]^{-1}(I_q \otimes X)'(\Sigma^{-1} \otimes I_n)y$. With $(A \otimes B)(C \otimes D) = AC \otimes BD$:

$$(I_q \otimes X)'(\Sigma^{-1} \otimes I_n)(I_q \otimes X) = \Sigma^{-1} \otimes X'X$$

$$(I_q \otimes X)'(\Sigma^{-1} \otimes I_n)\, y = (\Sigma^{-1} \otimes X')\, y$$

$$\hat{\beta}_{GLS} = (\Sigma \otimes (X'X)^{-1})(\Sigma^{-1} \otimes X')\, y = (I_q \otimes (X'X)^{-1}X')\, y, \tag{12}$$

which is (11) stacked. Σ **drops out because every equation has the same regressors.** The SUR gain of Greenberg §9.1 appears only when the X_j differ — for a VAR, when some coefficients are restricted to zero (the "refined" VAR of the site page, §6 of L09-02-vma-irf-fevd). The sampling covariance follows from (12) and $\mathrm{Cov}(y) = \Sigma \otimes I_n$:

$$\mathrm{Cov}(\hat{\beta}) = (I_q \otimes (X'X)^{-1}X')(\Sigma \otimes I_n)(I_q \otimes X(X'X)^{-1}) = \Sigma \otimes (X'X)^{-1}. \tag{13}$$

4.4 · Worked — the site's VAR(3) for UK/US/CA GDP growth (VAR-Example.html §2)

Quarterly growth of real GDP, standardised, 1980Q2–2011Q2 (Tsay 2014, Ex. 2.3), $q = 3$, $p = 3$, x_t of dimension 9, no constant (standardised data). The script builds $X = (\mathrm{lag1}, \mathrm{lag2}, \mathrm{lag3})$, computes `bhat = solve(t(X)%*%X) %*% t(X)%*%y` — that is (11) — and `Cov = kronecker(Shat, iXtX)`, which is (13). Printed:

$\hat{B}_1$	UK	US	CA
UK	0.4308	0.0587	0.1274
US	0.4380	0.2405	0.1747
CA	0.2938	0.4887	0.3231

and $\hat{\Sigma}$ with diagonal (0.5272, 0.5851, 0.4952). MTS's `VAR(rates,3)` (with a constant) prints $\hat{B}_1$ row 1 as (0.430, 0.0576, 0.128): the same coefficients to the third decimal, since the standardised series have mean zero. Its residual covariance differs (0.4918 for UK) because the script divides by a hand-typed 114 (`Shat = t(A)%*%A/(123-(3+1)*2-1)`) while MTS reports the ML version: **the divisor changes $\hat{\Sigma}$ and the standard errors, never $\hat{B}$** (by (11), $\hat{B}$ does not involve Σ).

Reading row 2 of $\hat{B}_1$: last quarter's UK growth enters US growth with 0.438 — larger than US growth's own coefficient 0.2405. **The equations are fitted separately, but they are read together**: this lead–lag pattern is what the impulse responses of L09-02-vma-irf-fevd turn into a picture.

---

5 · C — companion form and stationarity (slides 8–9; T eq. 8.15, pp. 404–405)

Stack $Y_t = (y_t', y_{t-1}', \ldots, y_{t-p+1}')'$ ($qp \times 1$). Then (1) without μ is the VAR(1)

$$Y_t = AY_{t-1} + U_t, \qquad A = \begin{pmatrix} B_1 & B_2 & \cdots & B_{p-1} & B_p \\ I_q & 0 & \cdots & 0 & 0 \\ 0 & I_q & \cdots & 0 & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & I_q & 0 \end{pmatrix}, \qquad U_t = \begin{pmatrix} u_t \\ 0 \\ \vdots \\ 0 \end{pmatrix}. \tag{14}$$

The first block row is (1); the others are the identities $y_{t-j} = y_{t-j}$. Everything proved for a VAR(1) in L01 §9 now applies to (14): it is stationary iff **all eigenvalues of A have modulus below one** (slide 9).

The equivalent root condition (slide 9: all z with $|I_q - B_1 z - \cdots - B_p z^p| = 0$ lie outside the unit circle). For $p = 2$, $A = \begin{pmatrix} B_1 & B_2 \\ I & 0 \end{pmatrix}$ and

$$\lambda I_{2q} - A = \begin{pmatrix} \lambda I - B_1 & -B_2 \\ -I & \lambda I \end{pmatrix}.$$

The block-determinant rule $\det \begin{pmatrix} P & Q \\ R & S \end{pmatrix} = \det(S)\det(P - QS^{-1}R)$ (for $\lambda \neq 0$) with $P = \lambda I - B_1$, $Q = -B_2$, $R = -I$, $S = \lambda I$ gives

$$\det(\lambda I - A) = \det(\lambda I_q)\, \det\left(\lambda I - B_1 - (-B_2)(\lambda I)^{-1}(-I)\right)$$

$$= \lambda^q \det\left(\lambda I - B_1 - \lambda^{-1} B_2\right)$$

$$= \lambda^{2q} \det\left(I - B_1\lambda^{-1} - B_2\lambda^{-2}\right). \tag{15}$$

With $z = 1/\lambda$, a nonzero eigenvalue λ of A is exactly the reciprocal of a root z of $|I - B_1 z - B_2 z^2| = 0$. Hence $|\lambda| < 1 \iff |z| > 1$. (The same Schur step, repeated, gives the general p; T p. 405: "$|I - \Phi B| = |\Phi(B)|$".)

> **Stationarity is a property of the whole system, not of any coefficient.** A diagonal entry above one can sit in a stationary VAR (T Ex. 8.3, $\Phi_{22} = 1.1$, L01 §9.1); all entries small can still sum to a unit root.

5.1 · Worked — the slide-13 simulated VAR(2)

$$B_1 = \begin{pmatrix} 0.7 & 0.1 & 0 \\ 0 & 0.4 & 0.1 \\ 0.9 & 0 & 0.8 \end{pmatrix}, \quad B_2 = \begin{pmatrix} -0.2 & 0 & 0 \\ 0 & 0.1 & 0.1 \\ 0 & 0 & 0 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} 0.26 & 0.03 & 0 \\ 0.03 & 0.09 & 0 \\ 0 & 0 & 0.81 \end{pmatrix}, \quad \mu = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$$

(slide 13, "based on Lütkepohl's (2007) Problem 2.3"). The 6×6 companion matrix (14) has eigenvalues

$$0.9099, \quad 0.4118 \pm 0.3382i \text{ (modulus } 0.5329), \quad 0.3456, \quad -0.1792, \quad 0,$$

all inside the unit circle: **stationary**, but with a dominant root of 0.91 (a half-life of $\ln 0.5 / \ln 0.9099 = 7.3$ periods). The zero root comes from the zero third row of B_2.

$\Phi(1) = I - B_1 - B_2 = \begin{pmatrix} 0.5 & -0.1 & 0 \\ 0 & 0.5 & -0.2 \\ -0.9 & 0 & 0.2 \end{pmatrix}$, $|\Phi(1)| = 0.032$: small, because $z = 1$ is close to a root. If slide 13's μ is the intercept (as the constant in Sims' code), (2) gives the mean. Solve $\Phi(1)m = \mu$ row by row:

- row 3: $-0.9m_1 + 0.2m_3 = 0 \Rightarrow m_3 = 4.5m_1$;
- row 2: $0.5m_2 - 0.2(4.5m_1) = 1 \Rightarrow m_2 = 2 + 1.8m_1$;
- row 1: $0.5m_1 - 0.1(2 + 1.8m_1) = 2 \Rightarrow 0.32m_1 = 2.2 \Rightarrow m_1 = 6.875$;

so $m = (6.875,\ 14.375,\ 30.94)'$. (A seeded 60-period simulation used in L09-04-bvar-minnesota-ssvs has sample means 6.81, 14.19, 30.24.) **A small $|\Phi(1)|$ magnifies the intercept** into a large mean, the multivariate version of $\mu = \phi_0/(1 - \phi_1)$ with ϕ_1 near one.

5.2 · Worked — the UK/US/CA VAR(3)

The 9×9 companion matrix of §4.4's $\hat{B}_1, \hat{B}_2, \hat{B}_3$ has moduli 0.6107 (×2), 0.5842 (×2), 0.5620, 0.5165 (×2), 0.3288 (×2) all inside, so the fitted VAR(3) is stationary and its impulse responses die out (they are below 0.005 by $h = 20$, L09-02-vma-irf-fevd §3).

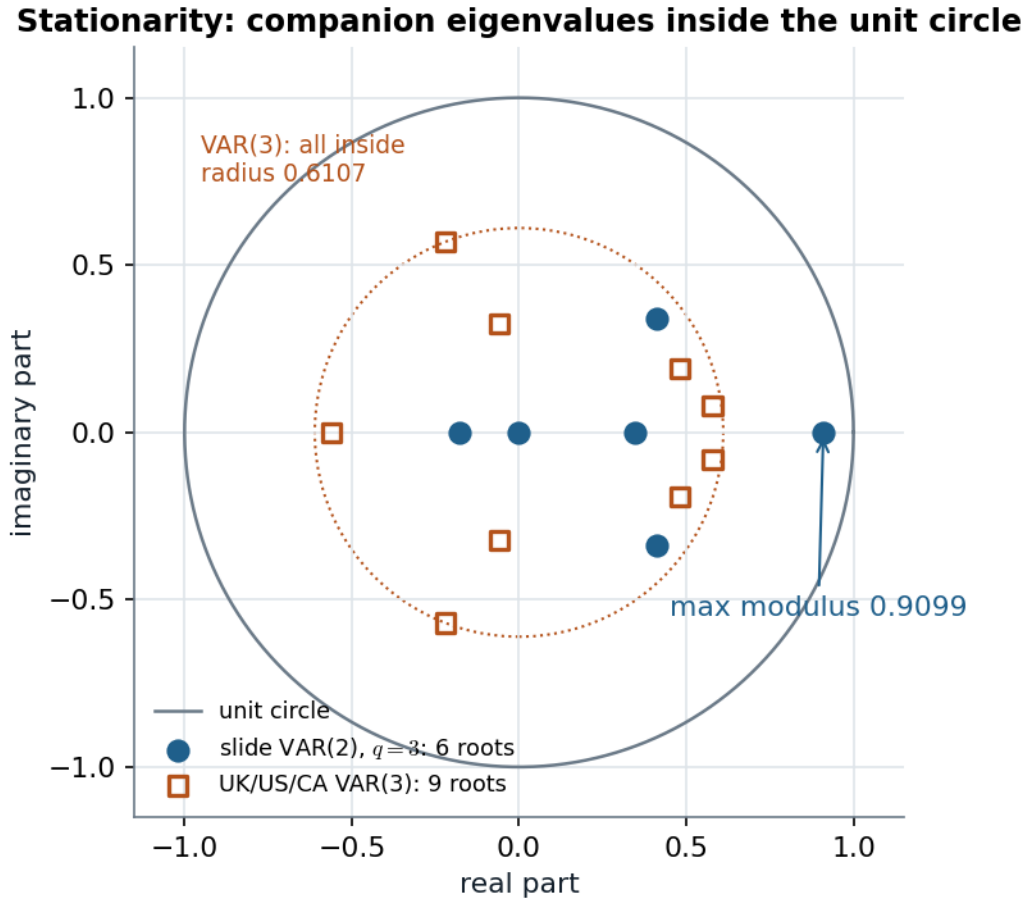


Figure — eigenvalues of the companion matrix (14) for the slide-13 VAR(2) and the site's OLS VAR(3); slide 9.

Both sets lie inside the unit circle. The VAR(2)'s root at 0.91 sits close to the boundary — the system is persistent — while the GDP VAR(3) has no root above 0.61. Complex pairs mean damped oscillation in the responses; the VAR(3) has four such pairs.

6 · C — forecasting (slide 8; T p. 409)

From (14), $Y_{t+h} = A^h Y_t + \sum_{i=0}^{h-1} A^i U_{t+h-i}$. Taking $E(\cdot \mid F_t)$ and reading the first q rows with $J = (I_q, 0, \ldots, 0)$:

$$y_t(h) = E(y_{t+h} \mid F_t) = JA^h Y_t \tag{16}$$

(slide 8: $F(A, y_t, h) = A^h y_t$, "a highly nonlinear function of $B_1, \ldots, B_p$"). The forecast error uses $JA^i J' = \Psi_i$ (the VMA weights, L09-02-vma-irf-fevd (3)):

$$e_t(h) = y_{t+h} - y_t(h) = \sum_{i=0}^{h-1} \Psi_i u_{t+h-i} \tag{17}$$

and, the u's being uncorrelated across dates,

$$\mathrm{MSE}(h) = \mathrm{Cov}(e_t(h)) = \Sigma + \Psi_1 \Sigma \Psi_1' + \cdots + \Psi_{h-1} \Sigma \Psi_{h-1}' \tag{18}$$

(slide 11; for $h = 2$, T p. 409: $e_h(2) = a_{h+2} + \Phi_1 a_{h+1}$, covariance $\Sigma + \Phi_1 \Sigma \Phi_1'$). Under stationarity, as $h \to \infty$, $y_t(h) \to m$ and $\mathrm{MSE}(h) \to \Gamma_0 = \mathrm{Cov}(y_t)$.

Bayesian forecast (slide 8). With a posterior for (B, Σ):

$$\hat{y}_t(h) = \int F(A, y_t, h)\, p(B, \Sigma \mid \text{data})\, d(B, \Sigma). \tag{19}$$

Because F is a power of A, $E[A^h] \neq (EA)^h$: plugging the posterior mean into (16) is not the posterior mean forecast. Posterior draws of B pushed through (16) give it, and its bands.

6.1 · Worked — Tsay's VAR(1) for IBM and S&P 500 (pp. 414–416)

Fitted: $\phi_0 = (1.0614,\ 0.4087)'$, $\Phi = \begin{pmatrix} -0.0320 & 0.1503 \\ -0.0223 & 0.1020 \end{pmatrix}$, residual sds 7.0078, 5.5247. By (2):

$$I - \Phi = \begin{pmatrix} 1.0320 & -0.1503 \\ 0.0223 & 0.8980 \end{pmatrix}, \qquad |I - \Phi| = 1.0320(0.8980) - (-0.1503)(0.0223) = 0.92674 + 0.00335 = 0.93009,$$

$$m_1 = \frac{0.8980(1.0614) + 0.1503(0.4087)}{0.93009} = \frac{0.95314 + 0.06143}{0.93009} = 1.0908, \qquad m_2 = \frac{-0.0223(1.0614) + 1.0320(0.4087)}{0.93009}$$

Tsay's 6-step forecasts are 1.0909 and 0.4280: **the forecast has already converged to the mean**, because the eigenvalues of Φ are 0.0013 and 0.0687. The forecast standard errors converge to $\sqrt{\mathrm{diag}\,\Gamma_0}$; with Γ_0 from the vec formula of L01 (25) and the innovation correlation 0.642 (§2 of L09-02-vma-irf-fevd), $\sqrt{\mathrm{diag}\,\Gamma_0} = (7.0436,\ 5.5455)$ against Tsay's printed 7.0436, 5.5454; at $h = 2$, (18) gives (7.0434, 5.5454) against the printed 7.0434, 5.5453.

7 · C — choosing p (T §8.2.4, pp. 405–407)

Fit VAR(0), VAR(1), … by OLS; with $\tilde{\Sigma}_i$ the ML residual covariance (8.19) of the VAR(i) and T the sample size:

$$M(i) = -\Big(T - k - i - \tfrac{3}{2}\Big) \ln \frac{|\tilde{\Sigma}_i|}{|\tilde{\Sigma}_{i-1}|} \overset{a}{\sim} \chi^2_{k^2} \quad (H_0 : \Phi_i = 0) \tag{20}$$

$$\mathrm{AIC}(i) = \ln|\tilde{\Sigma}_i| + \frac{2k^2 i}{T}, \qquad \mathrm{BIC}(i) = \ln|\tilde{\Sigma}_i| + \frac{k^2 i \ln T}{T}, \qquad \mathrm{HQ}(i) = \ln|\tilde{\Sigma}_i| + \frac{2k^2 i \ln\ln T}{T}. \tag{21}$$

Each lag adds k^2 coefficients, hence the k^2 in every penalty and in the χ^2 degrees of freedom. BIC's penalty $\ln T$ exceeds AIC's 2 once $T > e^2 \approx 7.4$, so **BIC never picks a longer lag than AIC**.

7.1 · Worked — `VARorder(rates)` on the GDP data (VAR-Example.html §3.1)

p	AIC	BIC	HQ	$M(p)$	p-value
0	−1.6743	−1.6743	−1.6743	—	—
1	−2.6013	**−2.3976**	**−2.5185**	115.13	0.0000
2	**−2.6825**	−2.2752	−2.5171	23.54	0.0051
3	−2.6418	−2.0309	−2.3936	10.49	0.3126
4	−2.6154	−1.8008	−2.2845	11.58	0.2382

"selected order: aic = 2, bic = 1, hq = 1". With $k = 3$, $M(p) \sim \chi^2_9$, 5% critical value 16.92: $M(2) = 23.54$ rejects $\Phi_2 = 0$, $M(3) = 10.49$ does not, so the sequential test stops at $p = 2$ (with a stray rejection again at $p = 8$, $M = 24.48$). HQ is a near-tie: −2.5185 at $p = 1$ vs −2.5171 at $p = 2$. The lecturer fits $p = 3$ anyway, one lag beyond AIC — a choice the criteria do not support but that costs only 9 coefficients here; L09-04-bvar-minnesota-ssvs replaces the lag-length decision by a prior that shrinks long lags toward zero.

---

8 · C — the curse of dimensionality (slides 27, 45)

Without intercepts, a VAR(p) has $q^2 p$ coefficients plus $q(q+1)/2$ in Σ:

VAR(1) (slide 45, Bańbura–Giannone–Reichlin sizes)	q	$q^2 + q(q+1)/2$
Small	3	$9 + 6 = 15$
Medium	20	$400 + 210 = 610$
Large	131	$17{,}161 + 8{,}646 = 25{,}807$

Kadiyala–Karlsson's revisit of Litterman (slide 27): $q = 7$, $p = 6$, "$7 + 6(7^2) = 301$ parameters" (intercepts and coefficients, Σ excluded) from 133 quarterly observations — **43 coefficients per equation from 133 data points**. OLS is still defined, but its variance (13) scales with $(X'X)^{-1}$ in a 43-dimensional regression: forecasts are poor. That is the opening of the Bayesian half of the lecture.

---

9 · U — the site's R and its Python twin

`VAR-Example.html` does (11) and (13) by hand, then `MTS::VAR`, `VARorder`, `refVAR`, `FEVdec`. Python equivalents (used in L09-10-python): `numpy.linalg.lstsq` for (11), `np.kron` for (13), `statsmodels.tsa.api.VAR(y).fit(p, trend="n")` for the fit, `.select_order(maxlags)` for (21) (statsmodels normalises the criteria differently; only the arg-min matters, as T p. 414 says of S-Plus vs SCA), `np.linalg.eigvals` on (14).

Links

- Neighbours: L09-02-vma-irf-fevd (the Ψ_i of (17)–(18)) · L09-04-bvar-minnesota-ssvs (the answer to §8).
- Unit: L09-00-map · L09-08-formulas · L09-09-pitfalls · L09-10-python.
- Class 1: L01 §8–§9 (AR(p) as a regression; VAR(1) mean, Γ_0 by the vec formula, T Ex. 8.3) · L01-06 (OLS algebra).
- Rules: rules/05.
- **Practice (number and printed page only):** T 8.1 (p. 462); T 8.2 (pp. 462–463); T 8.3, 8.4, 8.5 (p. 463); G 10.4 (p. 167) for the SUR structure of §3.

Slides 10–12 · Impulse responses and variance decompositions: the Ψ_s recursion, the Cholesky factor, and what the ordering decides

> **Claim.** A stationary VAR is a moving average of its innovations, $y_t = \sum_s \Psi_s u_{t-s}$, with $\Psi_s = B_1\Psi_{s-1} + \cdots + B_p\Psi_{s-p}$. $\Psi_s(i,j)$ **is the response of variable** i**,** s **periods later, to a unit innovation in variable** j — but innovations are correlated, so "holding the others constant" is not an experiment the data contain. A Cholesky factor P of Σ makes the shocks orthogonal, $\Theta_s = \Psi_s P$, and then the h-step forecast-error variance splits exactly into shares, one per shock. **Both objects depend on the ordering** of the variables, which is an identifying assumption, not a detail (L09-03-structural-var).

Prev: L09-01-var-p-estimation-forecasting · Next: L09-03-structural-var · Map: L09-00-map

Notation. As in L09-01-var-p-estimation-forecasting, plus:

Symbol	Meaning
Ψ_s	$q \times q$ VMA weight / reduced-form impulse response at horizon s; $\Psi_0 = I_q$
P	lower-triangular Cholesky factor, $PP' = \Sigma$, positive diagonal
L, G	unit-lower-triangular L and diagonal G with $\Sigma = LGL'$ (Tsay); $P = LG^{1/2}$
$\varepsilon_t = P^{-1}u_t$	orthogonalised shocks, $N(0, I_q)$
$\Theta_s = \Psi_s P$	orthogonalised impulse response (response to a one-standard-deviation shock)
$\omega_{ij}(h)$	share of variable i's h-step forecast-error variance due to shock j

---

1 · C — the VMA(∞) representation (slide 10; T eq. 8.21, p. 412)

Write the VAR(p) with the lag operator L: $\Phi(L)y_t = u_t$, $\Phi(L) = I - B_1L - \cdots - B_pL^p$. If the companion eigenvalues lie inside the unit circle, $\Phi(L)$ is invertible and

$$y_t = \Phi(L)^{-1}u_t = \Psi(L)u_t = \sum_{i=0}^{\infty} \Psi_i u_{t-i}. \tag{1}$$

The Ψ_i solve $\Phi(L)\Psi(L) = I$:

$$(I - B_1L - \cdots - B_pL^p)(\Psi_0 + \Psi_1L + \Psi_2L^2 + \dots) = I. \tag{2}$$

Collect the coefficient of L^s on the left and set it equal to the right-hand side's (I for $s = 0$, 0 for $s \geq 1$):

- L^0: $\Psi_0 = I_q$;
- L^s, $s \geq 1$: $\Psi_s - B_1\Psi_{s-1} - \cdots - B_p\Psi_{s-p} = 0$, with $\Psi_{s-i} = 0$ when $s - i < 0$.

$$\boxed{\Psi_0 = I_q, \qquad \Psi_s = B_1\Psi_{s-1} + B_2\Psi_{s-2} + \cdots + B_p\Psi_{s-p}, \quad s = 1, 2, \dots} \tag{3}$$

(slide 10). Equivalently $\Psi_s = JA^sJ'$ with the companion matrix A and $J = (I_q, 0, \dots, 0)$, so $\Psi_s \to 0$ geometrically at the rate of the largest companion eigenvalue. For a VAR(1), $\Psi_s = B_1^s$.

---

2 · C — the impulse-response function (slide 12; T p. 412)

Differentiating (1):

$$\frac{\partial y_{t+s}}{\partial u_t'} = \Psi_s. \tag{4}$$

Slide 12: "the (i, j) element of Ψ_s identifies the consequences of a one-unit increase in the innovation of variable j at time t for the value of variable i at time $t + s$, holding all other innovations at all dates constant." Plotted against s it is the **impulse-response function**, "highly nonlinear on $B_1, \dots, B_p$" (as the forecast was: L09-01-var-p-estimation-forecasting (16)).

The problem (T p. 413): "since the components of a_t are often correlated, the interpretation of elements in Ψ_i … is not straightforward." If $\Sigma_{12} \neq 0$, a unit u_{1t} comes on average with $E(u_{2t} \mid u_{1t} = 1) = \Sigma_{21}/\Sigma_{11}$, so "holding u_{2t} constant" describes a shock pattern the data rarely produce.

2.1 · Worked — Ψ_2 of the UK/US/CA VAR(3)

By (3), $\Psi_1 = B_1$ and $\Psi_2 = B_1\Psi_1 + B_2 = B_1^2 + B_2$. Entry (UK, UK), with $\hat{B}_1, \hat{B}_2$ from L09-01-var-p-estimation-forecasting §4.4 and VAR-Example.html §2.1:

$$\Psi_2(1,1) = \underbrace{0.4308(0.4308) + 0.0587(0.4380) + 0.1274(0.2938)}_{(B_1^2)_{11}} + \underbrace{0.0008}_{B_2(1,1)} = 0.18559+0.02571+0.03743+0.0008 = 0.2495$$

The full matrices:

	Ψ_2: UK	US	CA	Ψ_3: UK	US	CA
UK	0.2495	0.0578	0.2545	0.2754	0.3335	−0.1409
US	0.1354	0.3399	0.0597	0.1639	0.1141	0.0078
CA	0.2690	0.3155	0.0357	0.1506	0.1013	0.1204

Errata in the site script. The loop plots `Psi[2:H,i,j]` under the title "Response of `names[j]` to shock in `names[i]`". By (4), $\Psi_s(i, j)$ is the response of i to a shock in j, so every off-diagonal title is transposed. The script also starts the plot at $s = 1$ (`Psi[2:H,,]`), which hides $\Psi_0 = I$.

---

3 · C — orthogonalised responses (slide 11; T eq. 8.22, p. 413)

Σ is positive definite, so it has a Cholesky factorisation $\Sigma = PP'$ with P lower triangular and positive diagonal (Tsay's $\Sigma = LGL'$ with $P = LG^{1/2}$). Define

$$\varepsilon_t = P^{-1}u_t \qquad \Longrightarrow \qquad \mathrm{Cov}(\varepsilon_t) = P^{-1}\Sigma(P^{-1})' = P^{-1}PP'(P')^{-1} = I_q. \tag{5}$$

Insert $u_{t-i} = P\varepsilon_{t-i}$ into (1):

$$y_t = \sum_{i=0}^{\infty} \Psi_i P\, \varepsilon_{t-i} = \sum_{i=0}^{\infty} \Theta_i \varepsilon_{t-i}, \qquad \Theta_i = \Psi_i P. \tag{6}$$

$\Theta_s(i,j)$ is the response of $y_{i,t+s}$ to a **one-standard-deviation** orthogonal shock ε_{jt}. At impact, $\Theta_0 = P$ is lower triangular: **shock 1 moves every variable at once; shock q moves only variable q at impact.** Tsay (p. 413): "the result depends on the ordering of the components of r_t. In particular, $b_{1t} = a_{1t}$ so that a_{1t} is not transformed." Slide 11 writes the same with $\varepsilon_t = A^{-1}u_t \sim N(0, D)$, $\Sigma = ADA'$ (unit-lower A, diagonal D, "via singular value decomposition or Cholesky decomposition").

3.1 · Worked — Tsay's VAR(1) for IBM and the S&P 500 (pp. 414–417)

S-Plus `impRes` prints, responses in rows, innovations in columns (IBM first):

$$\Theta_0 = \begin{pmatrix} 6.9973 & 0 \\ 3.5432 & 4.2280 \end{pmatrix}, \qquad \Theta_1 = \begin{pmatrix} 0.3088 & 0.6353 \\ 0.2050 & 0.4312 \end{pmatrix}.$$

Check $\Theta_1 = \Phi\Theta_0$ with $\Phi = \begin{pmatrix} -0.0320 & 0.1503 \\ -0.0223 & 0.1020 \end{pmatrix}$ (p. 415):

$$\Theta_1(1,1) = -0.0320(6.9973) + 0.1503(3.5432) = -0.22391 + 0.53254 = 0.3086$$
$$\Theta_1(1,2) = -0.0320(0) + 0.1503(4.2280) = 0.6355$$
$$\Theta_1(2,1) = -0.0223(6.9973) + 0.1020(3.5432) = -0.15604 + 0.36141 = 0.2054$$
$$\Theta_1(2,2) = 0.1020(4.2280) = 0.4313$$

— the printed values to the rounding of the four-decimal Φ. **The implied $\Sigma = \Theta_0\Theta_0'$:** $\Sigma_{11} = 6.9973^2 = 48.96$, $\Sigma_{21} = 6.9973(3.5432) = 24.79$, $\Sigma_{22} = 3.5432^2 + 4.2280^2 = 12.554 + 17.876 = 30.43$, correlation $3.5432/\sqrt{30.43} = 0.642$ — the "concurrent correlation … 0.63" of Tsay's VAR(5) (p. 408). The reading: a one-sd S&P innovation (4.23 points, orthogonal to IBM's) raises next month's IBM return by 0.64 — the "unidirectional dynamic relationship from the monthly S&P 500 index return to the IBM return" (p. 408) — while the IBM shock moves the S&P **at impact** by 3.54, only because IBM was ordered first.

3.2 · Worked — the UK/US/CA VAR(3)

Cholesky of the script's $\hat{\Sigma}$ (VAR-Example.html §2.1):

$$P = \begin{pmatrix} 0.7261 & 0 & 0 \\ 0.2126 & 0.7348 & 0 \\ 0.1077 & 0.2800 & 0.6365 \end{pmatrix}, \qquad P_{11} = \sqrt{0.52719} = 0.7261, \quad P_{21} = \frac{0.15433}{0.7261} = 0.2126,$$

$P_{22} = \sqrt{0.58511 - 0.2126^2} = \sqrt{0.58511 - 0.04520} = \sqrt{0.53991} = 0.7348$. Then $\Theta_1 = \hat{B}_1 P$; e.g. Canada's response one quarter after a one-sd US shock:

$$\Theta_1(3,2) = 0.2938(0) + 0.4887(0.7348) + 0.3231(0.2800) = 0.35910 + 0.09047 = 0.4496.$$

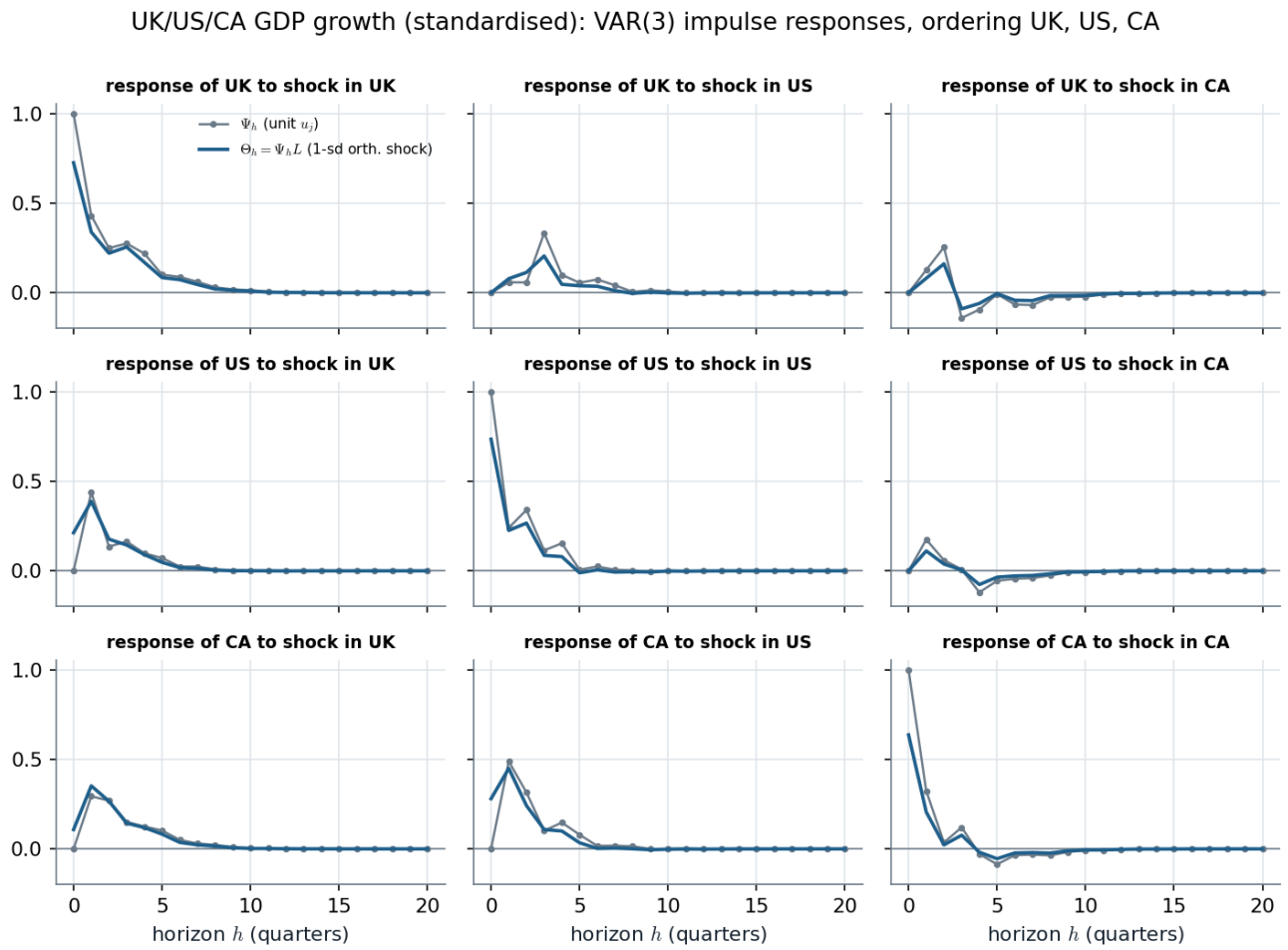


Figure — Ψ_h from (3) (grey, unit innovation) and $\Theta_h = \Psi_h P$ from (6) (blue, one-sd orthogonal shock), OLS VAR(3) of VAR-Example.html §2.1, ordering UK, US, CA.

Above the diagonal the blue curves start at zero (a later-ordered shock cannot move an earlier variable at impact); below it they start at $P_{ij} > 0$. The largest cross-effects run **from the US to Canada** (blue peak about 0.45 at $h = 1$) and from the UK to the US. Every response is negligible after ten quarters, as the companion moduli (at most 0.61) imply.

---

4 · C — forecast-error variance decomposition (slide 11)

Start from the forecast MSE (L09-01-var-p-estimation-forecasting (18)) and substitute $\Sigma = PP' = \sum_j p_j p_j'$, where p_j is column j of P:

$$\mathrm{MSE}(h) = \sum_{s=0}^{h-1} \Psi_s \Sigma \Psi_s' = \sum_{s=0}^{h-1} \Psi_s \Big(\sum_{j=1}^{q} p_j p_j' \Big) \Psi_s' = \sum_{j=1}^{q} \Big[\sum_{s=0}^{h-1} (\Psi_s p_j)(\Psi_s p_j)' \Big]. \tag{7}$$

The bracket is the contribution of orthogonal shock j (slide 11 writes it $d_j(a_j a_j' + \Psi_1 a_j a_j' \Psi_1' + \cdots + \Psi_{h-1} a_j a_j' \Psi_{h-1}')$ with $p_j = \sqrt{d_j}\, a_j$). Its (i,i) element is $\sum_s \Theta_s(i,j)^2$, so the share of variable i's h-step forecast-error variance due to shock j is

$$\boxed{\omega_{ij}(h) = \frac{\sum_{s=0}^{h-1} \Theta_s(i,j)^2}{\sum_{k=1}^{q} \sum_{s=0}^{h-1} \Theta_s(i,k)^2}}, \qquad \sum_j \omega_{ij}(h) = 1. \tag{8}$$

At $h = 1$, $\Theta_0 = P$, so $\omega_{ij}(1) = P_{ij}^2/\Sigma_{ii}$; for $i = 2, j = 1$: $P_{21}^2/\Sigma_{22} = (\Sigma_{21}/\sqrt{\Sigma_{11}})^2/\Sigma_{22} = \Sigma_{21}^2/(\Sigma_{11}\Sigma_{22}) = \rho_{21}^2$, the squared innovation correlation. **The first-ordered variable takes all of the shared variation.**

4.1 · Worked — reproducing the site's `FEVdec` table (VAR-Example.html §3.2)

The site first runs `refVAR`, which drops insignificant coefficients (e.g. $\hat{B}_1(1,2) = 0$, the whole third row of $\hat{B}_3$ set to zero), so the model now has **different regressors per equation** and is estimated by GLS — the case where SUR matters (L09-01-var-p-estimation-forecasting §4.3). Its residual covariance has $\Sigma_{11} = 0.49497$, $\Sigma_{22} = 0.55162$, $\Sigma_{21} = 0.14385$, $\Sigma_{33} = 0.46841$, $\Sigma_{31} = 0.07301$.

$h = 1$, by the formula just derived:

$$\omega_{21}(1) = \frac{0.14385^2}{0.49497(0.55162)} = \frac{0.020692}{0.273037} = 0.07579, \qquad \omega_{31}(1) = \frac{0.07301^2}{0.49497(0.46841)} = \frac{0.0053307}{0.231850} = 0.02299.$$

`FEVdec` prints 0.07578528 and 0.02299206. The first forecast-error sd is $\sqrt{0.49497} = 0.7035$ (printed 0.7035426). Running (3), (6) and (8) on the refined matrices reproduces the whole table; at $h = 20$:

share of … due to shock in →	UK	US	CA
UK	0.8774	0.0676	0.0550
US	0.2657	0.7051	0.0292
CA	0.2379	0.3263	0.4359

(printed values; (8) on the three-decimal refined coefficients gives each to within 0.001). **Canada's growth is mostly explained by foreign shocks** (56% at 20 quarters), the UK's almost entirely by its own (88%) — partly because the UK is ordered first.

Errata in the site script. It plots `vd.can = OmegaR[2,]` and `vd.us = OmegaR[3,]` and puts the legend "UK, CAN, US" on the shocks, but the series order is (uk, us, ca) (`cor(rates)` prints `uk us ca`): row 2 is the **US** and row 3 is **Canada**. The table above uses the correct labels.

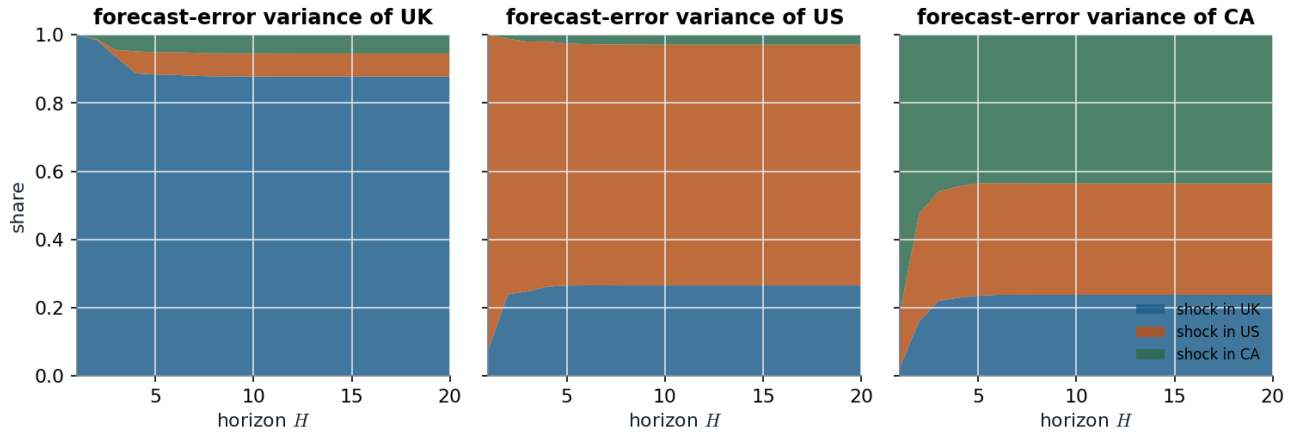


Figure — FEVD (8) of the refined VAR(3) (VAR-Example.html §3.2), ordering UK, US, CA; asserted against the printed `FEVdec` *values at* $h = 1$ *and* $h = 20$.

The shares move only in the first four or five quarters and are flat afterwards: once the responses die out, extra horizons add nothing to either numerator or denominator of (8). At $h = 1$ the panels are triangular — UK 100% own, Canada 82% own — which is the Cholesky ordering, not a finding.

---

5 · U — uncertainty bands, two ways

- **The site's bands** (VAR-Example.html §4.1): draw $\beta^{(r)} = \hat{\beta} + \text{chol}(\hat{\Sigma} \otimes (X'X)^{-1})'z$, $z \sim N(0, I)$, $r = 1, \ldots, 1000$, recompute (3), report the 2.5/50/97.5% quantiles. This is the normal approximation (13) of L09-01-var-p-estimation-forecasting — equivalently the posterior under a flat prior with Σ fixed at $\hat{\Sigma}$.
- **The slides' bands** (slides 14–17): posterior draws of (μ, B_1, B_2, Σ) for the simulated VAR(2) ("Posterior summaries based on 1000 draws", Sims' code), each pushed through (3) and (6). With 1000 observations the posteriors of B_1, B_2 concentrate near the true values of slide 13, and the IRF panels (slide 17) show, e.g., the response of Y_3 to its own shock starting near $\sqrt{0.81} = 0.9$.
- **Zivot–Wang example** (slides 18–21): monthly real stock returns, real interest rate, real IP growth and inflation, 1947.1–1987.12; posterior means of B_1, B_2 on slide 19, IRFs and the variance decomposition on slides 20–21 (figures only).

Because the IRF is nonlinear in B, the band of Ψ_s is **not** $\Psi_s(\hat{B}) \pm 2$ se's of anything simple: simulate.

Links

- Neighbours: L09-01-var-p-estimation-forecasting (MSE (18), companion form) · L09-03-structural-var (what makes an ordering defensible).
- Unit: L09-00-map · L09-08-formulas · L09-09-pitfalls · L09-10-python.
- Class 1: L01 §9 (the VAR(1) whose $\Psi_s = \Phi^s$ these generalise).
- Rules: rules/05.
- **Interactive companion.** The Ordering Dial (`companions/companion-var-irf-ordering.html`) — move the four coefficients and the innovation correlation of a bivariate VAR(1), flip the ordering, watch Θ_h and the FEVD change.
- **Practice (number and printed page only):** T 8.5 (p. 463); T 8.7(a)–(c) (p. 463: IRFs and forecasts; parts (d)–(e) are cointegration, outside Class 9).

Slides 31–37 · Structural VARs: the reduced form cannot tell orthogonal rotations apart, so $q(q-1)/2$ restrictions must come from economics

> **Claim.** A structural VAR $A_0 y_t = A_1 y_{t-1} + \cdots + u_t$ with orthogonal unit-variance shocks maps to the reduced form by $B_i = A_0^{-1} A_i$, $\Sigma = (A_0' A_0)^{-1}$. Any orthogonal Q turns (A_0, A) into (QA_0, QA) **with exactly the same** (B, Σ), so the data identify Σ's $q(q+1)/2$ numbers and nothing more; A_0 has q^2. The gap, $q(q-1)/2$, must be filled by restrictions. **The Cholesky ordering of L09-02-vma-irf-fevd is one such set — a recursive SVAR —** and Tsay's Example 8.3 shows two orderings giving two different, equally valid "structural" equations from one reduced form.

Prev: L09-02-vma-irf-fevd · Next: L09-04-bvar-minnesota-ssvs · Map: L09-00-map

Notation.

Symbol	Meaning
A_0	$q \times q$ contemporaneous structural matrix, invertible
A_i, $A = (A_1, \ldots, A_p)$	structural lag matrices
u_t	structural shocks, $N(0, I_q)$ (slide 33)
B_i, Σ	reduced-form coefficients and innovation covariance
$e_t = A_0^{-1} u_t$	reduced-form innovation (called u_t in L09-01-var-p-estimation-forecasting)
Q	$q \times q$ orthogonal matrix, $Q'Q = I$
L, G	Tsay's unit-lower-triangular L and diagonal G, $\Sigma = LGL'$

---

1 · U — the question

Slide 31 quotes Rubio-Ramírez, Waggoner and Zha (2010): "Since the seminal work by Sims (1980), identification of structural vector autoregressions (SVARs) has been an unresolved theoretical issue. Filling this theoretical gap is of vital importance because impulse responses based on SVARs have been widely used for policy analysis and to provide stylized facts for dynamic stochastic general equilibrium (DSGE) models." A reduced-form VAR forecasts; a **policy question** ("what does a monetary shock do to output?") needs a shock with an economic name, and naming it is identification.

---

2 · C — from structure to reduced form (slide 33)

$$A_0 y_t = A_1 y_{t-1} + \cdots + A_p y_{t-p} + u_t, \qquad u_t \overset{iid}{\sim} N(0, I_q). \tag{1}$$

Premultiply (1) by A_0^{-1}:

$$y_t = A_0^{-1} A_1 y_{t-1} + \cdots + A_0^{-1} A_p y_{t-p} + A_0^{-1} u_t. \tag{2}$$

Matching (2) with the reduced form $y_t = B_1 y_{t-1} + \cdots + B_p y_{t-p} + e_t$:

$$B_i = A_0^{-1} A_i, \qquad B = A_0^{-1} A, \qquad e_t = A_0^{-1} u_t. \tag{3}$$

The innovation covariance, using $\mathrm{Cov}(u_t) = I$ and $(MN)^{-1} = N^{-1}M^{-1}$:

$$\Sigma = \mathrm{Cov}(e_t) = A_0^{-1} I (A_0^{-1})' = A_0^{-1}(A_0')^{-1} = (A_0' A_0)^{-1}. \tag{4}$$

These are slide 33's "$B_i = A_0^{-1}A_i$, $B = A_0^{-1}A$, $\Sigma = (A_0'A_0)^{-1}$". The structural impulse response follows from L09-02-vma-irf-fevd (1) with $e_t = A_0^{-1}u_t$:

$$y_t = \sum_s \Psi_s A_0^{-1} u_{t-s}, \qquad \frac{\partial y_{t+s}}{\partial u_t'} = \Psi_s A_0^{-1}. \tag{5}$$

---

3 · C — why the reduced form cannot identify A_0

Slide 34 defines $g(A_0, A) = (A_0^{-1}A,\ (A_0'A_0)^{-1})$. Take any orthogonal Q and set $\tilde{A}_0 = QA_0$, $\tilde{A} = QA$. Then

$$\tilde{A}_0^{-1}\tilde{A} = (QA_0)^{-1}QA = A_0^{-1}Q^{-1}QA = A_0^{-1}A = B, \tag{6}$$

$$(\tilde{A}_0'\tilde{A}_0)^{-1} = (A_0'Q'QA_0)^{-1} = (A_0'A_0)^{-1} = \Sigma. \tag{7}$$

So $g(QA_0, QA) = g(A_0, A)$: **the likelihood is flat along the orbit** $\{QA_0\}$. Since B is pinned once A_0 is (by (3), $A = A_0 B$), everything reduces to recovering A_0 from Σ.

Order condition (counting). Σ is symmetric: $q(q+1)/2$ free numbers. A_0 has q^2. The shortfall is

$$q^2 - \frac{q(q+1)}{2} = \frac{q(q-1)}{2} \tag{8}$$

— the dimension of the orthogonal group. With $q = 2$: one restriction; $q = 3$: three; $q = 5$: ten; $q = 6$: fifteen. Restrictions are typically zeros in A_0 (a variable does not enter an equation contemporaneously).

Definition (slide 34, verbatim): "The SVAR is exactly identified if and only if, for almost any reduced-form parameter point (B, Σ), there exists a unique structural parameter point $(A_0, A) \in R$ such that $g(A_0, A) = (B, \Sigma)$." Counting is necessary, not sufficient: the zeros must also be placed so that no rotation other than $Q = I$ (up to sign) preserves them — the rank condition that Rubio-Ramírez, Waggoner and Zha make operational. "Much of the SVAR literature involves exactly identified models" (slide 33). Waggoner and Zha (2003) give an MCMC algorithm for a restricted A_0 (slide 34).

---

4 · C — recursive identification is the Cholesky factor

Restrict A_0 to be **lower triangular** with positive diagonal: $q(q-1)/2$ zeros above the diagonal, exactly (8). Then A_0^{-1} is lower triangular with positive diagonal and, by (4), $A_0^{-1}(A_0^{-1})' = \Sigma$. The Cholesky factor is unique, so

$$A_0^{-1} = P \quad \text{(the lower Cholesky factor of } \Sigma), \qquad \Psi_s A_0^{-1} = \Psi_s P = \Theta_s. \tag{9}$$

The orthogonalised IRF of L09-02-vma-irf-fevd (6) is the structural IRF of a recursive SVAR. The economic content of the ordering: variable 1's equation contains no other current variable (it reacts to nothing within the period); variable q's contains them all.

4.1 · Worked — Tsay Example 8.3 (pp. 400–401), both orderings

Reduced form: $\phi_0 = (0.2,\ 0.4)'$, $\Phi = \begin{pmatrix} 0.2 & 0.3 \\ -0.6 & 1.1 \end{pmatrix}$, $\Sigma = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ (the VAR(1) of L01 §9.1).

Ordering (r_1, r_2)**.** Factor $\Sigma = LGL'$ with $L = \begin{pmatrix} 1 & 0 \\ l & 1 \end{pmatrix}$, $G = \mathrm{diag}(g_1, g_2)$:

$$LGL' = \begin{pmatrix} g_1 & lg_1 \\ lg_1 & l^2 g_1 + g_2 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \Rightarrow g_1 = 2, \quad l = \tfrac{1}{2}, \quad g_2 = 1 - \tfrac{1}{4}(2) = 0.5.$$

$L^{-1} = \begin{pmatrix} 1 & 0 \\ -0.5 & 1 \end{pmatrix}$ (T: "$L^{-1} = \begin{pmatrix} 1.0 & 0.0 \\ -0.5 & 1.0 \end{pmatrix}$"). Premultiply the reduced form by L^{-1} (T eq. 8.9):

$$L^{-1}\phi_0 = \begin{pmatrix} 0.2 \\ -0.5(0.2) + 0.4 \end{pmatrix} = \begin{pmatrix} 0.2 \\ 0.3 \end{pmatrix}, \qquad L^{-1}\Phi = \begin{pmatrix} 0.2 & 0.3 \\ -0.5(0.2) - 0.6 & -0.5(0.3) + 1.1 \end{pmatrix} = \begin{pmatrix} 0.2 & 0.3 \\ -0.7 & 0.95 \end{pmatrix}.$$

Row 2 of $L^{-1}r_t$ is $r_{2t} - 0.5r_{1t}$; moving $-0.5r_{1t}$ to the right gives Tsay's structural equation (T eq. 8.10):

$$r_{2t} = 0.3 + 0.5r_{1t} - 0.7r_{1,t-1} + 0.95r_{2,t-1} + b_{2t}, \qquad \mathrm{Var}(b_{2t}) = 0.5. \tag{10}$$

Ordering (r_2, r_1)**.** Now $\Sigma^* = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$: $g_1 = 1$, $l = 1$, $g_2 = 2 - 1 = 1$, $L^{-1} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$, and (T p. 401)

$$r_{1t} = -0.2 + 1.0\, r_{2t} - 0.8r_{2,t-1} + 0.8r_{1,t-1} + c_{2t}, \qquad \mathrm{Var}(c_{2t}) = 1. \tag{11}$$

(Check the intercept: $-1(0.4) + 0.2 = -0.2$; the lag terms: $-1(1.1) + 0.3 = -0.8$ on $r_{2,t-1}$ and $-1(-0.6) + 0.2 = 0.8$ on $r_{1,t-1}$.)

> **(10) says r_1 drives r_2 within the period; (11) says r_2 drives r_1. Both are exact rewritings of the same reduced form.** The data are silent between them: the choice is an assumption about timing.

In SVAR form (1), $A_0 = G^{-1/2}L^{-1}$ (rows rescaled to unit-variance shocks). Impact responses $A_0^{-1} = LG^{1/2} = P$:

impact on → / of shock	ordering (r_1, r_2): shock 1	shock 2	ordering (r_2, r_1): r_1 shock	r_2 shock
r_1	$\sqrt{2} = 1.414$	0	1	1
r_2	$0.5\sqrt{2} = 0.707$	$\sqrt{0.5} = 0.707$	0	1

In general, with σ_1, σ_2, ρ: ordering (r_1, r_2) gives r_1 an own impact σ_1 and a cross impact 0; ordering (r_2, r_1) gives $\sigma_1\sqrt{1-\rho^2}$ and $\rho\sigma_1$. With $\sigma_1 = \sqrt{2}$ and $\rho = 1/\sqrt{2}$: $\sqrt{2}\sqrt{1/2} = 1$ and $(1/\sqrt{2})\sqrt{2} = 1$ — the table.

What it does to the variance decomposition. With Example 8.3's numbers, the share of r_1's 10-step forecast-error variance attributed to the r_2 shock is **13.6%** under ordering (r_1, r_2) and **48.5%** under (r_2, r_1) (the defaults of the Ordering Dial, verified in `companions/check_companions.py`): the same data, a story that changes by 35 percentage points.

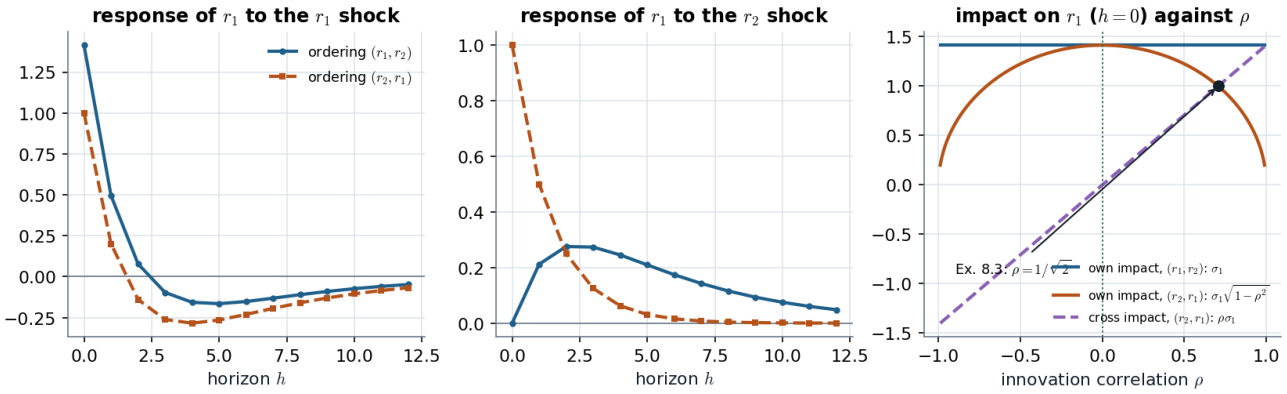


Figure — orthogonalised responses $\Phi^h A_0^{-1}$ of T Ex. 8.3 under the two recursive orderings, and the impact on r_1 against ρ with $\sigma_1^2 = 2$ (T p. 400; (9)).

The middle panel is the whole problem: "the effect of an r_2 shock on r_1" is zero on impact and humped, or one on impact and monotone, depending only on the ordering. The right panel shows when it does not matter: the curves meet at $\rho = 0$, and the gap grows with $|\rho|$ — at Example 8.3's $\rho = 0.707$ the own impact falls from 1.414 to 1.

---

5 · C — non-recursive identification on the slides

Slide 32 (commodity prices $P_{c,t}$, output Y_t, interest rate R_t, SVAR(1)): equation 1 is "monetary policy" with a 0.0 coefficient on $\log Y_t$ (the central bank does not see current output); equation 2 "characterizes behaviour of finished-goods producers" with 0.0 on R_t; equation 3, commodity prices "set in active competitive markets", is unrestricted. That is two zeros; (8) asks for three with $q = 3$, so the zeros shown are not enough by themselves — a third restriction (another zero, or a sign restriction) is needed to pin down A_0.

Illustrations 1–2 (slides 35–36), counted:

	q	needed, (8)	zeros in A_0	order condition
Illustration 1: $\log Y, \log P, R, \log M, \log P_c$ × PS, PS, MP, MD, Inf	5	10	1+2+2+2+4 = 11	met, 1 overidentifying
Illustration 2: PCOM, M2, R, Y, CPI, U × Inform, MP, MD, Prod, Prod, Prod	6	15	0+4+2+5+4+3 = 18	met, 3 overidentifying

(Row counts for Illustration 1 as printed, rows = variables: $\log Y$ 1 zero, $\log P$ 2, R 2, $\log M$ 2, $\log P_c$ 4; the slide reuses the labels a_{14}, a_{15} in several rows — read them as distinct free entries.) Illustration 2's monetary-policy row has free entries only on M2 and R: the Fed reacts within the month to money and the funds rate, not to output, prices or unemployment, which arrive with a lag. **Overidentified** models restrict Σ and can be tested; exactly identified ones fit Σ perfectly whatever the zeros mean. Slide 37 shows the resulting monetary-policy-shock responses (figure only).

---

6 · U — Python

`statsmodels.tsa.vector_ar.svar_model.SVAR(y, svar_type="A", A=mask)` estimates A_0 with `"E"` marking free entries; the recursive case needs no SVAR call at all — `VARResults.irf(h).orth_irfs` is (9). Both are used in L09-10-python §3.

Links

- Neighbours: L09-02-vma-irf-fevd (the Θ_s that (9) reinterprets) · L09-04-bvar-minnesota-ssvs.
- Later in the unit: L09-06-tvp-var-sv-state-space (Primiceri's A_t is a time-varying recursive A_0).
- Class 1: L01 §9.1 (Example 8.3's mean, Γ_0, stationarity).
- Rules: rules/05.
- **Interactive companion.** The Ordering Dial (`companions/companion-var-irf-ordering.html`) — set $\rho = 0$ and the two orderings coincide; push $|\rho|$ up and watch the impact matrix and the FEVD split.
- **Practice (number and printed page only):** T 8.1 (p. 462) and T 8.7(b) (p. 463) involve orthogonalised IRFs whose ordering must be stated.

Slides 22–30 and Part 2 slides 3–13, 56–60 · The Minnesota prior: shrink every equation toward a random walk, tighter for distant lags and other variables, and the posterior is ridge regression in closed form

> **Claim.** The Minnesota prior puts $\beta_i \sim N(\beta_{i0}, V_{i0})$ on each equation, centred on a random walk (own first lag 1, everything else 0), with prior sd λ/k on own lag k and $\theta\lambda\sigma_i/(k\sigma_j)$ on variable j's lag k. With Σ diagonal and known, **the posterior is equation-by-equation Gaussian:** $\bar{V}_i = (V_{i0}^{-1} + X'X/\sigma_i^2)^{-1}$, $\bar{\beta}_i = \bar{V}_i(V_{i0}^{-1}\beta_{i0} + X'y^{(i)}/\sigma_i^2)$ — a precision-weighted average of the prior centre and OLS. λ **is the dial**: $\lambda \to 0$ returns the random walk, $\lambda \to \infty$ returns OLS. Large VARs work because λ is tightened as the number of variables grows (Bańbura et al.).

Prev: L09-03-structural-var · Next: L09-05-factor-models-favar · Map: L09-00-map

Notation.

Symbol	Meaning	Range
β_i	coefficients of equation i (constant, then lags), row i of (μ, B); $\beta = (\beta_1', \dots, \beta_q')'$ (slide 22)	
$\beta_{i0},\ V_{i0}$	prior mean and (diagonal) prior covariance of β_i	
λ	overall tightness	> 0
θ	relative tightness of other variables' lags	$(0, 1]$
k	lag index	$1, \dots, p$
σ_i^2	error variance of equation i, fixed and known ($\Sigma = \mathrm{diag}(\sigma_1^2, \dots, \sigma_q^2)$, slide 24)	> 0
X, $y^{(i)}$	common regressor matrix, equation i's response vector	
$\bar{\beta}_i,\ \bar{V}_i$	posterior mean and covariance	
γ_j	inclusion indicator of coefficient j (SSVS)	$\{0, 1\}$

---

1 · U — why shrink

Part 2, slide 3: "VAR flexibility → over-parameterization → poor forecasts. First solution: Minnesota prior of Doan et al. (1984)." The count is in L09-01-var-p-estimation-forecasting §8: Kadiyala–Karlsson's 7-variable VAR(6) has 43 coefficients per equation and 133 observations. OLS is unbiased and useless; a prior that says "most of these are small, and distant lags are smaller" trades a little bias for a lot of variance — the MSE trade of L01-05 and the bias–variance logic of L02-01, with the prior as the regulariser.

---

2 · C — the Minnesota prior (slides 23–24)

Litterman's (1980, 1986) prior (slide 23): $\beta_i \sim N(\beta_{i0}, V_{i0})$, "with V_{i0} chosen to 'center' the individual equations around the random walk with drift"

$$y_{ti} = \mu_i + y_{t-1,i} + u_{ti}. \tag{1}$$

Prior mean. Coefficient on own first lag = 1; all other lag coefficients = 0; the constant gets a flat (very large variance) prior.

Prior variances (slide 24; V_{i0} diagonal):

$$\mathrm{Var}(B_k(i,i)) = \frac{\lambda^2}{k^2}, \qquad \mathrm{Var}(B_k(i,j)) = \frac{\theta^2\lambda^2}{k^2} \times \frac{\sigma_i^2}{\sigma_j^2}, \quad j \neq i. \tag{2}$$

Read each factor:

Factor	What it encodes (slide 23)
λ	how far any coefficient may stray from the random walk
$1/k^2$	"shrinking the number of lags p towards one": the prior sd falls like $1/k$
$\theta < 1$	"own lags should explain more of the variation than the lags of other variables"
σ_i^2/σ_j^2	units: $B_k(i,j)$ converts y_j into y_i, so its scale is σ_i/σ_j

"Shrinking the diagonal elements of B_1 toward one, shrinking the remaining coefficients of $B_1, \ldots, B_p$ toward zero" (slide 23). In practice σ_i^2 is replaced by the residual variance of a univariate AR(p) for series i (Litterman's convention), which is what "fixed and known" amounts to.

---

3 · C — the closed-form posterior

Factorisation. With Σ diagonal, the VAR likelihood (L09-01-var-p-estimation-forecasting (6)) splits by equation:

$$p(Y \mid \beta) \propto \prod_{i=1}^{q} \exp\Big\{ -\frac{1}{2\sigma_i^2}(y^{(i)} - X\beta_i)'(y^{(i)} - X\beta_i)\Big\}. \tag{3}$$

The prior is independent across equations, so the posterior is too: $p(\beta \mid Y) = \prod_i p(\beta_i \mid y^{(i)})$.

One equation. Drop the index i. Minus twice the log posterior kernel:

$$-2\log p(\beta \mid y) = \frac{(y - X\beta)'(y - X\beta)}{\sigma^2} + (\beta - \beta_0)'V_0^{-1}(\beta - \beta_0) + c. \tag{4}$$

Expand both quadratics and collect powers of β:

$$= \beta'\Big(\frac{X'X}{\sigma^2} + V_0^{-1}\Big)\beta - 2\beta'\Big(\frac{X'y}{\sigma^2} + V_0^{-1}\beta_0\Big) + c'. \tag{5}$$

Name $\bar{V}^{-1} = X'X/\sigma^2 + V_0^{-1}$ and $\bar{\beta} = \bar{V}(X'y/\sigma^2 + V_0^{-1}\beta_0)$; then (5) is

$$(\beta - \bar{\beta})'\bar{V}^{-1}(\beta - \bar{\beta}) - \bar{\beta}'\bar{V}^{-1}\bar{\beta} + c', \tag{6}$$

a Gaussian kernel in β:

$$\boxed{\beta_i \mid y \sim N(\bar{\beta}_i, \bar{V}_i), \qquad \bar{V}_i = \Big(V_{i0}^{-1} + \frac{X'X}{\sigma_i^2}\Big)^{-1}, \qquad \bar{\beta}_i = \bar{V}_i\Big(V_{i0}^{-1}\beta_{i0} + \frac{X'y^{(i)}}{\sigma_i^2}\Big)} \tag{7}$$

(slide 26, row "Minnesota": $\gamma_i \mid y \sim N(\tilde{\gamma}_i, \tilde{\Sigma}_i)$). Substituting $X'y^{(i)} = X'X\hat{\beta}_i$ (the OLS normal equations) into (7):

$$\bar{\beta}_i = \bar{V}_i\Big(V_{i0}^{-1}\beta_{i0} + \frac{X'X}{\sigma_i^2}\hat{\beta}_i\Big), \tag{8}$$

a **matrix-weighted average** of the prior mean and OLS, with weights the two precisions (which sum to $\bar{V}_i^{-1}$). It is generalised ridge regression: ridge with penalty $\sigma_i^2 V_{i0}^{-1}$ centred at β_{i0} instead of 0 (ridge itself is Classes 3+4).

The two limits in λ. By (2), $V_{i0} \propto \lambda^2$ entrywise (except the constant).

- $\lambda \to 0$: $V_{i0}^{-1} \to \infty$, $\bar{V}_i \to V_{i0} \to 0$, and $\bar{\beta}_i \to \beta_{i0}$ — **the random walk** (1).
- $\lambda \to \infty$: $V_{i0}^{-1} \to 0$, $\bar{V}_i \to \sigma_i^2(X'X)^{-1}$, and $\bar{\beta}_i \to \hat{\beta}_i$ — **OLS**, with the frequentist covariance.

The scalar case shows the weight. One regressor with $\sum_t x_t^2 = s_{xx}$, prior $N(b_0, v_0)$:

$$\bar{b} = \frac{v_0^{-1}b_0 + (s_{xx}/\sigma^2)\hat{b}}{v_0^{-1} + s_{xx}/\sigma^2} = w\,b_0 + (1-w)\hat{b}, \qquad w = \frac{\sigma^2/s_{xx}}{\sigma^2/s_{xx} + v_0} = \frac{\mathrm{Var}(\hat{b})}{\mathrm{Var}(\hat{b}) + v_0}. \tag{9}$$

The prior wins in proportion to how noisy OLS is relative to the prior variance. Distant lags ($v_0 = \lambda^2/k^2$ small) and short samples (s_{xx} small) shrink most.

3.1 · Worked — the slide-13 VAR(2), 58 observations

A seeded simulation (seed 9; 200 burn-in, 60 kept, $p = 2$ leaves $T = 58$) from slide 13's μ, B_1, B_2, Σ (L09-01-var-p-estimation-forecasting §5.1). Univariate AR(2) residual variances: $\hat{\sigma}^2 = (0.266,\ 0.126,\ 0.805)$ (true diagonal 0.26, 0.09, 0.81). Prior with $\theta = 0.5$ (the slides give no value; 0.5 is illustrative), constant variance 10^6.

Prior sds for equation 1 at $\lambda = 0.2$, by (2):

coefficient	formula	value
own lag 1, $B_1(1,1)$	$\lambda/1$	0.200
own lag 2, $B_2(1,1)$	$\lambda/2$	0.100

coefficient	formula	value
$B_1(1,2)$	$\theta\lambda\,\sigma_1/\sigma_2 = 0.5(0.2)\sqrt{0.266/0.126} = 0.1(1.455)$	0.145
$B_1(1,3)$	$0.5(0.2)\sqrt{0.266/0.805} = 0.1(0.575)$	0.057

Posterior means of equation 1 by (7) (computed in L09-10-python §4):

	$B_1(1,1)$	$B_1(1,2)$	$B_1(1,3)$	$B_2(1,1)$
truth (slide 13)	0.7	0.1	0.0	−0.2
prior mean	1	0	0	0
$\lambda = 0.05$	0.926	0.000	−0.010	−0.018
$\lambda = 0.2$	0.690	0.055	−0.043	−0.115
$\lambda = 1$	0.649	0.284	−0.038	−0.262
OLS ($\lambda \to \infty$)	0.647	0.379	−0.021	−0.292

With 58 observations OLS puts 0.379 on $B_1(1,2)$ (truth 0.1); at $\lambda = 0.2$ the posterior mean is 0.055 — **closer to the truth than OLS** for that coefficient, and the own-lag coefficient 0.690 is within 0.01 of 0.7. At $\lambda = 0.05$ the prior wins everywhere and the own lag sits at 0.926, near the random walk.

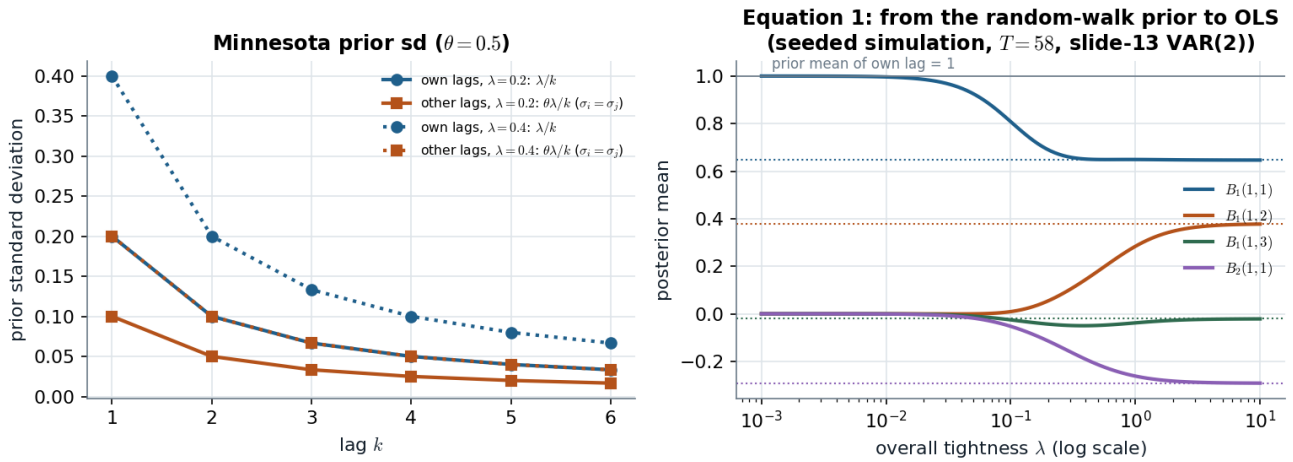


Figure — left: prior sds (2) with $\theta = 0.5$, $\sigma_i = \sigma_j$; right: posterior means (7) for equation 1 against λ, seeded simulation from the slide-13 VAR(2), $T = 58$; dotted lines are OLS.

The left panel is the prior's shape: sds fall like $1/k$ and cross-variable lags sit at half the own-lag sd. On the right, the transition happens between $\lambda \approx 0.03$ and $\lambda \approx 1$; each coefficient leaves its prior value at its own λ, where its data precision overtakes its prior precision ((9): $w = 1/2$ when $\mathrm{Var}(\widehat{b}) = v_0$) — the own lags first, $B_1(1,2)$ last. **Choosing λ is choosing where on these curves to stand**; Bańbura et al. choose it by in-sample fit relative to the cross-section (§6).

4 · C — other priors on (B, Σ) (slides 25–26)

Kadiyala and Karlsson (1993, 1997) extend the Minnesota prior (slide 25), which fixes Σ. In slide 26's notation (Γ = coefficient matrix, Z = regressors, Ψ = error covariance, m = number of series):

Prior	Prior	Posterior
Minnesota	$\gamma_i \sim N(\tilde{\gamma}_i, \tilde{\Sigma}_i)$, Ψ fixed and diagonal	$\gamma_i \mid y \sim N(\bar{\gamma}_i, \bar{\Sigma}_i)$ — (7)
Diffuse (Jeffreys)	$p(\gamma, \Psi) \propto \lvert\Psi\rvert^{-(m+1)/2}$	$\Gamma \mid y \sim MT(Z'Z, (Y - Z\hat{\Gamma})'(Y - Z\hat{\Gamma}), \hat{\Gamma}, T-k)$
Normal-Wishart	$\gamma \mid \Psi \sim N(\tilde{\gamma}, \Psi \otimes \tilde{\Omega})$, $\Psi \sim iW(\tilde{\Psi}, \alpha)$	$\Gamma \mid y \sim MT(\bar{\Omega}^{-1}, \bar{\Psi}, \bar{\Gamma}, T+\alpha)$
Normal-Diffuse	$\gamma \sim N(\tilde{\gamma}, \tilde{\Sigma})$, $p(\Psi) \propto \lvert\Psi\rvert^{-(m+1)/2}$	no closed form (kernel on slide 26) → Gibbs
Extended Natural Conjugate	slide 26	slide 26

The diffuse case, derived (it is the site's band recipe). Multiply the likelihood (L09-01-var-p-estimation-forecasting (6), (10)) by $\lvert\Sigma\rvert^{-(q+1)/2}$:

$$p(B, \Sigma \mid Y) \propto \lvert\Sigma\rvert^{-(n+q+1)/2} \exp\Big\{ -\tfrac{1}{2}\mathrm{tr}(\Sigma^{-1} S(\hat{B})) \Big\} \exp\Big\{ -\tfrac{1}{2}\mathrm{tr}(\Sigma^{-1}(\hat{B}-B)X'X(\hat{B}-B)') \Big\}. \quad (10)$$

For fixed Σ the second exponential is the matrix-normal kernel, i.e.

$$\mathrm{vec}(B') \mid \Sigma, Y \sim N(\mathrm{vec}(\hat{B}'),\ \Sigma \otimes (X'X)^{-1}), \quad (11)$$

whose normalising constant is $\propto \lvert\Sigma\rvert^{k/2}$ (k = regressors per equation). Integrating B out of (10) leaves $\lvert\Sigma\rvert^{-(n-k+q+1)/2} \exp\{-\frac{1}{2}\mathrm{tr}(\Sigma^{-1} S(\hat{B}))\}$:

$$\Sigma \mid Y \sim IW(S(\hat{B}),\ n-k). \quad (12)$$

Drawing Σ from (12) then B from (11) is exact posterior simulation; the site's VAR-Example bands fix $\Sigma = \hat{\Sigma}$ and draw only (11). Part 2 slides 8–9 give the same pair as full conditionals ($\beta \mid \Sigma, y \sim N(\mu_1, V_1)$ with $V_1^{-1} = V_0^{-1} + \sum_t \tilde{z}_t' \Sigma^{-1} \tilde{z}_t$; $\Sigma \mid \beta, y \sim IW(T, S)$), and Greenberg's SUR Gibbs sampler (Algorithm 9.1, p. 137) is the general version with equation-specific regressors.

Why Normal-Wishart cannot be Minnesota. In $\Psi \otimes \tilde{\Omega}$ the prior covariance of equation i's coefficients is $\psi_{ii}\tilde{\Omega}$: the **same** relative tightness across regressors in every equation. Minnesota's $\theta < 1$ makes "own" and "other" depend on the equation, which a Kronecker product cannot do — so NW imposes $\theta = 1$. Kadiyala–Karlsson's verdict (slide 30): "our preferred choice is the Normal-Wishart when the prior beliefs are of the Litterman type", and "for more general prior beliefs … the Normal-Diffuse and EN priors are strong alternatives".

The K&K example (slides 27–29): Litterman's seven US series (RGNPG, INFLA, UNEMP, M1, INVEST, CPRATE, CBI), quarterly 1948:1–1981:1, out-of-sample 1980:2–1986:4, $q = 7$, $p = 6$, 301 parameters; RMSE comparison on slide 29 (figure).

5 · C — variable selection: the ON/OFF switch and SSVS (Part 2, slides 5–13)

ON/OFF switch (slides 5–6). Write the VAR as a SUR, $y_t = z_t\beta + u_t$ with $z_t = I_q \otimes (y'_{t-1}, \dots, y'_{t-p})$ ($q \times s$, $s = q^2p$), and switch coefficients with $\Gamma = \text{diag}(\gamma_{11}, \dots, \gamma_{q,pq})$: $y_t = z_t\Gamma\beta + u_t$, so $B_{ij} = 0$ when $\gamma_{ij} = 0$. "2^s VAR models: variable selection = model selection." Priors (slide 7): $\beta \sim N(0, V_0)$, $p(\Sigma) \propto |\Sigma|^{-(q+1)/2}$, $\gamma_j \mid \gamma_{-j} \sim \text{Bernoulli}(\pi_{0j})$. Gibbs full conditionals (slides 8–10): with $\tilde{z}_t = z_t\Gamma$, β is Normal (as above), Σ is inverse Wishart, and

$$\gamma_j \mid \gamma_{-j}, \beta, \Sigma, y \sim \text{Bernoulli}\Big(\frac{A_j}{A_j + B_j}\Big), \quad A_j = p(y \mid \beta, \Sigma, \gamma_{-j}, \gamma_j = 1)\pi_{0j}, \quad B_j = p(y \mid \beta, \Sigma, \gamma_{-j}, \gamma_j = 0)(1-\pi_{0j}) \tag{13}$$

(the slide prints "$\gamma_j = 1$" inside B_j too; it must be $\gamma_j = 0$).

SSVS (slides 11–12; George and McCulloch 1993, 1997; Jochmann et al. 2011). Instead of an exact zero, a **spike-and-slab** prior:

$$\beta_j \mid \gamma_j \sim (1-\gamma_j)N(0, \tau_{0j}^2) + \gamma_j N(0, \tau_{1j}^2), \qquad \gamma_j \sim \text{Bernoulli}(q_j), \tag{14}$$

τ_{0j}^2 small ("coefficient is virtually zero"), τ_{1j}^2 large ("relatively non-informative"). Default semi-automatic choice (George et al. 2008, slide 12): $\tau_{kj} = c_k \cdot \widehat{\text{var}}(\beta_j)$ from an unrestricted VAR, $c_0 \ll c_1$; Jochmann et al. set $c_0 = 0.1$, $c_1 = 10$. Here γ_j given β_j needs only the two densities. By Bayes:

$$P(\gamma_j = 1 \mid \beta_j) = \frac{q_j\, \phi(\beta_j; 0, \tau_{1j}^2)}{q_j\, \phi(\beta_j; 0, \tau_{1j}^2) + (1-q_j)\, \phi(\beta_j; 0, \tau_{0j}^2)}. \tag{15}$$

Worked. Take $q_j = \frac{1}{2}$ and read the τ's as $c_0 = 0.1$ and $c_1 = 10$ times a coefficient's OLS standard error s_j; write $z = \beta_j/s_j$. With $\phi(\beta; 0, (cs)^2) = \frac{1}{cs}\phi(z/c)$, the s cancels in (15):

$$P(\gamma_j = 1 \mid z) = \frac{0.1\, \phi(z/10)}{0.1\, \phi(z/10) + 10\, \phi(10z)}.$$

At $z = 0.2$: $0.1\phi(0.02) = 0.1(0.39886) = 0.039886$, $10\phi(2) = 10(0.053991) = 0.53991$, so $P = 0.039886/0.579796 = 0.069$. At $z = 0.5$: $10\phi(5) = 10(1.4867 \times 10^{-6}) = 1.4867 \times 10^{-5}$, so $P = 0.039844/(0.039844 + 0.0000149) = 0.9996$. The two densities cross where $\frac{z^2}{2}(c_0^{-2} - c_1^{-2}) = \ln(c_1/c_0)$:

$$z^* = \sqrt{\frac{2\ln(c_1/c_0)}{c_0^{-2} - c_1^{-2}}} = \sqrt{\frac{2\ln 100}{100 - 0.01}} = \sqrt{\frac{9.2103}{99.99}} = 0.3035.$$

A draw of β_j beyond 0.30 OLS standard errors is classified "in" almost surely — the spike is so narrow that SSVS keeps far more than a t-test at 1.96 would. The ratio c_1/c_0, not a significance level, sets the threshold.

Prior on Ψ (slide 13): with $\Sigma^{-1} = \Psi\Psi'$ (Ψ lower triangular, the Cholesky root of the precision), $\psi_{jj}^2 \sim G(a_j, b_j)$ and spike-and-slab on the off-diagonal ψ_{ij} — the same idea applied to contemporaneous correlations.

---

6 · C — large BVARs work (Part 2, slides 50, 56–60)

- **Carriero, Kapetanios and Marcellino (2009)**, $q = 33$ exchange rates against the dollar: the BVAR improves on the random walk for most rates, gains "typically in the range of 2%–3%", "6%–9%" for euro–dollar and pound–dollar; a trading rule on BVAR forecasts beats the RW (slide 50).
- **Bańbura, Giannone and Reichlin (2010)** (slides 56–60): "When the degree of shrinkage is set in relation to the cross-sectional dimension, the forecasting performance of small monetary VARs can be improved by adding additional macroeconomic variables and sectoral information"; "Large VARs with shrinkage produce credible impulse responses". Data: SMALL (3), CEE (7), MEDIUM (20), LARGE (131 Stock–Watson indicators).

Table II (slide 58), MSFE relative to a random walk with drift, 1971–2003, $h = 1$:

	SMALL $p = 13$ OLS	SMALL BVAR	CEE $p = 13$ OLS	CEE p=BIC	CEE BVAR	LARGE BVAR
EMPL	1.14	1.14	**7.56**	0.76	0.67	**0.46**
CPI	0.89	0.89	5.61	0.55	0.52	0.50
FFR	1.86	1.86	6.39	1.21	0.89	0.75

Unshrunk OLS with 13 lags in 7 variables forecasts employment with **7.56 times** the RW's MSFE; the BVAR with all 131 variables reaches 0.46. More variables help **only with shrinkage**. Figure 1 (slide 60): the responses to a monetary-policy shock (employment falls, the federal funds rate jumps then decays) keep their shape from SMALL to LARGE, with the prior on the sum of coefficients ($\tau = 10\lambda$) added. FAVAR comparisons (Table III) are in L09-05-factor-models-favar §6.

---

7 · U — Python

(7) is ten lines of NumPy (`minnesota_posterior` in `figures/make_figures.py`, reused in L09-10-python §4). No PyMC is needed for the Minnesota or diffuse cases: (7), (11)–(12) are exact. The Normal-Diffuse prior and SSVS need a Gibbs loop over (13)/(15) and the Normal/IW conditionals — the same loop structure as Greenberg's Algorithm 9.1.

Links

- Neighbours: L09-03-structural-var · L09-05-factor-models-favar (the other answer to dimensionality).
- Earlier units: L01-04 (a first prior pulling an MLE) · L01-05 (biased but lower MSE) · L02-01 (bias–variance).
- Unit: L09-00-map · L09-08-formulas · L09-09-pitfalls · L09-10-python.
- Rules: rules/05.
- **Practice (number and printed page only):** G 10.4 (p. 167) — the SUR Gibbs structure of §4–§5 with autocorrelated errors.

Part 2 slides 14–37, 51–55 · Factor models: k common factors carry all the cross-correlation, $\Omega = \beta\beta^{'} + \Sigma$; a FAVAR puts them in a VAR

> **Claim.** In the k-factor model $y_t = \beta f_t + \varepsilon_t$ the $m \times m$ covariance is $\Omega = \beta\beta' + \Sigma$ with Σ diagonal: **common factors explain all the dependence**, and $m(m+1)/2$ covariances are described by $m(k+1) - k(k-1)/2$ parameters. Loadings are defined only up to an orthogonal rotation, so a block-lower-triangular β is imposed — the same $k(k-1)/2$ count as the SVAR's rotation problem. Given β and Σ, the factors have a Gaussian posterior and the Gibbs sampler cycles through three regressions. **A FAVAR** adds a few factors, extracted from hundreds of series, as regressors in a small VAR — the second answer (after shrinkage) to the curse of dimensionality.

Prev: L09-04-bvar-minnesota-ssvs · Next: L09-06-tvp-var-sv-state-space · Map: L09-00-map

Notation.

Symbol	Meaning	Range
y_t	$m \times 1$ observed vector (slides also write p for m)	
f_t	$k \times 1$ common factors, $N(0, I_k)$	$k < m$
β	$m \times k$ loadings	block lower triangular, $\beta_{ii} > 0$
$\Sigma = \text{diag}(\sigma_1^2, \dots, \sigma_m^2)$	idiosyncratic variances	> 0
Ω	$\text{Var}(y_t)$	
$h_i^2 = \sum_l \beta_{il}^2 / \Omega_{ii}$	communality of series i	$[0, 1]$
V_f	posterior covariance of f_t given (y_t, β, Σ)	
x_t	$r \times 1$ large panel of predictors (FAVAR)	r in the hundreds

---

1 · C — the basic model (slide 15; T eq. 9.2, p. 469)

"For any specified positive integer $k \leq m$, the standard k-factor model relates each y_t to an underlying k-vector of random variables f_t, the common factors, via"

$$y_t \mid f_t \sim N(\beta f_t, \Sigma), \qquad f_t \sim N(0, I_k), \qquad \Sigma \text{ diagonal.} \tag{1}$$

Equivalently $y_t = \beta f_t + \varepsilon_t$ with $\varepsilon_t \sim N(0, \Sigma)$ independent of f_t. Then, using $\text{Cov}(f_t) = I_k$ and $\text{Cov}(f_t, \varepsilon_t) = 0$:

$$\Omega = \text{Var}(y_t) = \beta \, \text{Var}(f_t) \, \beta' + \text{Var}(\varepsilon_t) = \beta\beta' + \Sigma, \tag{2}$$

element by element (slide 15):

$$\text{var}(y_{it}) = \sum_{l=1}^{k} \beta_{il}^2 + \sigma_i^2, \qquad \text{cov}(y_{it}, y_{jt}) = \sum_{l=1}^{k} \beta_{il}\beta_{jl} \quad (i \neq j). \tag{3}$$

Conditionally on f_t, $\text{var}(y_{it} \mid f) = \sigma_i^2$ and $\text{cov}(y_{it}, y_{jt} \mid f) = 0$: "Common factors explain all the dependence structure among the m variables." Tsay's general form (eq. 9.2) allows a mean α and correlated factors, $\text{Cov}(r_t) = \beta\Sigma_f\beta' + D$; when the factors are **observed** (the market return, macro series), it is a multivariate regression estimated by OLS (T §9.2, p. 470, $\hat{\xi} = (G'G)^{-1}G'R$; the market model eq. 9.5) — the same algebra as L09-01-var-p-estimation-forecasting §4.

---

2 · C — rotation, identification and the bound on k (slides 16–19)

Invariance. For any orthogonal $k \times k$ matrix P, set $\beta^* = \beta P'$ and $f_t^* = Pf_t$:

$$\beta^* f_t^* = \beta P' P f_t = \beta f_t, \qquad \text{Var}(f_t^*) = P I_k P' = I_k, \qquad \beta^*\beta^{*\prime} = \beta P' P \beta' = \beta\beta'. \tag{4}$$

The data cannot tell (β, f) from $(\beta P', Pf)$ — exactly the problem of L09-03-structural-var (6)–(7), now for loadings. "Classical approach: $\beta'\Sigma^{-1}\beta = I$ [diagonal]. Our approach: β is a block lower triangular" (slide 16): $\beta_{ij} = 0$ for $j > i$ in the first k rows, $\beta_{ii} > 0$. That removes $k(k-1)/2$ entries, the dimension of the rotation group, and "provides both identification and, often, useful interpretation" — series 1 loads only on factor 1, so factor 1 is named after series 1.

Parameter count (slide 17). β has mk entries minus $k(k-1)/2$ zeros; Σ adds m:

$$m(k+1) - \frac{k(k-1)}{2} \quad \text{vs} \quad \frac{m(m+1)}{2} \text{ in an unrestricted } \Omega. \tag{5}$$

A factor model must not have more parameters than Ω has entries:

$$\frac{m(m+1)}{2} - m(k+1) + \frac{k(k-1)}{2} \geq 0. \tag{6}$$

Multiply (6) by 2 and collect powers of k:

$$k^2 - (2m+1)k + (m^2 - m) \geq 0.$$

The roots are $k = \frac{(2m+1) \pm \sqrt{(2m+1)^2 - 4(m^2-m)}}{2} = \frac{(2m+1) \pm \sqrt{8m+1}}{2}$, and $k < m$ requires the smaller root:

$$k \leq \frac{2m + 1 - \sqrt{8m+1}}{2}. \tag{7}$$

m	$(2m+1-\sqrt{8m+1})/2$	$k_{\max}$ (slide 17)
6	$(13-7)/2 = 3.00$	3
12	$(25-9.849)/2 = 7.58$	7
20	$(41-12.69)/2 = 14.15$	14
50	$(101-20.02)/2 = 40.49$	40

At $m = 6$, $k = 3$: $21 - 6(4) + 3 = 0$, an exact fit with no restriction left on Ω. "Even for small m, the bound will often not matter as relevant k values will not be so large."

Rank deficiency (slide 18, G): if β has rank $r < k$ there are Q with $\beta Q = 0$, $Q'Q = I$, and $\beta\beta' + \Sigma = (\beta + MQ')(\beta + MQ')' + (\Sigma - MM')$ for orthogonal M — a second source of non-identification, tied to uncertainty about k (Frühwirth-Schnatter and Lopes 2009 use it to search over k, slides 30–33).

Ordering (slide 19): reordering the series (Ay_t) changes which series define the factors, but an orthonormal P with $A\beta P$ lower triangular always exists, so "the order of the variables in y_t is irrelevant assuming that k is properly chosen" (Lopes and West 2004).

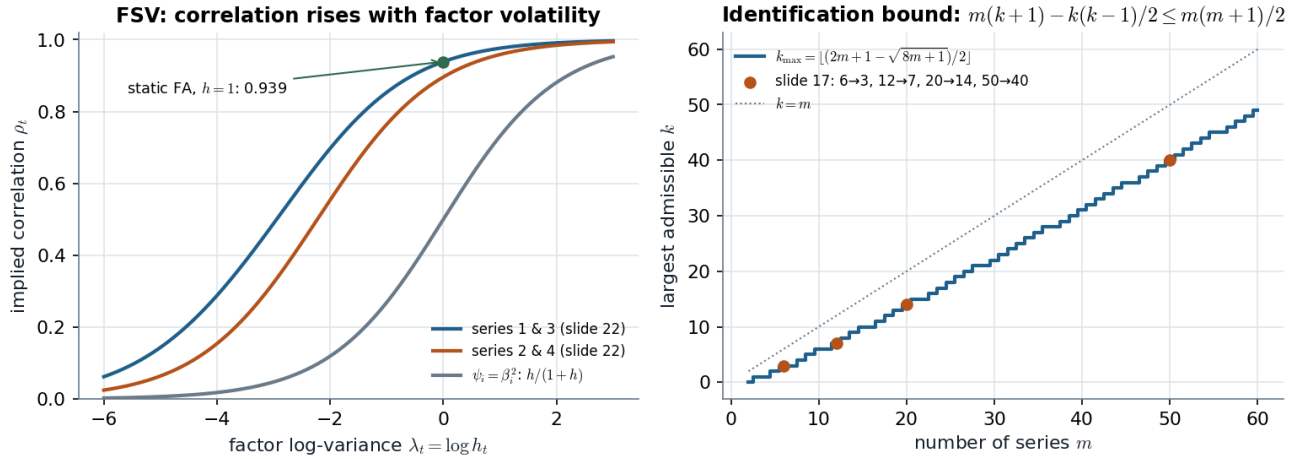


Figure — left: $\rho_{ij} = \beta_i\beta_j h / \sqrt{(\beta_i^2 h + \sigma_i^2)(\beta_j^2 h + \sigma_j^2)}$, the correlation (3) when $\mathrm{Var}(f) = h$, for slide 22's series pairs (1, 3) and (2, 4); right: bound (7) with slide 17's four cases.

Right panel: the bound grows almost linearly, $k_{\max} \approx m - \sqrt{2m}$, so it binds only for very small panels. Left panel: with the factor variance at 1 (the static model (1)), series 1 and 3 of the simulation have correlation 0.939; letting the factor variance move is the factor stochastic volatility model of L09-07-fsv-dcc-covariances, and the curve says what it implies: **correlations rise when the common factor is volatile and collapse toward zero when it is calm.**

---

3 · C — priors and full conditionals (slides 20–21)

Priors (slide 20): $\beta_{ij} \sim N(0, C_0)$ for free off-diagonal entries, $\beta_{ii} \sim N(0, C_0)\mathbb{1}(\beta_{ii} > 0)$, $\sigma_i^2 \sim IG(\nu/2, \nu s^2/2)$ with s^2 the prior mode. The improper $p(\sigma_i^2) \propto 1/\sigma_i^2$ is avoided: it "lead[s] to the Bayesian analogue of the so-called Heywood problem" (an idiosyncratic variance collapsing to zero).

Factor scores. For one t, the kernel of $p(f_t \mid y_t, \beta, \Sigma)$ is the likelihood times the $N(0, I)$ prior:

$$-2\log p(f_t \mid \cdot) = (y_t - \beta f_t)'\Sigma^{-1}(y_t - \beta f_t) + f_t' f_t + c = f_t'(I_k + \beta'\Sigma^{-1}\beta)f_t - 2f_t'\beta'\Sigma^{-1}y_t + c'. \quad (8)$$

Completing the square as in L09-04-bvar-minnesota-ssvs (5)–(7):

$$\boxed{f_t \mid y_t, \beta, \Sigma \sim N(V_f\beta'\Sigma^{-1}y_t,\ V_f), \qquad V_f = (I_k + \beta'\Sigma^{-1}\beta)^{-1}} \quad (9)$$

(slide 21). **Idiosyncratic variances**: given f, series i is a regression with T residuals $d_i = (y_i - f\beta_i')'(y_i - f\beta_i')$, so $\sigma_i^2 \mid \cdot \sim IG((\nu + T)/2, (\nu s^2 + d_i)/2)$. **Loadings**: row i is a Bayesian regression of y_i on the factors, $\beta_i \sim N(M_i, C_i)$ with $C_i^{-1} = C_0^{-1}I + \sigma_i^{-2} f'f$ and $M_i = C_i(C_0^{-1}\mu_0 + \sigma_i^{-2} f' y_i)$, truncated to $\beta_{ii} > 0$ in the first k rows. Cycling (9) → loadings → variances is the Gibbs sampler.

3.1 · Worked — slide 22's simulation exercise

$T = 500$, $m = 5$, $k = 2$,

$$\beta = \begin{pmatrix} 0.99 & 0 \\ 0 & 0.99 \\ 0.90 & 0 \\ 0 & 0.90 \\ 0.50 & 0.50 \end{pmatrix}, \qquad \Sigma = \mathrm{diag}(0.01,\ 0.05,\ 0.10,\ 0.15,\ 0.20).$$

By (3): $\Omega_{11} = 0.99^2 + 0.01 = 0.9901$, $\Omega_{13} = 0.99(0.90) = 0.891$, $\Omega_{15} = 0.99(0.50) = 0.495$, $\Omega_{55} = 0.25 + 0.25 + 0.20 = 0.70$, $\Omega_{12} = 0$ (series 1 and 2 load on different factors). Communalities $\sum_l \beta_{il}^2/\Omega_{ii}$: 0.990, 0.951, 0.890, 0.844, 0.714.

The factor-score precision in (9):

$$\beta'\Sigma^{-1}\beta = \begin{pmatrix} \frac{0.99^2}{0.01} + \frac{0.90^2}{0.10} + \frac{0.5^2}{0.20} & \frac{0.5(0.5)}{0.20} \\ \cdot & \frac{0.99^2}{0.05} + \frac{0.90^2}{0.15} + \frac{0.5^2}{0.20} \end{pmatrix} = \begin{pmatrix} 98.01 + 8.10 + 1.25 & 1.25 \\ 1.25 & 19.60 + 5.40 + 1.25 \end{pmatrix} = \begin{pmatrix} 107.36 & 1 \\ 1.25 & 26 \end{pmatrix}$$

Add I_2 and invert ($\det = 108.36(27.25) - 1.25^2 = 2951.5$):

$$V_f = \frac{1}{2951.5}\begin{pmatrix} 27.25 & -1.25 \\ -1.25 & 108.36 \end{pmatrix} = \begin{pmatrix} 0.00923 & -0.00042 \\ -0.00042 & 0.03671 \end{pmatrix}.$$

Posterior sds 0.096 and 0.192: with idiosyncratic variances this small, **the factors are almost observed** — factor 1 more precisely, because series 1's noise variance (0.01) is the smallest. The recovered loadings, variances and factors are on slides 23–25 (figures).

3.2 · Worked — exchange rates (slides 26–28; Lopes and West 2004)

Standardised first differences of monthly log exchange rates against the pound, 1/1975–12/1986: US, CAN, JAP, FRA, ITA, GER; $k = 2$. Posterior means, first ordering:

	US	CAN	JAP	FRA	ITA	GER
$\beta_{\cdot 1}$	0.99	0.95	0.46	0.39	0.41	0.40
$\beta_{\cdot 2}$	0.00	0.05	0.42	0.91	0.77	0.77
σ_i^2	0.05	0.13	0.62	0.04	0.25	0.28

Check against (3): the data are standardised, so $\sum_l \beta_{il}^2 + \sigma_i^2$ should be near 1. US: $0.9801 + 0 + 0.05 = 1.030$; JAP: $0.2116 + 0.1764 + 0.62 = 1.008$; FRA: $0.1521 + 0.8281 + 0.04 = 1.020$. Factor 1 is a **dollar-bloc factor** (US, CAN), factor 2 a **European factor** (FRA, ITA, GER); the yen loads on both and keeps 62% idiosyncratic. Implied correlation ITA–GER by (3): $0.41(0.40) + 0.77(0.77) = 0.164 + 0.593 = 0.757$. The second ordering (JAP second) gives loadings within 0.02 of these: slide 19's ordering result in practice.

The site page `sv-fsv-US-BR.html` (§3.2) runs `factanal(ret, factors=2)` on Apple, Microsoft, Amazon, Petrobras, Vale and CSN daily returns: the correlations of the two factor scores with the returns are $(0.3, 0.3, 0.2, 0.8, 0.9, 0.9)$ and $(0.8, 0.9, 0.8, 0.4, 0.4, 0.3)$ — a **Brazil factor** and a **US factor**.

4 · C — forecasting with many predictors: the FAVAR (slides 51–55)

Korobilis (2008): "the issue of improving the forecasting performance of VARs when the set of available predictors is inconveniently large", $x_t = (x_{t1}, \dots, x_{tr})$, r in the hundreds (slide 51). Reduce x_t to $k \ll r$ factors and put them in the VAR (slide 52):

$$(y_t \mid y_{t-1:t-p}, \Theta) \sim N\Big(\sum_{i=1}^{p} B_i y_{t-i} + \sum_{i=0}^{s} A_i f_{t-i},\ \Sigma\Big), \qquad (x_t \mid f_t, \Theta) \sim N(\beta f_t, \Omega), \tag{10}$$

$\Theta = (B_1, \dots, B_p, A_0, \dots, A_s, \Sigma, \beta, \Omega)$. The second block is (1) for the big panel; the first is a VAR in the few series of interest augmented by the factors — "closely related to the factor augmented VAR (FAVAR) of Bernanke et al. (2005)". The VAR stays small (a few y's, a few f's) while the information set is hundreds of series. Data (slide 53): Stock and Watson (2005), 132 US series, JAN60–DEC03 (output, employment, sales, housing, inventories, prices, money, interest rates, exchange rates, …).

Forecast evaluation (slide 54). For series i, horizon h, model j, with $\tilde{y}_{i,t+h,j}$ the posterior predictive mean and forecast origins 1982:12 to 2003:12$-h$ (N_h of them):

$$\text{RMSFE}_{ihj} = \sqrt{\frac{1}{N_h} \sum_{t=1982:12}^{2003:12-h} \left(y_{i,t+h} - \tilde{y}_{i,t+h,j}\right)^2}, \tag{11}$$

reported relative to a benchmark (slide 55). (The slide prints the sum without the $1/N_h$; in a ratio of two models over the same origins the factor cancels.)

FAVAR vs large BVAR — Bańbura et al., Table III (slide 59), MSFE relative to the random walk, 1971–2003, $h = 1$:

	FAVAR 1 factor, BVAR	FAVAR 3 factors, $p = 13$ OLS	FAVAR 3 factors, BVAR	LARGE BVAR
EMPL	0.70	3.02	0.65	**0.46**
CPI	0.65	2.39	0.58	**0.50**
FFR	0.89	2.40	0.85	**0.75**

The factors help (0.65 vs the 7-variable CEE BVAR's 0.67 for employment, L09-04-bvar-minnesota-ssvs §6) but **the shrunk 131-variable BVAR beats the FAVAR** at $h = 1$ for all three series; OLS with 13 lags is again far worse than the random walk. Large TVP-BVARs and fully Bayesian FAVARs (Koop–Korobilis 2012; Ahmadi–Uhlig 2009) are in L09-06-tvp-var-sv-state-space §6.

---

5 · G — named only

- **Parsimonious BFA** (Frühwirth-Schnatter and Lopes 2009, slides 30–33): generalised lower triangular loadings, birth/death of loadings and columns, inference on k.
- **British Cohort Study** (Conti, Heckman, Frühwirth-Schnatter, Lopes and Piatek 2011, slides 34–37): BFA on cognitive and health measurements; 5,105 women and 5,420 men.
- **Spatial dynamic factor models** (Lopes, Salazar and Gamerman 2008, slides 44–49): $y_t \sim N(\beta f_t, \Sigma)$, $f_t \sim N(\Gamma f_{t-1}, \Lambda)$, columns of β Gaussian processes over space; SO_2 at US monitoring stations, interpolation and forecasting.

- **Principal components** as the non-Bayesian factor extractor (Tsay §9.4) and Stock–Watson's two-step FAVAR: outside the slides' scope here.

Links

- Neighbours: L09-04-bvar-minnesota-ssvs (shrinkage, the other answer) · L09-07-fsv-dcc-covariances (factors with stochastic volatility).
- Unit: L09-00-map · L09-08-formulas · L09-09-pitfalls · L09-10-python.
- Rotation problem twin: L09-03-structural-var §3.
- Rules: rules/05.
- **Interactive companion.** The Factor-Volatility Dial (`companions/companion-fsv-correlation.html`) — the left panel of the figure, live: move loadings, idiosyncratic variances and the factor log-variance.
- **Practice (number and printed page only):** T 9.7 (p. 503).

Slides 38–44 · Letting the VAR move: GARCH and SV in the errors, random-walk coefficients, and the Kalman filter that makes them estimable

Claim. Three ways to relax a constant VAR, in increasing order of ambition: GARCH or SV in Σ_t (Pelloni–Polasek, Uhlig), and random-walk coefficients **and** random-walk log volatilities in a time-varying recursive structure $\Sigma_t = A_t^{-1} D_t A_t^{-1\prime}$ (Primiceri). Every time-varying coefficient is a **state** in a linear state-space model, and the Kalman filter is its exact Gaussian posterior updater: a precision-weighted average of the prediction and the new observation, with gain K_t. In the local-level model the gain settles at $K^* = p^*/(p^*+1)$, $p^* = (q + \sqrt{q^2+4q})/2$, and $1-K^*$ **is the MA coefficient of the equivalent ARIMA(0,1,1)** — 0.858 for Tsay's Alcoa volatility, from two different routes.

Prev: L09-05-factor-models-favar · Next: L09-07-fsv-dcc-covariances · Map: L09-00-map

Notation.

Symbol	Meaning	Range
Σ_t	conditional covariance of u_t	p.d.
$\mathrm{vech}(\cdot)$	stack the lower triangle, $d = q(q+1)/2$ entries	
B_t, β_t	time-varying VAR coefficients (matrix, stacked vector)	
A_t, α_t	unit-lower-triangular contemporaneous matrix, its free entries	
$D_t = \mathrm{diag}(\sigma_{1t}^2, \dots, \sigma_{qt}^2)$	structural variances	> 0
Q, S, W	innovation covariances of B_t, α_t, $\log \sigma_t$	p.d.
μ_t	state of the local-level model	
$\sigma_e^2,\ \sigma_\eta^2$	measurement and state innovation variances	> 0
$q = \sigma_\eta^2/\sigma_e^2$	signal-to-noise ratio	≥ 0
$\mu_{t\|j},\ \Sigma_{t\|j}$	$E(\mu_t \mid F_j)$, $\mathrm{Var}(\mu_t \mid F_j)$ (Tsay's notation)	
$v_t,\ V_t,\ K_t$	one-step forecast error, its variance, the Kalman gain	
δ	discount factor	$(0, 1]$

---

1 · C — VAR-GARCH (slides 38–39)

Pelloni and Polasek (2003):

$$y_t = \sum_{i=1}^{p} B_i y_{t-i} + u_t, \qquad u_t \sim N(0, \Sigma_t), \qquad \mathrm{vech}(\Sigma_t) = \alpha_0 + \sum_{i=1}^{r} A_i \,\mathrm{vech}(\Sigma_{t-i}) + \sum_{i=1}^{s} \Gamma_i \,\mathrm{vech}(u_{t-i} u_{t-i}'). \tag{1}$$

Each A_i, Γ_i is $d \times d$ with $d = q(q+1)/2$: for $q = 3$, $r = s = 1$, the variance equation alone has $6 + 36 + 36 = 78$ parameters, and positive definiteness of Σ_t is not guaranteed. This is the full VEC model whose restricted versions (diagonal VEC, BEKK) are Tsay §10.2; Class 9 uses (1) only to show why factor and correlation models (L09-07-fsv-dcc-covariances) are needed. Application (slide 39): German, US and UK quarterly employment shares, 1968–1998.

---

2 · C/G — VAR-SV (slide 40; Uhlig 1997)

$$y_t = \sum_{i=1}^{p} B_i y_{t-i} + u_t, \quad u_t \sim N(0, \Sigma_t), \quad \Sigma_t^{-1} = L_t L_t', \qquad \Sigma_{t+1}^{-1} = \frac{L_t \Theta_t L_t'}{\lambda}, \quad \Theta_t \sim B_q\Big(\frac{\nu + pq}{2}, \frac{1}{2}\Big). \tag{2}$$

The precision is multiplied each period by a random matrix-Beta shock and rescaled by λ. **The scalar case** ($q = 1$, G — derived here to read (2)). With $h_t = 1/\sigma_t^2$ and $L_t = \sqrt{h_t}$, (2) is $h_{t+1} = h_t \theta_t / \lambda$ with $\theta_t \sim \text{Beta}(a, \frac{1}{2})$, $a = (\nu + p)/2$. Since $E\theta_t = a/(a + \frac{1}{2})$:

$$E(h_{t+1} \mid h_t) = h_t \frac{a}{(a + \frac{1}{2})\lambda} = h_t \quad \iff \quad \lambda = \frac{a}{a + \frac{1}{2}} = \frac{\nu + p}{\nu + p + 1}. \tag{3}$$

So λ sets the drift of the precision: at the value (3) the precision is a martingale, a multiplicative random walk. The Beta–Wishart pair is conjugate, which is why Uhlig's model has a closed-form filter — the multiplicative cousin of the discounting of §6.

---

3 · C — TVP-VAR-SV (slides 41–44; Primiceri 2005)

$$y_t = \sum_{i=1}^{p} B_{it} y_{t-i} + u_t, \qquad u_t \sim N(0, \Sigma_t), \qquad \Sigma_t = A_t^{-1} D_t (A_t^{-1})', \tag{4}$$

$$A_t = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ \alpha_{21,t} & 1 & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ \alpha_{q1,t} & \cdots & \alpha_{q,q-1,t} & 1 \end{pmatrix}, \qquad D_t = \text{diag}(\sigma_{1t}^2, \ldots, \sigma_{qt}^2), \tag{5}$$

with three independent random walks (slide 42):

$$B_t = B_{t-1} + \nu_t,\ \nu_t \sim N(0, Q); \qquad \alpha_t = \alpha_{t-1} + \xi_t,\ \xi_t \sim N(0, S); \qquad \log \sigma_t = \log \sigma_{t-1} + \eta_t,\ \eta_t \sim N(0, W). \tag{6}$$

Reading (4)–(5). Multiply u_t by A_t: $\text{Cov}(A_t u_t) = A_t \Sigma_t A_t' = D_t$, diagonal. So $A_t u_t = D_t^{1/2} \varepsilon_t$ with $\varepsilon_t \sim N(0, I)$: **a recursive SVAR (L09-03-structural-var §4) whose contemporaneous matrix and shock variances drift over time.** Ordering still matters, now at every t. The decomposition is $\Sigma_t = LGL'$ of Tsay with $L = A_t^{-1}$, $G = D_t$.

Why random walks in $\log \sigma$: positivity for free, and $\log \sigma_{it}^2$ enters linearly once the model is rewritten (the SV linearisation of Class 8). Each of (6) is a state equation, and conditional on the other two blocks each block is a linear Gaussian state-space model (or becomes one after the log-χ^2 mixture

approximation for $\log \sigma_t$) — so the Gibbs sampler draws $B_{1:T}$, $\alpha_{1:T}$ and $\log \sigma_{1:T}$ with the Kalman filter of §4 run forward and sampled backward.

Counting states: $q^2 p$ in B_t (plus intercepts), $q(q-1)/2$ in α_t, q in $\log \sigma_t$ — for Primiceri's three US variables (inflation, unemployment, interest rate) with two lags, $21 + 3 + 3 = 27$ states per period (with intercepts). Slide 43 shows the posterior mean and 16/84% bands of the standard deviations of the residuals of the inflation, unemployment and interest-rate equations: large peaks in the mid-1970s and around 1980, calm in the 1990s. Slide 44: Nakajima, Kasuya and Watanabe (2011) apply it to Japan.

---

4 · C — state space and the Kalman filter (Tsay §11.1, pp. 558–563)

The local trend (local level) model (T eqs. 11.1–11.2):

$$y_t = \mu_t + e_t, \quad e_t \sim N(0, \sigma_e^2); \qquad \mu_{t+1} = \mu_t + \eta_t, \quad \eta_t \sim N(0, \sigma_\eta^2). \tag{7}$$

(7) is the scalar skeleton of (6): an unobserved random walk seen through noise. "Equation (11.1) provides the link between the data y_t and the state μ_t and is called the observation equation … Equation (11.2) governs the time evolution of the state variable and is the state equation" (T p. 558).

Derivation of the filter (T pp. 561–563). Suppose $\mu_t \mid F_{t-1} \sim N(\mu_{t|t-1}, \Sigma_{t|t-1})$. The one-step forecast of y_t is $\mu_{t|t-1}$, so

$$v_t = y_t - \mu_{t|t-1} = (\mu_t - \mu_{t|t-1}) + e_t, \qquad V_t = \mathrm{Var}(v_t \mid F_{t-1}) = \Sigma_{t|t-1} + \sigma_e^2, \tag{8}$$

$$\mathrm{Cov}(\mu_t, v_t \mid F_{t-1}) = E[(\mu_t - \mu_{t|t-1})(\mu_t - \mu_{t|t-1} + e_t) \mid F_{t-1}] = \Sigma_{t|t-1} \tag{9}$$

(T eq. 11.9; e_t is independent of μ_t). (μ_t, v_t) is jointly normal given F_{t-1}, and $F_t = \{F_{t-1}, v_t\}$. Theorem 11.1 ($E(x \mid y) = \mu_x + \Sigma_{xy}\Sigma_{yy}^{-1}(y - \mu_y)$, $\mathrm{Var}(x \mid y) = \Sigma_{xx} - \Sigma_{xy}\Sigma_{yy}^{-1}\Sigma_{yx}$) with $x = \mu_t$, $y = v_t$:

$$\mu_{t|t} = \mu_{t|t-1} + \frac{\Sigma_{t|t-1}}{V_t} v_t = \mu_{t|t-1} + K_t v_t, \qquad K_t = \frac{\Sigma_{t|t-1}}{V_t}, \tag{10}$$

$$\Sigma_{t|t} = \Sigma_{t|t-1} - \frac{\Sigma_{t|t-1}^2}{V_t} = \Sigma_{t|t-1}(1 - K_t). \tag{11}$$

Predict with the state equation: $\mu_{t+1|t} = \mu_{t|t}$ and

$$\Sigma_{t+1|t} = \Sigma_{t|t} + \sigma_\eta^2 = \Sigma_{t|t-1}(1 - K_t) + \sigma_\eta^2 \tag{12}$$

(T eqs. 11.12–11.14). K_t **is the regression coefficient of the state on the surprise**: the share of each forecast error that moves the state. The likelihood comes free: $v_t \sim N(0, V_t)$ independently, so $\ln L = -\frac{T}{2}\ln 2\pi - \frac{1}{2}\sum_t (\ln V_t + v_t^2/V_t)$ (T eq. 11.25), maximised over (σ_e, σ_η).

Note that (11)–(12) do not involve y: the variance and gain path is deterministic.

4.1 · The steady state

If $\Sigma_{t|t-1} \to P$, then (12) and $K = P/(P+\sigma_e^2)$ give the fixed point

$$P = P\Big(1 - \frac{P}{P+\sigma_e^2}\Big) + \sigma_\eta^2 = \frac{P\sigma_e^2}{P+\sigma_e^2} + \sigma_\eta^2.$$

Multiply by $P+\sigma_e^2$:

$$P^2 + P\sigma_e^2 = P\sigma_e^2 + \sigma_\eta^2 P + \sigma_\eta^2\sigma_e^2 \quad \Longrightarrow \quad P^2 - \sigma_\eta^2 P - \sigma_\eta^2\sigma_e^2 = 0.$$

Divide by σ_e^4, write $p = P/\sigma_e^2$ and $q = \sigma_\eta^2/\sigma_e^2$: $p^2 - qp - q = 0$, positive root

$$p^* = \frac{q + \sqrt{q^2+4q}}{2}, \qquad K^* = \frac{p^*}{p^*+1}. \tag{13}$$

Comparative statics. $q \to 0$ (a nearly constant state): $p^* \approx \sqrt{q} \to 0$, $K^* \to 0$ — the filter averages the whole past. $q \to \infty$ (noise-free data): $p^* \approx q$, $K^* \to 1$ — the state is the last observation. $dK^*/dq > 0$ throughout.

The ARIMA link (T pp. 558–559). In steady state, (10) with $\mu_{t+1|t} = \mu_{t|t}$ reads

$$\mu_{t+1|t} = \mu_{t|t-1} + K^*(y_t - \mu_{t|t-1}) = (1-K^*)\mu_{t|t-1} + K^* y_t, \tag{14}$$

exponential smoothing with weight K^*, which is the forecast function of $(1-B)y_t = (1-\theta B)a_t$ with $\theta = 1 - K^*$ (Tsay: (7) "is also an ARIMA(0,1,1) model … which is the simple exponential smoothing model of Chapter 2").

4.2 · Worked — Tsay Example 11.1: Alcoa realised volatility (pp. 559–560)

Log daily realised volatility, 340 days. ARIMA fit: $(1-B)y_t = (1-0.858B)a_t$, $\hat\sigma_a = 0.5184$ (T eq. 11.6). Local-trend MLEs: $\hat\sigma_\eta = 0.0735$, $\hat\sigma_e = 0.4803$.

Route 1 — the filter's steady state, (13):

$$q = \Big(\frac{0.0735}{0.4803}\Big)^2 = 0.15303^2 = 0.023418,$$

$$p^* = \frac{0.023418 + \sqrt{0.023418^2 + 4(0.023418)}}{2} = \frac{0.023418 + \sqrt{0.000548 + 0.093672}}{2} = \frac{0.023418 + 0.306953}{2} = 0.165186$$

$$K^* = \frac{0.165186}{1.165186} = 0.14177, \qquad 1 - K^* = 0.8582.$$

Route 2 — the ARIMA fit: $\hat\theta = 0.858$. The two agree to the third decimal. Tsay checks the converse with T eqs. 11.4–11.5 ($\theta\sigma_a^2 = \sigma_e^2$, $(1+\theta^2)\sigma_a^2 = 2\sigma_e^2 + \sigma_\eta^2$): $\sigma_e = \sqrt{0.858(0.5184^2)} = 0.4802$ and $\sigma_\eta^2 = (1+0.858^2)0.26874 - 2(0.23058) = 0.46658 - 0.46116 = 0.00542$, $\sigma_\eta = 0.0736$ — "close to the

MLE". **Measurement noise is 6.5 times the state innovation sd, so only 14% of each day's surprise is believed.**

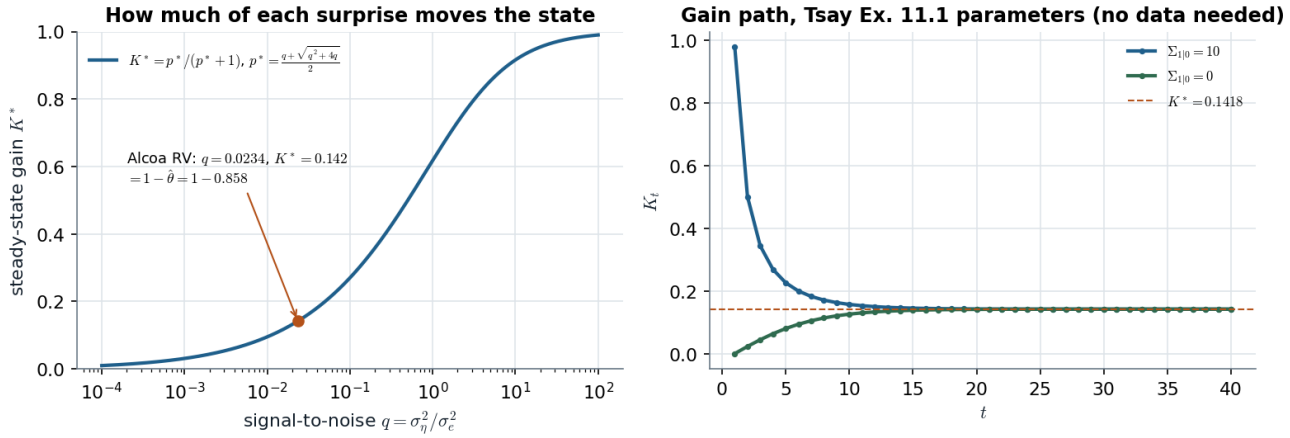


Figure — left: K^* from (13) against q, with T Ex. 11.1's $q=0.0234$; right: the deterministic gain path (11)–(12) for $\sigma_e=0.4803$, $\sigma_\eta=0.0735$ from $\Sigma_{1|0}=10$ and from 0.

Left: the gain is an S-curve in $\log q$, steepest around $q\approx 1$; Alcoa's volatility sits in the flat lower part, where the filter is heavily smoothing. Right: from a diffuse start the first observation gets almost full weight ($K_1=10/10.23=0.98$), from a zero-variance start none ($K_1=0$); both paths reach 0.1418 within about 15 days — **initialisation matters only at the start** (T §11.1.6).

---

5 · C — time-varying regression coefficients

TVP-CAPM (T §11.3.1, pp. 577–578):

$$r_t=\alpha_t+\beta_t r_{M,t}+e_t,\qquad \alpha_{t+1}=\alpha_t+\eta_t,\qquad \beta_{t+1}=\beta_t+\epsilon_t, \tag{15}$$

state $s_t=(\alpha_t,\beta_t)'$, transition I_2, observation row $Z_t=(1,r_{M,t})$ — **time-varying regressors live in the observation matrix**, and the Kalman filter of §4 runs unchanged with vectors. A TVP-VAR is (15) for each equation with Z_t the VAR's lagged y's: the SUR form $y_t=z_t\beta_t+u_t$ of L09-04-bvar-minnesota-ssvs §5 with β_t a state.

Bayesian TVP regression (G §10.3, pp. 161–164): $y_t=x_t'\beta_t+u_t$, $\beta_t=\beta_{t-1}+\eta_t$, $u_t\sim N(0,\sigma^2)$, $\eta_t\sim N_K(0,\Omega)$ (G eqs. 10.10–10.11). Iterating the state equation from a known β_0:

$$\beta_t=\beta_0+\sum_{s=1}^{t}\eta_s\quad\Longrightarrow\quad \mathrm{Var}(\beta_t\mid\beta_0)=t\,\Omega, \tag{16}$$

the prior uncertainty about the coefficient **grows linearly with time** — the random-walk signature of L01 §7 in parameter space. Greenberg stacks $\beta=(\beta_T',\dots,\beta_1')'$ and writes the transition as $H\beta=\tilde{\beta}_0+\eta$, which gives a Gaussian prior $\beta\sim N(H^{-1}\tilde{\beta}_0,H^{-1}(I_T\otimes\Omega)H^{-1\prime})$ and a Gibbs sampler with

$$\beta\mid y,\sigma^2,\Omega\sim N_{TK}(\bar{\beta},B_1),\quad B_1=\left[\frac{X'X}{\sigma^2}+H'(I_T\otimes\Omega^{-1})H\right]^{-1},\quad \Omega^{-1}\mid\cdot\sim W_K\Big(\alpha_0+T,\Big[S_0^{-1}+\sum_t(\beta_t-\beta_{t-1})(\beta_t-\beta_{t-1}\Big) \tag{17}$$

"This algorithm must be modified for large KT because B_1 is the inverse of a $KT \times KT$ matrix" — hence the recursive (Kalman) sampler. **Greenberg's Phillips-curve example** (Table 10.3): the TVP model's $\log_{10}$ marginal likelihood is -92.506, and "the AR(1) model has a larger marginal likelihood despite the large variation in the values of β_t": time variation is not free — it must earn its extra states.

---

6 · C — large TVP-VARs without MCMC (Part 2, slide 61)

Koop and Korobilis (2012): $y_t = z_t\beta_t + u_t$, $u_t \sim N(0, \Sigma)$, $\beta_t = \beta_{t-1} + v_t$, $v_t \sim N(0, V_t)$, and "the old discount strategy of West and Harrison (1997) to avoid MCMC altogether". Discounting chooses V_t so that the prediction step inflates the filtered variance by a fixed factor (G — the standard West–Harrison form):

$$P_{t|t-1} = P_{t-1|t-1} + V_t \stackrel{!}{=} \frac{P_{t-1|t-1}}{\delta} \quad \Longrightarrow \quad V_t = \Big(\frac{1}{\delta} - 1\Big) P_{t-1|t-1}. \tag{18}$$

No Q to estimate: one number δ replaces a $K \times K$ matrix. Information j periods old is down-weighted by about δ^j, an effective memory of $1/(1-\delta)$ observations ($\delta = 0.99$: 100 periods; $\delta = 1$: constant coefficients). See Ahmadi and Uhlig (2009) for a fully Bayesian FAVAR alternative (slide 61).

Links

- Neighbours: L09-05-factor-models-favar · L09-07-fsv-dcc-covariances (time-varying covariances without a VAR mean).
- Structural link: L09-03-structural-var §4 (A_t is a drifting recursive A_0).
- Class 1: L01 §7 (random walks: variance $t\sigma^2$).
- Rules: rules/05 (state space, TVP).
- **Practice (number and printed page only):** G 10.4 (p. 167).

Part 2 slides 38–43 and the two site pages · Time-varying covariances: DCC models the correlations directly, factor SV gets them from the factor's volatility

Claim. A k-dimensional conditional covariance has $k(k+1)/2$ moving entries, so every usable model imposes structure. **DCC** splits $\Sigma_t = D_t R_t D_t$: univariate GARCH for D_t, then one scalar recursion with two parameters for all the correlations. **Factor SV** keeps $y_t = \beta f_t + \varepsilon_t$ and lets the factor and idiosyncratic log-variances follow AR(1)s; then $\Sigma_t = \beta H_t \beta' + \Sigma_t^{\varepsilon}$ and, with one factor, $\rho_{ij,t} = \text{sign}(\beta_i\beta_j)/\sqrt{(1+a_{it}/h_t)(1+a_{jt}/h_t)}$: **correlations rise exactly when the common factor is volatile** — a prediction, not a free parameter. On the site's W&T Offshore / S&P 500 data, the DCC correlation shock has a half-life of 7 days (Engle) or 2 days (Tse–Tsui), and the S&P GARCH persistence is above one.

Prev: L09-06-tvp-var-sv-state-space · Next: L09-08-formulas · Map: L09-00-map

Notation.

Symbol	Meaning	Range
a_t	$k \times 1$ return innovation, $E(a_t \mid F_{t-1}) = 0$	
$\Sigma_t = \text{Cov}(a_t \mid F_{t-1})$	conditional covariance	p.d.
$D_t = \text{diag}(\sigma_{11,t}^{1/2}, \ldots, \sigma_{kk,t}^{1/2})$	conditional sds	
R_t (ρ_t in Tsay)	conditional correlation matrix	unit diagonal
$\varepsilon_t = D_t^{-1} a_t$	standardised innovations	
θ_1, θ_2	DCC scalars	≥ 0, $\theta_1 + \theta_2 < 1$
λ	EWMA decay	$(0, 1)$
f_t, β	factors and loadings (as in L09-05-factor-models-favar)	
$h_t = \sigma_{p+1,t}^2$	factor variance (one factor), $\lambda_t = \log h_t$	> 0
σ_{it}^2, $\eta_{it} = \log \sigma_{it}^2$	idiosyncratic variances and log-variances	
$a_{it} = \sigma_{it}^2/\beta_i^2$	idiosyncratic-to-loading ratio	≥ 0

---

1 · C — the problem (T p. 506)

"Multivariate volatility modeling is concerned with the time evolution of Σ_t … the curse of dimensionality quickly becomes a major obstacle in applications because there are $k(k+1)/2$ quantities in Σ_t for a k-dimensional return series. To illustrate, there are 15 conditional variances and covariances in Σ_t for a five-dimensional return series." The full VEC/VAR-GARCH (L09-06-tvp-var-sv-state-space (1)) gives each of them its own equation; everything below is a way not to.

---

2 · C — the exponentially weighted benchmark (T §10.1, eq. 10.2, p. 506)

$$\hat{\Sigma}_t = \frac{1-\lambda}{1-\lambda^{t-1}} \sum_{j=1}^{t-1} \lambda^{j-1} a_{t-j} a_{t-j}', \qquad 0 < \lambda < 1. \tag{1}$$

The weights sum to one: $\sum_{j=1}^{t-1}\lambda^{j-1} = (1-\lambda^{t-1})/(1-\lambda)$. For large t ($\lambda^{t-1}\approx 0$), split off the $j=1$ term and re-index the rest ($j = i+1$):

$$\hat{\Sigma}_t = (1-\lambda)a_{t-1}a_{t-1}' + \lambda\Big[(1-\lambda)\sum_{i=1}^{t-2}\lambda^{i-1}a_{t-1-i}a_{t-1-i}'\Big] = (1-\lambda)a_{t-1}a_{t-1}' + \lambda\hat{\Sigma}_{t-1}. \tag{2}$$

One parameter for all $k(k+1)/2$ entries, positive definite by construction. It is an IGARCH: no mean reversion, the persistence is exactly one.

---

3 · C — correlations separately: $\Sigma_t = D_tR_tD_t$ (T eq. 10.7, p. 516)

$$\Sigma_t = D_tR_tD_t, \qquad [\Sigma_t]_{ij} = \sigma_{ii,t}^{1/2}\,\rho_{ij,t}\,\sigma_{jj,t}^{1/2}. \tag{3}$$

The k variances can be modelled one series at a time (GARCH, SV — Class 8). What remains is R_t: $k(k-1)/2$ correlations, symmetric, unit diagonal, positive definite.

---

4 · C — dynamic conditional correlation (T §10.4.3, pp. 531–532)

Tse and Tsui (2002):

$$R_t = (1-\theta_1-\theta_2)\bar{R} + \theta_1 R_{t-1} + \theta_2\Psi_{t-1}, \tag{4}$$

with $\bar{R}$ a fixed correlation matrix (often the sample one) and Ψ_{t-1} the sample correlation of the standardised shocks over the last m days.

Engle (2002):

$$Q_t = (1-\theta_1-\theta_2)\bar{Q} + \theta_1\varepsilon_{t-1}\varepsilon_{t-1}' + \theta_2 Q_{t-1}, \qquad R_t = J_tQ_tJ_t, \quad J_t = \mathrm{diag}(q_{11,t}^{-1/2},\dots,q_{kk,t}^{-1/2}), \tag{5}$$

in Tsay's order (θ_1 on the news term). J_t rescales Q_t into a correlation matrix; $0<\theta_1+\theta_2<1$ keeps it positive definite. "An obvious drawback of the prior two models is that θ_1 and θ_2 are scalar so that all the conditional correlations have the same dynamics" (T p. 531).

Mean reversion. Write $s = \theta_1+\theta_2$. Take $E(\cdot \mid F_t)$ of (5) at $t+2$:

$$E(Q_{t+2}\mid F_t) = (1-s)\bar{Q} + \theta_1 E(\varepsilon_{t+1}\varepsilon_{t+1}'\mid F_t) + \theta_2 Q_{t+1}.$$

$E(\varepsilon_{t+1}\varepsilon_{t+1}'\mid F_t) = R_{t+1}$, not Q_{t+1}; with the usual approximation $R_{t+1}\approx Q_{t+1}$:

$$E(Q_{t+2}\mid F_t) - \bar{Q} \approx s\,(Q_{t+1}-\bar{Q}) \quad\Longrightarrow\quad E(Q_{t+h}\mid F_t) - \bar{Q} \approx s^{h-1}(Q_{t+1}-\bar{Q}). \tag{6}$$

The same algebra is exact for (4) in expectation if $E\Psi_t \approx R_t$, and exact for a GARCH(1,1) variance: $E(\sigma_{t+h}^2\mid F_t) - \bar{\sigma}^2 = (\alpha+\beta)^{h-1}(\sigma_{t+1}^2-\bar{\sigma}^2)$, with $\bar{\sigma}^2 = \omega/(1-\alpha-\beta)$. Half-life: $h_{1/2} = \ln 0.5/\ln s$.

4.1 · Worked — W&T Offshore vs the S&P 500 (wti-sp500-timevaryingcovariance.html)

Daily percentage returns, 2014-04-05 to 2023-04-04, 2263 observations after dropping missing values. (The ticker "WTI" here is **W&T Offshore, Inc.**, an oil-and-gas stock, not the crude-oil benchmark.)

Step 1, univariate GARCH(1,1)-t (`dccPre`, `cond.dist="std"`):

	ω	α_1	β_1	ν	$\alpha_1+\beta_1$
W&T Offshore	1.2985	0.1061	0.8432	5.46	0.9493
S&P 500	0.0203	0.1964	0.8065	5.60	**1.0029**

W&T: $\bar{\sigma}^2 = 1.2985/(1-0.9493) = 1.2985/0.0507 = 25.6$, a long-run daily sd of 5.06%; half-life $\ln 0.5/\ln 0.9493 = 13.3$ days. **S&P:** $\alpha_1+\beta_1 = 1.0029 > 1$ — no finite unconditional variance, the fitted variance does not revert (the 2020 crash dominates the sample). With $\nu \approx 5.5$ the Student-t kurtosis $3(\nu-2)/(\nu-4) = 3(3.46)/1.46 = 7.1$: heavy tails.

Step 2, correlations (`dccFit` on the standardised residuals; MTS prints the two scalars and the t degrees of freedom):

model	first θ	second θ	ν	sum s	half-life $\ln 0.5/\ln s$
Tse–Tsui (default)	0.7018	0.0500	6.11	0.7518	2.43 days
Engle	0.8704	0.0381	6.14	0.9085	7.22 days

By magnitude the first printed scalar is the weight on the lagged correlation term (it plays θ_2 in Tsay's (5) ordering) and the second the weight on the news term; the sum s and the half-life do not depend on the labelling. **Engle's correlation shocks persist three times as long as Tse–Tsui's.** Diagnostics (`MCHdiag`): the portmanteau $Q(m)$ on e_t has p-values 0.23 and 0.26, but the rank-based test rejects (0.0022 and 0.0018), and the robust $Q_k(m)$ rejects for Engle at 5% (0.033) — some dependence is left.

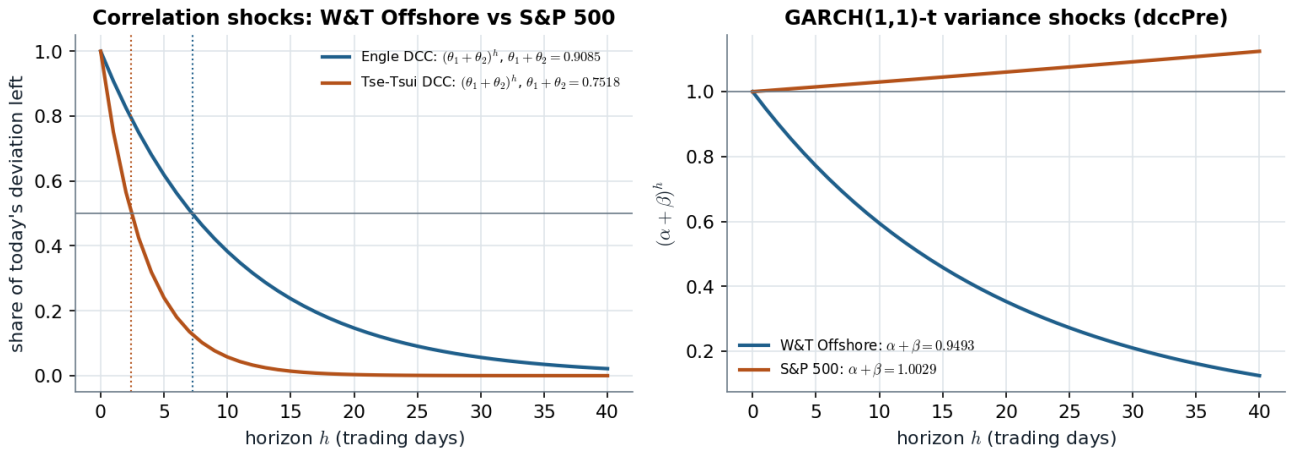


Figure — reversion multipliers from (6) with the dccFit and dccPre estimates of the site page; dotted lines mark the half-lives.

The left panel says how fast a correlation surprise is forgotten: after two weeks (10 days) Engle keeps $0.9085^{10} = 0.38$ of it, Tse–Tsui $0.7518^{10} = 0.06$. The right panel shows the S&P variance curve **above**

one and rising — a fitted model that predicts today's volatility shock never fades, which is the IGARCH edge of (2).

---

5 · C — factor stochastic volatility (Part 2, slide 38; Pitt and Shephard 1999)

$$y_t \mid f_t \sim N(\beta f_t, \Sigma_t), \qquad f_t \sim N(0, H_t), \tag{7}$$

$$\Sigma_t = \mathrm{diag}(\sigma_{1t}^2, \dots, \sigma_{pt}^2), \qquad H_t = \mathrm{diag}(\sigma_{p+1,t}^2, \dots, \sigma_{p+k,t}^2), \tag{8}$$

$$\eta_{it} = \log \sigma_{it}^2 \sim N(\alpha_i + \gamma_i \eta_{i,t-1}, \xi_i^2), \qquad \lambda_{jt} = \log \sigma_{p+j,t}^2 \sim N(\mu_j + \phi_j \lambda_{j,t-1}, \tau_j^2). \tag{9}$$

It is the static factor model of L09-05-factor-models-favar (1) with $p + k$ univariate SV-AR(1) processes (Class 8) for the variances. Exactly as there, (2) of that note becomes

$$\mathrm{Var}(y_t \mid H_t, \Sigma_t) = \beta H_t \beta' + \Sigma_t. \tag{10}$$

One factor, two series. With $h_t = H_t$ (scalar), by (10):

$$[\mathrm{Var}]_{ii} = \beta_i^2 h_t + \sigma_{it}^2, \qquad [\mathrm{Var}]_{ij} = \beta_i \beta_j h_t,$$

$$\rho_{ij,t} = \frac{\beta_i \beta_j h_t}{\sqrt{(\beta_i^2 h_t + \sigma_{it}^2)(\beta_j^2 h_t + \sigma_{jt}^2)}}. \tag{11}$$

Divide numerator and denominator by $|\beta_i \beta_j| h_t$ and write $a_{it} = \sigma_{it}^2 / \beta_i^2$:

$$\rho_{ij,t} = \frac{\mathrm{sign}(\beta_i \beta_j)}{\sqrt{(1 + a_{it}/h_t)(1 + a_{jt}/h_t)}}. \tag{12}$$

Both factors under the root fall as h_t rises, so $|\rho_{ij,t}|$ **is increasing in** h_t, from 0 ($h_t \to 0$) to 1 ($h_t \to \infty$). Correlations go up in turbulent markets **because** the common factor's variance goes up — the FSV account of "correlations rise in crises", with no correlation parameter at all.

Worked — slide 22's series 1 and 3 ($\beta = 0.99,\ 0.90$; $\sigma^2 = 0.01,\ 0.10$): $a_1 = 0.01/0.9801 = 0.010203$, $a_3 = 0.10/0.81 = 0.123457$.

- $h = 1$: $\rho = 1/\sqrt{1.010203 \times 1.123457} = 1/\sqrt{1.134920} = 1/1.065326 = 0.9387$.
- $h = 0.05$ ($\lambda_t = \log 0.05 = -3.0$): $\rho = 1/\sqrt{(1 + 0.20406)(1 + 2.46914)} = 1/\sqrt{1.20406 \times 3.46914} = 1/\sqrt{4.17706} = 0.4893$.

A twentyfold fall in the factor variance halves the correlation. The curve is the left panel of the figure in L09-05-factor-models-favar §2.

> **Interactive companion.** The Factor-Volatility Dial (`companions/companion-fsv-correlation.html`) — move the factor log-variance λ_t and watch ρ_t travel along (11); change the loadings and idiosyncratic variances to see which pairs are most sensitive.

Extensions on the slides.

- **Correlated factor volatilities** (Aguilar and West 2000, slide 39): $\lambda_t \sim N(\mu + \Phi\lambda_{t-1}, U)$ with U a full covariance matrix, $\Phi = \text{diag}(\phi_1, \ldots, \phi_k)$.
- **Time-varying loadings** (Lopes and Carvalho 2007, slide 40): the $d = pk - k(k-1)/2$ free loadings follow AR(1)s, $\beta_{ij,t} \sim N(\mu_{ij} + \phi_{ij}\beta_{ij,t-1}, \sigma^2_{ij})$.
- **Daily exchange rates** (slides 41–43): DEM, GBP, JPY, FRF, CAD, ESP against the dollar, 1/1/1992–10/31/1995, $k = 3$ with time-varying loadings and vague priors; slide 43 plots, per currency, the share of variance due to each factor and to the idiosyncratic term over time.
- **Tsay §10.6** ("factor–volatility models"): pick the first few principal components of a_t and model their volatilities — the non-Bayesian route to the same structure.

Parameter counts for k series (one factor, AR(1) log-variances with 3 parameters each):

model	parameters	grows like
full VEC GARCH(1,1)	$d + 2d^2$, $d = k(k+1)/2$	k^4
DCC (Engle) with GARCH(1,1) margins	$3k + 2$ (+ $\bar{Q}$ estimated by moments)	k (+ k^2 moments)
one-factor FSV	$3(k+1) + k$ loadings	k

---

6 · C — the site's two FSV pages

Six stocks, US and Brazil (`sv-fsv-US-BR.html`): Apple, Microsoft, Amazon, Petrobras, Vale, CSN, 1/1/2006–3/24/2021, $n = 3831$ demeaned log returns. Univariate SV-AR(1) by `stochvol`, $y_t = e^{h_t/2}\varepsilon_t$, $h_t = \mu + \phi(h_{t-1} - \mu) + \sigma\eta_t$, priors $\mu \sim N(0, 100^2)$, $(\phi + 1)/2 \sim \text{Beta}(5, 1.5)$, $\sigma^2 \sim IG(1/2, 1/2)$ (as printed), $h_0 \sim N(\mu, \sigma^2/(1-\phi^2))$. Then 50-day rolling correlations, a static two-factor `factanal` (a Brazil and a US factor, L09-05-factor-models-favar §3.2), and `fsvsample` with 2 factors on all six and 1 factor on each country's three. Correlations between the factor paths:

	6-d FSV factor 1	6-d FSV factor 2
3-d US FSV factor	**0.989**	0.073
3-d BR FSV factor	0.636	**0.858**

(The two country-level factors correlate 0.562 with each other.) The joint model rediscovers the two country factors (in the opposite order to `factanal`: FSV identifies factors by the lower-triangular β, whose first row is Apple). The page then compares FSV's time-varying correlations with the 50-day rolling ones and the static sample correlation, and the 6-d, 3-d and 1-d volatility estimates (figures only).

W&T Offshore and the S&P 500 (§4.1 page): `svtlsample` (SV with leverage and t errors) per series, `fsvsample(return, factors=1)` for the pair, posterior densities of the two loadings (between 0.4 and 2.0), FSV sds against the SV ones, and the FSV correlation against Engle's DCC correlation on $[0, 1]$ with guides at 0.2, 0.5, 0.8 (figures only).

---

7 · C — DCC vs FSV, side by side

	DCC	FSV
object modelled	R_t directly, (4)–(5)	H_t, Σ_t; R_t implied by (11)
what moves a correlation	products of yesterday's standardised shocks	the factor's volatility (and the idiosyncratic ones)
all pairs share dynamics?	yes (scalar θ's)	no: each pair's sensitivity is set by a_{it}, a_{jt}
positive definiteness	by $\theta_1 + \theta_2 < 1$ and $\bar{Q}$ p.d.	by construction, (10)
estimation	two-step QMLE (dccPre, dccFit)	MCMC (fsvsample)
dimension	$\bar{Q}$ has $k(k-1)/2$ entries	linear in k
interpretation	statistical	economic (market/country factors)

Links

- Neighbours: L09-06-tvp-var-sv-state-space (VAR-GARCH, SV in a VAR) · L09-08-formulas.
- Factor algebra: L09-05-factor-models-favar §1–§3.
- Unit: L09-00-map · L09-09-pitfalls · L09-10-python.
- Rules: rules/05 (GARCH, SV, factor SV).
- **Practice (number and printed page only):** T 10.1, 10.2, 10.3, 10.4, 10.5, 10.6 (p. 554).

Class 9 · Formulas — the exam specification, every result, every number

Prev: L09-07-fsv-dcc-covariances · Next: L09-09-pitfalls · Map: L09-00-map

Equations are cited by the **section note's own numbering**: "L09-02 (8)" is equation (8) of L09-02-vma-irf-fevd. Every number in §2 is asserted in L09-10-python (script `L09_python_checks.py`) or in `figures/make_figures.py`.

---

0 · The exam specification, mapped

0.1 · The site's description of Class 9 (`Aula/COURSE.md`)

"Introduction to multivariate time series econometrics", six items:

#	Site item (verbatim)	Where it is settled
i	"Factor SV models"	L09-05-factor-models-favar §1–§3 (factor algebra); L09-07-fsv-dcc-covariances §5–§6
ii	"Dynamic Conditional Correlation (DCC) models"	L09-07-fsv-dcc-covariances §3–§4, §7
iii	"Vector autoregressive (VAR) models, impulses-response function, structural VAR (SVAR)"	L09-01-var-p-estimation-forecasting; L09-02-vma-irf-fevd; L09-03-structural-var
iv	"Large BVAR, Factor-augmented VAR (FAVAR)"	L09-04-bvar-minnesota-ssvs §6; L09-05-factor-models-favar §4
v	"Time-varying parameter (TVP)-VAR"	L09-06-tvp-var-sv-state-space §3–§6
vi	"Bayesian VAR (BVAR) and Bayesian FAVAR (BFAVAR)"	L09-04-bvar-minnesota-ssvs §2–§5; L09-05-factor-models-favar §3–§4

0.2 · The class plan's goals (`Map/class-plan.md`, Class 9)

Goal	Note
VAR estimation	L09-01 §3–§4, §7
impulse responses	L09-02 §1–§3
structural VAR	L09-03
large Bayesian VAR (Minnesota-type shrinkage)	L09-04
factor-augmented VAR	L09-05 §4
time-varying-parameter VAR with SV	L09-06 §2–§3
factor stochastic volatility	L09-07 §5
DCC vs FSV for time-varying covariances	L09-07 §4, §7

The final exam has no coding: expect derivations (OLS = MLE for a VAR, Ψ_s recursion, FEVD at $h = 1$, the Minnesota posterior, $\Omega = \beta\beta' + \Sigma$, the Kalman update, DCC reversion) and the reading of printed output (VARorder, FEVdec, dccFit).

---

1 · Definitions and results

1.1 · VAR(p) (L09-01)

#	Statement	Note
VAR(p)	$y_t = \mu + B_1 y_{t-1} + \cdots + B_p y_{t-p} + u_t$, $u_t \sim N(0, \Sigma)$ i.i.d.	L09-01 (1)
Mean	$m = (I - B_1 - \cdots - B_p)^{-1}\mu$	L09-01 (2)
Regression form	$y_t = Bx_t + u_t$; $Y = XB' + U$, $U \sim N(0, I_n, \Sigma)$; $\mathrm{vec}(Y) = (I_q \otimes X)\beta + u$, $u \sim N(0, \Sigma \otimes I_n)$	L09-01 (3)–(5)
OLS = MLE	$S(B) = S(\hat{B}) + (\hat{B} - B)X'X(\hat{B} - B)' \Rightarrow \hat{B}' = (X'X)^{-1}X'Y$ for every Σ; $\tilde{\Sigma} = S(\hat{B})/n$	L09-01 (9)–(11)
GLS = OLS	same regressors in every equation $\Rightarrow$ $\hat{\beta}_{GLS} = (I_q \otimes (X'X)^{-1}X')y$	L09-01 (12)
Sampling covariance	$\mathrm{Cov}(\hat{\beta}) = \Sigma \otimes (X'X)^{-1}$	L09-01 (13)
Companion form	$Y_t = AY_{t-1} + U_t$, first block row $(B_1, \dots, B_p)$, identities below	L09-01 (14)
Stationarity	all eigenvalues of A inside the unit circle $\Leftrightarrow$ all roots of $\lvert I - B_1 z - \cdots - B_p z^p \rvert = 0$ outside; $\det(\lambda I - A) = \lambda^{qp}\det(I - B_1\lambda^{-1} - \dots)$	L09-01 (15)
Forecast	$y_t(h) = JA^h Y_t$; $e_t(h) = \sum_{i<h} \Psi_i u_{t+h-i}$; $\mathrm{MSE}(h) = \sum_{i<h} \Psi_i \Sigma \Psi_i'$	L09-01 (16)–(18)
Bayesian forecast	$\int F(A, y_t, h)\, p(B, \Sigma \mid \text{data})$, F nonlinear in B	L09-01 (19)
Order selection	$M(i) = -(T - k - i - \frac{3}{2})\ln(\lvert\tilde{\Sigma}_i\rvert / \lvert\tilde{\Sigma}_{i-1}\rvert) \sim \chi^2_{k^2}$; AIC/BIC/HQ with penalties $2k^2 i/T$, $k^2 i \ln T/T$, $2k^2 i \ln\ln T/T$	L09-01 (20)–(21)

1.2 · Impulse responses and FEVD (L09-02)

#	Statement	Note
VMA(∞)	$y_t = \sum_i \Psi_i u_{t-i}$, $\Psi_0 = I$, $\Psi_s = \sum_{i=1}^{p} B_i \Psi_{s-i}$	L09-02 (1)–(3)
IRF	$\partial y_{t+s}/\partial u_t' = \Psi_s$; (i,j) = response of i to a shock in j	L09-02 (4)
Orthogonalised IRF	$\Sigma = PP'$ (lower Cholesky), $\varepsilon_t = P^{-1}u_t \sim N(0,I)$, $\Theta_s = \Psi_s P$, $\Theta_0 = P$	L09-02 (5)–(6)
FEVD	$\omega_{ij}(h) = \sum_{s<h} \Theta_s(i,j)^2 / \sum_k \sum_{s<h} \Theta_s(i,k)^2$; $\omega_{21}(1) = \rho_{21}^2$	L09-02 (7)–(8)

1.3 · Structural VAR (L09-03)

#	Statement	Note
SVAR	$A_0 y_t = A_1 y_{t-1} + \cdots + u_t$, $u_t \sim N(0,I)$	L09-03 (1)
Reduced form	$B_i = A_0^{-1}A_i$, $\Sigma = (A_0'A_0)^{-1}$; structural IRF $\Psi_s A_0^{-1}$	L09-03 (3)–(5)
Observational equivalence	(QA_0, QA), Q orthogonal, gives the same (B,Σ)	L09-03 (6)–(7)
Order condition	$q^2 - q(q+1)/2 = q(q-1)/2$ restrictions	L09-03 (8)
Exact identification	unique $(A_0, A) \in R$ with $g(A_0, A) = (B,\Sigma)$ for almost every (B,Σ) (slide 34)	L09-03 §3
Recursive = Cholesky	A_0 lower triangular $\Rightarrow A_0^{-1} = P$, structural IRF $= \Theta_s$	L09-03 (9)

1.4 · Bayesian VAR (L09-04)

#	Statement	Note
Minnesota prior	mean: own lag 1 = 1, else 0; $\mathrm{Var}(B_k(i,i)) = \lambda^2/k^2$, $\mathrm{Var}(B_k(i,j)) = \theta^2\lambda^2\sigma_i^2/(k^2\sigma_j^2)$; Σ diagonal, known	L09-04 (1)–(2)
Posterior	$\bar{V}_i = (V_{i0}^{-1} + X'X/\sigma_i^2)^{-1}$, $\bar{\beta}_i = \bar{V}_i(V_{i0}^{-1}\beta_{i0} + X'y^{(i)}/\sigma_i^2)$; $\lambda \to 0$: prior; $\lambda \to \infty$: OLS	L09-04 (7)–(8)
Scalar weight	$\bar{b} = wb_0 + (1-w)\hat{b}$, $w = \mathrm{Var}(\hat{b})/(\mathrm{Var}(\hat{b}) + v_0)$	L09-04 (9)
Diffuse prior	$\mathrm{vec}(B') \mid \Sigma, Y \sim N(\mathrm{vec}(\hat{B}'), \Sigma \otimes (X'X)^{-1})$; $\Sigma \mid Y \sim IW(S(\hat{B}), n-k)$	L09-04 (11)–(12)
Normal-Wishart	$\gamma \mid \Psi \sim N(\tilde{\gamma}, \Psi \otimes \tilde{\Omega})$ forces $\theta = 1$	L09-04 §4

#	Statement	Note
ON/OFF switch	$\gamma_j \mid \cdot \sim \text{Bernoulli}(A_j/(A_j+B_j))$	L09-04 (13)
SSVS	$\beta_j \mid \gamma_j \sim$ $(1-\gamma_j)N(0,\tau_{0j}^2)+\gamma_j N(0,\tau_{1j}^2)$; $P(\gamma_j=1 \mid \beta_j)$ by Bayes	L09-04 (14)–(15)

1.5 · Factor models and FAVAR (L09-05)

#	Statement	Note
Model	$y_t \mid f_t \sim N(\beta f_t, \Sigma)$, $f_t \sim N(0, I_k)$, Σ diagonal	L09-05 (1)
Covariance	$\Omega = \beta\beta' + \Sigma$; $\text{var}(y_{it}) = \sum_l \beta_{il}^2 + \sigma_i^2$, $\text{cov} = \sum_l \beta_{il}\beta_{jl}$	L09-05 (2)–(3)
Rotation	$(\beta P', Pf_t)$ observationally equivalent; block lower triangular β, $\beta_{ii} > 0$	L09-05 (4)
Bound on k	$m(k+1) - k(k-1)/2 \leq m(m+1)/2 \Leftrightarrow$ $k \leq (2m+1-\sqrt{8m+1})/2$	L09-05 (5)–(7)
Factor scores	$f_t \mid y_t \sim N(V_f\beta'\Sigma^{-1}y_t, V_f)$, $V_f = (I_k + \beta'\Sigma^{-1}\beta)^{-1}$	L09-05 (9)
FAVAR	$y_t \mid \cdot \sim$ $N(\sum B_i y_{t-i} + \sum A_i f_{t-i}, \Sigma)$, $x_t \mid f_t \sim N(\beta f_t, \Omega)$	L09-05 (10)
RMSFE	$\sqrt{N_h^{-1}\sum_t (y_{i,t+h} - \tilde{y}_{i,t+h,j})^2}$	L09-05 (11)

1.6 · Time-varying VARs and the Kalman filter (L09-06)

#	Statement	Note
VAR-GARCH	$\text{vech}(\Sigma_t) =$ $\alpha_0 + \sum A_i \text{vech}(\Sigma_{t-i}) +$ $\sum \Gamma_i \text{vech}(u_{t-i}u_{t-i}')$	L09-06 (1)
VAR-SV (Uhlig)	$\Sigma_{t+1}^{-1} = L_t\Theta_t L_t'/\lambda$, $\Theta_t \sim B_q((\nu+pq)/2, 1/2)$; scalar martingale iff $\lambda = (\nu+p)/(\nu+p+1)$	L09-06 (2)–(3)
TVP-VAR-SV (Primiceri)	$\Sigma_t = A_t^{-1}D_t A_t^{-1\prime}$; random walks for B_t, α_t, $\log\sigma_t$	L09-06 (4)–(6)
Local level	$y_t = \mu_t + e_t$, $\mu_{t+1} = \mu_t + \eta_t$	L09-06 (7)
Kalman filter	$v_t = y_t - \mu_{t\|t-1}$, $V_t = \Sigma_{t\|t-1} + \sigma_e^2$, $K_t = \Sigma_{t\|t-1}/V_t$, $\mu_{t\|t} = \mu_{t\|t-1} + K_t v_t$, $\Sigma_{t\|t} = \Sigma_{t\|t-1}(1-K_t)$, $\Sigma_{t+1\|t} = \Sigma_{t\|t} + \sigma_\eta^2$	L09-06 (8)–(12)

#	Statement	Note
Steady state	$p^* = (q + \sqrt{q^2+4q})/2$, $K^* = p^*/(p^*+1)$, $q = \sigma_\eta^2/\sigma_e^2$; $\theta_{\text{ARIMA(0,1,1)}} = 1 - K^*$	L09-06 (13)–(14)
TVP regression	$\text{Var}(\beta_t \mid \beta_0) = t\Omega$; Greenberg's Gibbs (17)	L09-06 (16)–(17)
Discounting	$V_t = (\delta^{-1} - 1)P_{t-1\|t-1}$	L09-06 (18)

1.7 · Multivariate volatility (L09-07)

#	Statement	Note
EWMA	$\hat{\Sigma}_t = (1-\lambda)a_{t-1}a_{t-1}' + \lambda\hat{\Sigma}_{t-1}$	L09-07 (1)–(2)
Correlation split	$\Sigma_t = D_t R_t D_t$	L09-07 (3)
Tse–Tsui DCC	$R_t = (1-\theta_1-\theta_2)\bar{R} + \theta_1 R_{t-1} + \theta_2 \Psi_{t-1}$	L09-07 (4)
Engle DCC	$Q_t = (1-\theta_1-\theta_2)\bar{Q} + \theta_1\varepsilon_{t-1}\varepsilon_{t-1}' + \theta_2 Q_{t-1}$, $R_t = J_t Q_t J_t$	L09-07 (5)
Reversion	$E(Q_{t+h} \mid F_t) - \bar{Q} \approx (\theta_1+\theta_2)^{h-1}(Q_{t+1} - \bar{Q})$; half-life $\ln 0.5 / \ln(\theta_1+\theta_2)$	L09-07 (6)
FSV	$y_t \mid f_t \sim N(\beta f_t, \Sigma_t)$, $f_t \sim N(0, H_t)$, AR(1) log-variances; $\text{Var}(y_t) = \beta H_t \beta' + \Sigma_t$	L09-07 (7)–(10)
FSV correlation	$\rho_{ij,t} = \text{sign}(\beta_i\beta_j)/\sqrt{(1+a_{it}/h_t)(1+a_{jt}/h_t)}$, increasing in h_t	L09-07 (11)–(12)

---

2 · Every number

Number	What it is	Note
0.9099; 0.5329	largest companion moduli, slide-13 VAR(2)	L09-01 §5.1
$\|\Phi(1)\| = 0.032$; $m = (6.875, 14.375, 30.94)$	slide-13 VAR(2), if μ is the intercept	L09-01 §5.1
0.6107	largest companion modulus, UK/US/CA VAR(3)	L09-01 §5.2
0.4308, 0.4380, 0.4887	$\hat{B}_1$ entries (UK own; UK→US; US→CA)	L09-01 §4.4
1.0908 / 0.4280; 7.0436 / 5.5454	Tsay VAR(1) mean = 6-step forecast; forecast-sd limits	L09-01 §6.1

Number	What it is	Note
AIC $\to$ 2, BIC $\to$ 1, HQ $\to$ 1; $M(3) = 10.49$ (p 0.31)	VARorder on the GDP data	L09-01 §7.1
15, 610, 25,807; 301	parameter counts (slides 45, 27)	L09-01 §8
$\Psi_2(1,1) = 0.2495$	VAR(3)	L09-02 §2.1
$\Theta_1 =$ $(0.3088, 0.6353; 0.2050, 0.4312)$; corr 0.642	Tsay IBM/S&P orthogonalised IRF	L09-02 §3.1
$\Theta_1(\text{CA}, \text{US}) = 0.4496$	VAR(3)	L09-02 §3.2
0.0758, 0.0230; 0.8774 / 0.0676 / 0.0550	FEVD at $h = 1$; UK at $h = 20$	L09-02 §4.1
$L^{-1} = (1, 0; -0.5, 1)$, $G = \text{diag}(2, 0.5)$; $r_{2t} = 0.3 + 0.5 r_{1t} - ...$	T Ex. 8.3, ordering (r_1, r_2)	L09-03 §4.1
$r_{1t} = -0.2 + 1.0 r_{2t} - ...$	T Ex. 8.3, ordering (r_2, r_1)	L09-03 §4.1
13.6% vs 48.5%	share of r_1's 10-step variance due to the r_2 shock, two orderings, Ex. 8.3 (Ordering Dial)	L09-03 §4.1
11 zeros vs 10; 18 vs 15	Illustrations 1 and 2	L09-03 §5
0.647 $\to$ 0.690 (λ=0.2) $\to$ 0.926 (λ=0.05)	Minnesota posterior mean of $B_1(1,1)$, $T = 58$	L09-04 §3.1
0.379 $\to$ 0.055	$B_1(1,2)$: OLS vs λ=0.2 (truth 0.1)	L09-04 §3.1
0.069, 0.9996, $z^* = 0.3035$	SSVS inclusion at $z = 0.2, 0.5$; crossing	L09-04 §5
7.56 vs 0.46	relative MSFE, CEE OLS $p = 13$ vs LARGE BVAR, EMPL $h = 1$	L09-04 §6
3, 7, 14, 40	$k_{\max}$ for $m = 6, 12, 20, 50$	L09-05 §2
0.9901, 0.891, 0.70; 0.990 … 0.714	slide-22 Ω entries; communalities	L09-05 §3.1
V_f diag 0.00923, 0.03671	slide-22 factor-score posterior variances	L09-05 §3.1
1.030, 1.008, 1.020; 0.757	exchange-rate communality + uniqueness; ITA–GER correlation	L09-05 §3.2
0.70 / 0.65 vs 0.46	FAVAR (1, 3 factors, BVAR) vs LARGE BVAR, EMPL $h = 1$	L09-05 §4
$q = 0.0234$, $p^* = 0.1652$, $K^* = 0.1418$, $1 - K^* = 0.858$	Tsay Ex. 11.1	L09-06 §4.2
$\sigma_e = 0.4802$, $\sigma_\eta = 0.0736$	from $\hat{\theta} = 0.858$, $\hat{\sigma}_a = 0.5184$	L09-06 §4.2
-92.506	$\log_{10}$ marginal likelihood, Greenberg's TVP Phillips curve	L09-06 §5
0.9493; 25.6; 13.3 days	W&T GARCH persistence, long-run variance, half-life	L09-07 §4.1
1.0029	S&P GARCH persistence (above one)	L09-07 §4.1
0.7518 / 2.43 days; 0.9085 / 7.22 days	Tse–Tsui and Engle DCC persistence and half-life	L09-07 §4.1

Number	What it is	Note
0.9387; 0.4893	FSV correlation at $h = 1$ and $h = 0.05$ (slide-22 pair)	L09-07 §5
0.989; 0.858	6-d FSV factors vs the 3-d US and BR factors	L09-07 §6

Class 9 · Pitfalls — the confusions, each with its minimal counterexample, and an original True/False self-test

Prev: L09-08-formulas · Next: L09-10-python · Map: L09-00-map

Every number below is asserted in L09-10-python or in the companion check. Equations are cited by the section note's own numbering.

Part I — the confusions this material produces

1 · A coefficient above one vs a non-stationary VAR

The confusion. "$\Phi_{22} = 1.1 > 1$, so the VAR is explosive."

Counterexample (T Ex. 8.3; L01 §9.1). The eigenvalues of $\Phi = \begin{pmatrix} 0.2 & 0.3 \\ -0.6 & 1.1 \end{pmatrix}$ are 0.8 and 0.5: stationary. Conversely, the slide-13 VAR(2) has no entry above 0.9 but a companion root of 0.91 and $|\Phi(1)| = 0.032$.

> **Stationarity is read off the companion eigenvalues (L09-01 (14)–(15)), never off an entry.**

2 · Reduced-form vs orthogonalised impulse responses

The confusion. $\Psi_s(i,j)$ is "the effect of a shock to j".

Counterexample (Tsay IBM/S&P, L09-02 §3.1). Innovations correlate 0.642; a unit IBM innovation "holding the S&P innovation constant" is a pattern the data rarely produce. The orthogonalised $\Theta_0 = \begin{pmatrix} 6.9973 & 0 \\ 3.5432 & 4.2280 \end{pmatrix}$ attributes the common part to whichever series comes first.

> **Ψ_s answers a counterfactual with correlated shocks held fixed; $\Theta_s = \Psi_s P$ answers one with orthogonal shocks — and needs an ordering (L09-02 (6)).**

3 · The ordering as a detail vs an identifying assumption

The confusion. "Cholesky is just a numerical device."

Counterexample (T Ex. 8.3, L09-03 §4.1). Ordering (r_1, r_2): $r_{2t} = 0.3 + 0.5 r_{1t} - ...$; ordering (r_2, r_1): $r_{1t} = -0.2 + 1.0 r_{2t} - ....$ The share of r_1's 10-step variance due to the r_2 shock is 13.6% or 48.5%.

> **A recursive ordering is a statement about who reacts to whom within the period (L09-03 (9)). At $\rho = 0$, and only then, it is harmless.**

4 · The order condition vs identification

The confusion. "I have $q(q-1)/2$ zeros, so A_0 is identified."

Counterexample. Counting is necessary, not sufficient: the zeros must leave no rotation other than $Q = I$ (up to sign) that preserves them (slide 34's definition). And slide 32's commodity-price example shows only two zeros for $q = 3$ — one short.

> **Count first (L09-03 (8)), then check that the restrictions actually pin the rotation down.**

5 · OLS equation by equation vs "ignoring the correlation between equations"

The confusion. "Equation-by-equation OLS throws away Σ; SUR must be better."

Counterexample (L09-01 (12)). With the same regressors in every equation, GLS on the stacked system returns OLS exactly. SUR gains only after restrictions make the regressors differ (the site's `refVAR`).

> **Σ drops out of $\hat{B}$ when X is common (L09-01 (11)–(12)); it still enters $\mathrm{Cov}(\hat{\beta}) = \Sigma \otimes (X'X)^{-1}$.**

6 · AIC vs BIC lag choice

The confusion. "The criteria disagree, so one is wrong."

Counterexample (VARorder on the GDP data, L09-01 §7.1). AIC picks 2, BIC and HQ pick 1; $M(i)$ stops at 2. BIC's penalty $\ln T$ exceeds AIC's 2 once $T > 7.4$, so BIC can never choose more lags than AIC.

> **They answer different questions (prediction vs consistent selection) with different penalties (L09-01 (21)).**

7 · Shrinkage as bias vs shrinkage as a better estimator

The confusion. "The Minnesota posterior is biased toward a random walk, so OLS is safer."

Counterexample (L09-04 §3.1, $T = 58$). OLS puts 0.379 on $B_1(1,2)$ (truth 0.1); the posterior mean at $\lambda = 0.2$ is 0.055. Bańbura et al.: CEE OLS with 13 lags has MSFE 7.56 times the random walk's, the 131-variable BVAR 0.46.

> **Shrinkage trades bias for variance; with many coefficients per observation, it wins (L09-04 (9)).**

8 · λ as "prior strength" vs a variance scale

The confusion. "Large λ means a strong prior."

Counterexample (L09-04 (2), (7)). λ multiplies the prior **standard deviations**: $\lambda \to \infty$ returns OLS, $\lambda \to 0$ returns the random walk.

> **Small λ = tight prior.**

9 · Normal-Wishart vs Minnesota

The confusion. "Normal-Wishart is the Minnesota prior with Σ unknown."

Counterexample (L09-04 §4). $\Psi \otimes \tilde{\Omega}$ gives every equation the same relative tightness across regressors, so it cannot make other variables' lags tighter than own lags: $\theta = 1$.

10 · Factors identified vs loadings identified

The confusion. "The estimated loadings are the factor structure."

Counterexample (L09-05 (4)). $(\beta P', P f_t)$ fits identically for any orthogonal P; only $\beta\beta'$ is identified. The site's FSV names the US factor first because Apple is row 1; `factanal` names the Brazil factor first.

Lower-triangular β (or another normalisation) is a convention that names the factors.

11 · More factors always fit better vs the identification bound

The confusion. "With $m = 6$ series, use as many factors as you like."

Counterexample (L09-05 (7)). $k \leq 3$ at $m = 6$; at $k = 3$ the factor model has 21 parameters, as many as an unrestricted Ω — no restriction left.

12 · The Kalman gain as a tuning knob vs a derived quantity

The confusion. "Pick K to smooth more or less."

Counterexample (Tsay Ex. 11.1, L09-06 §4.2). K^* is fixed by $q = \sigma_\eta^2/\sigma_\varepsilon^2$: 0.1418, which equals $1 - \hat{\theta} = 0.142$ from the ARIMA(0,1,1) fit.

The filter's smoothing weight is the model's signal-to-noise ratio (L09-06 (13)).

13 · Time-varying parameters as "more realistic, hence better"

The confusion. "Coefficients surely move, so a TVP model must win."

Counterexample (G §10.3). Greenberg's Phillips curve: the constant-coefficient AR(1) has the larger marginal likelihood. Each random-walk state adds variance $t\Omega$ (L09-06 (16)).

14 · DCC's parameters as pair-specific

The confusion. "DCC lets each correlation have its own dynamics."

Counterexample (T p. 531). θ_1, θ_2 are scalars: every pair reverts at the same rate $(\theta_1 + \theta_2)^h$. FSV pairs differ through a_{it}, a_{jt} (L09-07 (12)).

15 · Correlations rising in crises as a separate phenomenon

The confusion. "Correlations and volatilities need separate models."

Counterexample (L09-07 (12)). In a one-factor model $|\rho_t|$ increases with the factor variance automatically: 0.939 at $h = 1$, 0.489 at $h = 0.05$ for the slide-22 pair.

16 · GARCH persistence below one by assumption

The confusion. "Fitted GARCH models always mean-revert."

Counterexample (site page, L09-07 §4.1). The S&P 500 fit has $\alpha_1 + \beta_1 = 1.0029$: no finite unconditional variance, shocks never fade.

17 · Ticker vs asset

The confusion. "WTI" on the site page is crude oil.

Counterexample. It is W&T Offshore, Inc. (`names = c("W&T Offshore, Inc.","S&P 500")`).

18 · Labels in the site scripts

The confusion. Taking the plot titles of `VAR-Example.html` at face value.

Counterexample (L09-02 §2.1, §4.1). The IRF titles swap shock and response, and the FEVD plot labels row 2 "CAN" although the order is (uk, us, ca).

Part II — True/False self-test (original)

Scoring (rules/00): V/F score = $\max\{0,\ \text{correct} - \text{wrong}/2\}\times$ item value; blank items do not count as wrong. Difficulty: **[E]** easy · **[M]** medium · **[P]** hard. Numbers are illustrative unless a note is cited. Answers follow each block.

Block A — VAR, IRF, FEVD

Setting (items 1–6). A central-bank analyst fits a VAR(2) to quarterly Brazilian inflation, the Selic rate and output growth (in that order), 100 quarters, by OLS equation by equation.

1. **[E]** Because each equation was estimated separately, the coefficient estimates would change if the analyst re-estimated the system by GLS using the residual covariance matrix.
2. **[M]** If one coefficient of $\hat{B}_1$ equals 1.05, the fitted VAR is necessarily explosive.
3. **[M]** The fitted VAR is stationary if and only if all six eigenvalues of its companion matrix have modulus below one.
4. **[E]** The $(2,1)$ element of Ψ_4 is the response of the Selic rate, four quarters later, to a unit inflation innovation, other innovations held fixed.
5. **[M]** Under the Cholesky ordering above, the one-quarter-ahead forecast-error variance of inflation is entirely attributed to the inflation shock.
6. **[P]** Re-ordering the variables as (Selic, inflation, output) leaves the orthogonalised impulse responses unchanged whenever the residual correlation between inflation and the Selic rate is positive.

Answers. 1. **F.** Same regressors in every equation ⇒ GLS = OLS (L09-01 (12)). 2. **F.** Stationarity depends on the companion eigenvalues, not on single entries (Pitfall 1). 3. **V.** $qp = 3\times 2 = 6$ eigenvalues; L09-01 (14)–(15). 4. **V.** L09-02 (4): $\Psi_s(i,j)$ is the response of i to j. 5. **V.** $\Theta_0 = P$ is lower triangular: the first variable's 1-step error loads only on shock 1 (L09-02 §4). 6. **F.** The orderings coincide only when the correlation is zero (Pitfall 3; L09-03 §4.1).

Block B — structural identification

Setting (items 7–10). A four-variable SVAR $A_0 y_t = A_1 y_{t-1} + u_t$, $u_t \sim N(0, I)$.

7. **[E]** The reduced-form covariance is $\Sigma = (A_0' A_0)^{-1}$.
8. **[M]** At least six restrictions on A_0 are needed for exact identification.
9. **[M]** Any two matrices A_0 and QA_0, with Q orthogonal, imply different likelihood values.
10. **[P]** Imposing that A_0 be lower triangular with a positive diagonal makes A_0^{-1} the Cholesky factor of Σ, and the structural IRFs coincide with the orthogonalised IRFs.

Answers. 7. **V.** L09-03 (4). 8. **V.** $q(q-1)/2 = 6$ (L09-03 (8)). 9. **F.** They give the same (B, Σ), hence the same likelihood (L09-03 (6)–(7)). 10. **V.** L09-03 (9).

Block C — Bayesian VAR

Setting (items 11–15). A Minnesota-prior BVAR for 20 Brazilian macro series, $p = 4$, with Σ fixed at univariate AR residual variances.

11. [**E**] As the overall tightness λ goes to zero, the posterior mean of each equation approaches a random walk for its own variable.
12. [**M**] The prior standard deviation of the fourth own lag is one quarter of that of the first own lag.
13. [**M**] With $\theta < 1$, the prior can be written as Normal-Wishart, $\beta \mid \Sigma \sim N(\beta_0, \Sigma \otimes \Omega_0)$.
14. [**P**] For a single coefficient with an orthogonal regressor, the weight on the prior mean is $\mathrm{Var}(\hat{b})/(\mathrm{Var}(\hat{b}) + v_0)$, so it rises with the sample size.
15. [**M**] Under SSVS with $c_0 = 0.1$ and $c_1 = 10$ times the OLS standard error, a draw of a coefficient equal to 0.5 standard errors is classified as included with probability above 0.99.

Answers. 11. **V.** L09-04 (7): $V_{i0} \to 0$, $\bar{\beta}_i \to \beta_{i0}$. 12. **V.** sd $= \lambda/k$: $\lambda/4$ vs λ (L09-04 (2)). 13. **F.** Kronecker structure forces $\theta = 1$ (Pitfall 9). 14. **F.** $\mathrm{Var}(\hat{b}) = \sigma^2/s_{xx}$ falls with T, so the prior weight falls (L09-04 (9)). 15. **V.** 0.9996 (L09-04 §5); the crossing is at 0.30.

Block D — factors, TVP, covariances

Setting (items 16–24). A risk desk models daily returns of Itaú, Bradesco, Petrobras and Vale.

16. [**E**] In a one-factor model with $f_t \sim N(0, 1)$ and diagonal idiosyncratic covariance, the covariance between Itaú and Bradesco equals the product of their loadings.
17. [**M**] With four series, a three-factor model is admissible under the parameter-count bound.
18. [**M**] Rotating the loadings by an orthogonal matrix changes the implied covariance matrix of the returns.
19. [**M**] In a one-factor SV model, if the factor's log-variance falls sharply on a calm day, the implied correlation between two positively loaded banks falls.
20. [**E**] In an Engle DCC with $\theta_1 + \theta_2 = 0.9085$, half of a correlation shock is gone after about 7 trading days.
21. [**M**] In a DCC, the bank–bank and bank–oil correlations can revert at different speeds.
22. [**P**] For a local-level model of log realised volatility with $\sigma_\eta^2/\sigma_e^2 = 0.0234$, the steady-state Kalman filter puts about 86% weight on the latest observation.
23. [**M**] A random-walk TVP coefficient has prior variance growing linearly with time.
24. [**P**] A fitted GARCH(1,1) with $\alpha_1 + \beta_1 = 1.003$ has a finite unconditional variance $\omega/(1 - \alpha_1 - \beta_1)$.

Answers. 16. **V.** L09-05 (3). 17. **F.** $k \le (2m + 1 - \sqrt{8m + 1})/2 = (9 - \sqrt{33})/2 = 1.63$, so $k \le 1$ at $m = 4$ (L09-05 (7)). 18. **F.** $\beta P' P \beta' = \beta\beta'$ (L09-05 (4)). 19. **V.** $|\rho_t|$ increases in h_t (L09-07 (12)). 20. **V.** $\ln 0.5 / \ln 0.9085 = 7.22$ (L09-07 §4.1). 21. **F.** Scalar θ's: one speed for all pairs (Pitfall 14). 22. **F.** $K^* = 0.142$: about 14% on the latest observation, 86% on the previous estimate (L09-06 §4.2). 23. **V.** $\mathrm{Var}(\beta_t \mid \beta_0) = t\Omega$ (L09-06 (16)). 24. **F.** $1 - \alpha_1 - \beta_1 < 0$: no finite unconditional variance (Pitfall 16).

Part III — the distinctions that cost most points

Pair	The difference
Ψ_s vs Θ_s	unit correlated innovation vs one-sd orthogonal shock
Order condition vs identification	counting restrictions vs pinning down the rotation

Pair	The difference
OLS vs SUR in a VAR	identical with common regressors; differ after zero restrictions
λ small vs large	prior (random walk) vs OLS
Minnesota vs Normal-Wishart	$\theta < 1$ allowed vs $\theta = 1$ forced
Loadings vs $\beta\beta'$	identified only up to rotation vs identified
K_t vs K^*	initial path vs steady state set by q
DCC vs FSV correlation	own recursion with scalar θ's vs implied by factor volatility

Class 9 · Computation in Python — every routine, with the assert that must pass

> Every computation the unit uses, written in Python (numpy, scipy, statsmodels; no R, no PyMC), each checked with an `assert` against the number printed in the slides, the site pages or Tsay. The code below is the script `Notas/L09/L09_python_checks.py`, split into its sections. Run `python Notas/L09/L09_python_checks.py` (about 20 s). It prints one `ok` line per section and ends with `ALL L09 CHECKS PASSED`, which is the output of the run on 2026-10-25. Prev: L09-09-pitfalls · Map: L09-00-map · Formulas: L09-08-formulas

What is reused, not rewritten.

Already in the unit	What it computes	How this note uses it
`figures/make_figures.py`	the printed matrices (slide 13's VAR(2), the site's OLS and refined VAR(3), $\hat{\Sigma}$'s); `companion`, `psi`, `fevd`, `minnesota_posterior`, `lagmat`	**imported** (`import make_figures as mf`; importing draws nothing). Every routine below that the figures also use is the same function
`companions/check_companions.py`	the two pages' read-outs recomputed in Python and compared with their JavaScript under `node`	**cited**: it checks the pages; this script checks the formulas
`Notas/L01/L01_python_checks.py` §9	VAR(1) mean and Γ_0 by the vec formula (T Ex. 8.3)	**cited**: §2 below uses the same formula for Tsay's IBM/S&P VAR(1)

Not installed and not needed. `arch` and `PyMC` are absent. GARCH/DCC estimates are taken from the site page's printed output (§7 checks their implications, not a re-fit); the Minnesota and diffuse posteriors are closed form (§4); the Kalman filter is twenty lines (§6).

Conventions. `near(x, printed, tol)` passes when `x` matches a printed value within `tol` (default 5e-4). Equation numbers "(k)" are those of the section note named in each heading.

---

0 · Setup

```
"""Class 9 (L09) reference implementation, with assert self-checks against the
 ↪  sources' numbers.

Run:  python Notas/L09/L09_python_checks.py      (stops at the first failed assert)

Sections §1-§7 are the code blocks of L09-10-python.md, in the same order. The model
 ↪  routines
(companion, psi, fevd, minnesota_posterior, lagmat) and the printed matrices are
 ↪  imported from
figures/make_figures.py (reuse, not a copy; importing it draws nothing). Equation
 ↪  numbers "(k)"
refer to the section note named in each header. Simulations are seeded.
"""
```

```
import pathlib
import sys

import numpy as np
from scipy import stats
import statsmodels.api as sm
from statsmodels.tsa.api import VAR
from statsmodels.tsa.vector_ar.var_model import VARProcess
from statsmodels.tsa.vector_ar.svar_model import SVAR

HERE = pathlib.Path(__file__).resolve().parent
sys.path.insert(0, str(HERE / "figures"))
import make_figures as mf  # noqa: E402  printed matrices + model routines

sys.stdout.reconfigure(encoding="utf-8")


def near(x, printed, tol=5e-4):
    """True when x matches a printed number to within tol (default: half a unit in
     ↪  the 3rd decimal)."""
    return abs(float(x) - float(printed)) <= tol


def ok(msg):
    print("ok", msg)
```

§1 · VAR(p) as a multivariate regression: OLS = MLE = SUR/GLS (L09-01 (11)-(13))

Simulates 1000 observations from slide 13's VAR(2) (seed 2025), fits it by (11) with `numpy.linalg.solve`, and checks: the coefficients equal `statsmodels VAR(...).fit(2, trend="c")`; GLS on the stacked system (5) returns OLS, i.e. Σ drops out (12); the standard errors are $\sqrt{\mathrm{diag}(\hat{\Sigma} \otimes (X'X)^{-1})}$ (13), as the site's `kronecker(Shat, iXtX)`.

```
rng = np.random.default_rng(2025)
Lc = np.linalg.cholesky(mf.SIG_S)
Y = np.zeros((1200, 3))
for t in range(2, 1200):
    Y[t] = mf.MU_S + mf.B1_S @ Y[t - 1] + mf.B2_S @ Y[t - 2] + Lc @
 ↪  rng.standard_normal(3)
Y = Y[200:]                                          # 1000 observations, as on
 ↪  slide 13
X, Yr = mf.lagmat(Y, 2)                              # constant, lag 1, lag 2
Bhat_T = np.linalg.solve(X.T @ X, X.T @ Yr)          # (11): (X'X)^{-1} X'Y, one
 ↪  column per equation
res = VAR(Y).fit(2, trend="c")
assert np.allclose(Bhat_T, res.params, atol=1e-10)      # statsmodels params: rows =
 ↪  regressors
U = Yr - X @ Bhat_T
S_ml = U.T @ U / len(Yr)
```

```
Sig_inv = np.linalg.inv(S_ml)
n = 300                                                  # GLS on the first 300 rows
 ↪  (the n x n Kronecker is big)
Xg, Yg = X[:n], Yr[:n]
Bg = np.linalg.solve(Xg.T @ Xg, Xg.T @ Yg)
y_vec = Yg.T.reshape(-1)                                 # vec(Y): equation 1's n
 ↪  values, then 2's, 3's
Z = np.kron(np.eye(3), Xg)
gls = np.linalg.solve(Z.T @ np.kron(Sig_inv, np.eye(n)) @ Z, Z.T @ np.kron(Sig_inv,
 ↪  np.eye(n)) @ y_vec)
assert np.allclose(gls, Bg.T.reshape(-1), atol=1e-8)     # (12): GLS = OLS, Sigma
 ↪  drops out
cov = np.kron(res.sigma_u, np.linalg.inv(X.T @ X))       # (13): Sigma (x) (X'X)^{-1}
assert np.allclose(np.sqrt(np.diag(cov)), res.bse.T.reshape(-1), rtol=1e-8)
assert np.max(np.abs(Bhat_T[1:4].T - mf.B1_S)) < 0.1     # T = 1000 recovers slide
 ↪  13's B1
ok("§1 OLS = statsmodels VAR = GLS on (5); se = sqrt diag(Sigma (x) (X'X)^-1)")
```

---

§2 · Companion form, stationarity, means, forecasts (L09-01 (2), (14)-(18))

Companion eigenvalues of the slide-13 VAR(2) (max modulus 0.9099) and of the site's VAR(3) (0.6107); every nonzero eigenvalue λ makes $|I - B_1/\lambda - B_2/\lambda^2| = 0$ (L09-01 (15)); the implied mean $(6.875, 14.375, 30.9375)$; Tsay's IBM/S&P VAR(1): mean = the printed 6-step forecasts, forecast sds converging to 7.0436/5.5454 and 7.0434/5.5453 at $h = 2$ (L09-01 (18)); the parameter counts of slides 27 and 45.

```
ev = np.linalg.eigvals(mf.companion([mf.B1_S, mf.B2_S]))
assert near(np.abs(ev).max(), 0.9099, 5e-5) and near(np.sort(np.abs(ev))[-2], 0.5329,
 ↪  5e-5)
# (15): nonzero eigenvalues are reciprocals of the roots of |I - B1 z - B2 z^2| = 0
for lam in ev[np.abs(ev) > 1e-9]:
    z = 1 / lam
    assert abs(np.linalg.det(np.eye(3) - mf.B1_S * z - mf.B2_S * z**2)) < 1e-9
m = np.linalg.solve(np.eye(3) - mf.B1_S - mf.B2_S, mf.MU_S)
assert np.allclose(m, [6.875, 14.375, 30.9375])
assert near(np.abs(np.linalg.eigvals(mf.companion([mf.G1, mf.G2, mf.G3]))).max(),
 ↪  0.6107, 5e-5)
# Tsay VAR(1), IBM and S&P 500 (pp. 414-416)
Phi = np.array([[-0.0320, 0.1503], [-0.0223, 0.1020]])
phi0 = np.array([1.0614, 0.4087])
mt = np.linalg.solve(np.eye(2) - Phi, phi0)
assert near(mt[0], 1.0909, 5e-4) and near(mt[1], 0.4280, 5e-4)     # 6-step forecasts
 ↪  printed
r = 3.5432 / np.hypot(3.5432, 4.2280)                              # innovation
 ↪  correlation 0.642
s1, s2 = 7.0078, 5.5247                                            # residual
 ↪  scales printed
Sig_t = np.array([[s1**2, r * s1 * s2], [r * s1 * s2, s2**2]])
```

```
G0 = np.linalg.solve(np.eye(4) - np.kron(Phi, Phi), Sig_t.reshape(-1,
 ↪  order="F")).reshape(2, 2, order="F")
assert near(np.sqrt(G0[0, 0]), 7.0436) and near(np.sqrt(G0[1, 1]), 5.5454, 2e-4)
mse2 = Sig_t + Phi @ Sig_t @ Phi.T                                  # (18), h = 2
assert near(np.sqrt(mse2[0, 0]), 7.0434) and near(np.sqrt(mse2[1, 1]), 5.5453, 2e-4)
# parameter counts (slides 27, 45)
assert [q * q + q * (q + 1) // 2 for q in (3, 20, 131)] == [15, 610, 25807] and 7 + 6
 ↪  * 49 == 301
ok("§2 companion roots, mean (6.875, 14.375, 30.94), Tsay forecast limits 7.0436 /
 ↪  5.5454")
```

---

§3 · VMA weights, orthogonalised IRFs, FEVD, recursive SVAR (L09-02 (3), (6), (8); L09-03 (9)-(11))

The Ψ_s recursion (L09-02 (3)) against `VARProcess.ma_rep`, Θ_s against `orth_ma_rep`; Tsay's $\Theta_1 = \Phi\Theta_0$; the FEVD (8) against the site's `FEVdec` table at $h = 1$ (to 10^{-6}) and $h = 20$ (to 10^{-3}, the refined coefficients being printed to three decimals); both orderings of Tsay Ex. 8.3 (L09-03 (10)–(11)); and `statsmodels` `SVAR` with a lower-triangular A on 2900 simulated observations, whose A^{-1} matches the Cholesky factor of $\hat{\Sigma}$ (L09-03 (9)).

```
P = mf.psi([mf.G1, mf.G2, mf.G3], 6)
proc = VARProcess(np.array([mf.G1, mf.G2, mf.G3]), np.zeros(3), mf.SH)
assert np.allclose(np.array(P), proc.ma_rep(5))                     # (3) =
 ↪  statsmodels ma_rep
assert near(P[2][0, 0], 0.2495, 5e-5) and near(P[3][0, 1], 0.3335, 5e-5)
Th = proc.orth_ma_rep(5)
assert np.allclose(Th[1], P[1] @ np.linalg.cholesky(mf.SH)) and near(Th[1][2, 1],
 ↪  0.4496, 5e-5)
L0 = np.array([[6.9973, 0], [3.5432, 4.2280]])                      # Tsay impRes
 ↪  lag 0
T1 = Phi @ L0
assert all(near(a, b, 5e-4) for a, b in zip(T1.ravel(), [0.3088, 0.6353, 0.2050,
 ↪  0.4312]))
f1, f20 = mf.fevd([mf.R1, mf.R2, mf.R3], mf.SR, 1), mf.fevd([mf.R1, mf.R2, mf.R3],
 ↪  mf.SR, 20)
assert near(f1[1, 0], 0.07578528, 1e-6) and near(f1[2, 1], 0.1599804, 1e-6)
assert near(f20[0, 0], 0.8774260, 1e-3) and near(f20[1, 0], 0.2657297, 1e-3) and
 ↪  near(f20[2, 2], 0.43585314, 1e-3)
# Tsay Ex. 8.3: both recursive orderings
Phi83, phi083, S83 = np.array([[0.2, 0.3], [-0.6, 1.1]]), np.array([0.2, 0.4]),
 ↪  np.array([[2.0, 1.0], [1.0, 1.0]])
Lunit = np.array([[1.0, 0.0], [0.5, 1.0]])                          # unit-diagonal L,
 ↪  ordering (r1, r2)
Gd = np.linalg.inv(Lunit) @ S83 @ np.linalg.inv(Lunit).T
assert np.allclose(Gd, np.diag([2.0, 0.5]))
assert np.allclose(np.linalg.inv(Lunit) @ phi083, [0.2, 0.3]) and
 ↪  np.allclose(np.linalg.inv(Lunit) @ Phi83, [[0.2, 0.3], [-0.7, 0.95]])
Pm = np.array([[0, 1], [1, 0]])
Linv_b = np.array([[1.0, 0.0], [-1.0, 1.0]])
```

```
assert np.allclose(Linv_b @ Pm @ S83 @ Pm @ Linv_b.T, np.eye(2))
assert np.allclose(Linv_b @ Pm @ phi083, [0.4, -0.2]) and np.allclose(Linv_b @ Pm @
 ↪  Phi83 @ Pm, [[1.1, -0.6], [-0.8, 0.8]])
# statsmodels SVAR with a lower-triangular A recovers A^{-1} = Cholesky of Sigma-hat
 ↪  (L09-03 (9))
rng = np.random.default_rng(83)
Ls = np.linalg.cholesky(S83)
Ysv = np.zeros((3000, 2))
for t in range(1, 3000):
    Ysv[t] = phi083 + Phi83 @ Ysv[t - 1] + Ls @ rng.standard_normal(2)
mask = np.array([["E", 0], ["E", "E"]], dtype=object)
sres = SVAR(Ysv[100:], svar_type="A", A=mask).fit(maxlags=1, trend="c",
 ↪  A_guess=np.array([1.0, -0.5, 1.0]))
Ainv = np.linalg.inv(sres.A)
Pchol = np.linalg.cholesky(VAR(Ysv[100:]).fit(1, trend="c").sigma_u)
assert np.allclose(np.abs(Ainv), np.abs(Pchol), atol=0.02)
ok("§3 Psi recursion = ma_rep; Tsay Theta_1; FEVD = MTS table; Ex. 8.3 orderings;
 ↪  SVAR A^-1 = chol")
```

---

§4 · Minnesota posterior, limits in lambda, SSVS inclusion probability (L09-04 (2), (7)-(9), (15))

The Minnesota posterior (L09-04 (7)) with the figure's seed-9 simulation: the univariate AR(2) variances, the OLS and posterior-mean table of L09-04 §3.1, the two limits in λ, the prior sds 0.145 and 0.057, the scalar weight (9) as a special case of (7), and the SSVS inclusion probabilities (15): 0.069 at $z = 0.2$, 0.9996 at $z = 0.5$, crossing at $z^* = 0.3035$.

```
rng = np.random.default_rng(9)                          # the figure's simulation,
 ↪  same seed
Ym = np.zeros((260, 3))
for t in range(2, 260):
    Ym[t] = mf.MU_S + mf.B1_S @ Ym[t - 1] + mf.B2_S @ Ym[t - 2] + Lc @
 ↪  rng.standard_normal(3)
Ym = Ym[200:]
Xm, Yrm = mf.lagmat(Ym, 2)
s2 = []
for i in range(3):
    Xi, yi = mf.lagmat(Ym[:, [i]], 2)
    e = yi[:, 0] - Xi @ np.linalg.lstsq(Xi, yi, rcond=None)[0][:, 0]
    s2.append(e.var(ddof=3))
s2 = np.array(s2)
assert np.allclose(s2, [0.266, 0.1257, 0.8046], atol=5e-4)


def minnesota_eq(i, lam, theta=0.5, p=2):
    """Posterior mean of equation i under (2), constant variance 1e6, theta = 0.5
     ↪  (illustrative)."""
    g0 = np.zeros(1 + 3 * p)
    g0[1 + i] = 1.0
```

```
    v0 = [1e6] + [(lam / k) ** 2 * (1 if j == i else theta**2 * s2[i] / s2[j]) for k
 ↪  in range(1, p + 1) for j in range(3)]
    return mf.minnesota_posterior(Yrm[:, i], Xm, g0, np.array(v0), s2[i])[0]


ols = np.linalg.lstsq(Xm, Yrm[:, 0], rcond=None)[0]
assert np.allclose(ols[1:5], [0.647, 0.379, -0.021, -0.292], atol=5e-4)
for lam, row in ((0.05, [0.926, 0.000, -0.010, -0.018]), (0.2, [0.690, 0.055, -0.043,
 ↪  -0.115]),
                 (1.0, [0.649, 0.284, -0.038, -0.262])):
    assert np.allclose(minnesota_eq(0, lam)[1:5], row, atol=5e-4), (lam,
     ↪  minnesota_eq(0, lam)[1:5])
assert np.allclose(minnesota_eq(0, 1e-6)[1:], [1, 0, 0, 0, 0, 0], atol=1e-6)      #
 ↪  lambda -> 0: prior
assert np.allclose(minnesota_eq(0, 1e4)[1:], ols[1:], atol=1e-5)                    #
 ↪  lambda -> inf: OLS
assert near(0.1 * np.sqrt(s2[0] / s2[1]), 0.145) and near(0.1 * np.sqrt(s2[0] /
 ↪  s2[2]), 0.057)
# (9): scalar shrinkage weight equals (7) with one regressor
x1 = Xm[:, 1]
y1 = Yrm[:, 0] - Yrm[:, 0].mean()
x1 = x1 - x1.mean()
v0, b0, sig2 = 0.04, 1.0, s2[0]
post = mf.minnesota_posterior(y1, x1[:, None], np.array([b0]), np.array([v0]),
 ↪  sig2)[0][0]
bhat = x1 @ y1 / (x1 @ x1)
w = (sig2 / (x1 @ x1)) / (sig2 / (x1 @ x1) + v0)
assert near(post, w * b0 + (1 - w) * bhat, 1e-12)
# SSVS (15) with c0 = 0.1, c1 = 10, q = 1/2
incl = lambda z: 0.1 * stats.norm.pdf(z / 10) / (0.1 * stats.norm.pdf(z / 10) + 10 *
 ↪  stats.norm.pdf(10 * z))
assert near(incl(0.2), 0.069) and near(incl(0.5), 0.9996, 5e-5)
zstar = np.sqrt(2 * np.log(100) / (100 - 0.01))
assert near(zstar, 0.3035, 5e-5) and near(incl(zstar), 0.5, 1e-12)
ok("§4 Minnesota table, both limits, scalar weight (9), SSVS 0.069 / 0.9996 / z* =
 ↪  0.3035")
```

---

§5 · Factor model algebra (L09-05 (2)-(9))

Slide 22's Ω, communalities and V_f (L09-05 (2), (9)); rotation invariance (4) with a random orthogonal matrix; the bound (7) for $m = 6, 12, 20, 50$; the exchange-rate check that communality plus uniqueness is near one.

```
beta = np.array([[0.99, 0], [0, 0.99], [0.90, 0], [0, 0.90], [0.50, 0.50]])
psi_ = np.array([0.01, 0.05, 0.10, 0.15, 0.20])
Om = beta @ beta.T + np.diag(psi_)
assert near(Om[0, 0], 0.9901, 1e-12) and near(Om[0, 2], 0.891, 1e-12) and near(Om[4,
 ↪  4], 0.70, 1e-12)
comm = (beta**2).sum(1) / np.diag(Om)
assert np.allclose(comm, [0.990, 0.951, 0.890, 0.844, 0.714], atol=5e-4)
```

```
Vf = np.linalg.inv(np.eye(2) + beta.T @ np.diag(1 / psi_) @ beta)
assert np.allclose(Vf, [[0.00923, -0.00042], [-0.00042, 0.03671]], atol=5e-6)
Qr, _ = np.linalg.qr(np.random.default_rng(5).standard_normal((2, 2)))           # random rotation
assert np.allclose((beta @ Qr.T) @ (beta @ Qr.T).T, beta @ beta.T)                 # (4)
for mm, kk in {6: 3, 12: 7, 20: 14, 50: 40}.items():
    assert int(np.floor((2 * mm + 1 - np.sqrt(8 * mm + 1)) / 2)) == kk            # (7)
fx_load = {"US": (0.99, 0.00, 0.05), "JAP": (0.46, 0.42, 0.62), "FRA": (0.39, 0.91, 0.04)}
assert [round(a * a + b * b + s, 3) for a, b, s in fx_load.values()] == [1.030, 1.008, 1.020]
assert near(0.41 * 0.40 + 0.77 * 0.77, 0.757)
ok("§5 Omega, communalities, V_f, rotation invariance, k bound, exchange-rate check")
```

---

§6 · Kalman filter, steady state, ARIMA link, discounting (L09-06 (8)-(14), (18); (3))

Tsay Ex. 11.1: q, p^*, $K^* = 0.14177$ and $1 - K^* = 0.858 = \hat{\theta}$ (L09-06 (13)); T eqs. 11.4–11.5 back to $\sigma_e = 0.4802$, $\sigma_\eta = 0.0736$; a local-level filter written from (8)–(12) that matches `statsmodels` `UnobservedComponents(level="llevel")` state by state and in log-likelihood (statsmodels' `llf` drops the first observation, so `llf_obs.sum()` is compared); the gain within 0.005 of K^* by day 15; the discount identity (18); Uhlig's scalar Beta mean (3).

```
se, sn = 0.4803, 0.0735
q = (sn / se) ** 2
pstar = (q + np.sqrt(q * q + 4 * q)) / 2
Kstar = pstar / (pstar + 1)
assert near(q, 0.023418, 5e-7) and near(pstar, 0.165186, 5e-7) and near(Kstar, 0.14177, 5e-6)
assert near(1 - Kstar, 0.858)                                                     # = theta-hat of (11.6)
sa, th = 0.5184, 0.858
assert near(np.sqrt(th * sa**2), 0.4802) and near(np.sqrt((1 + th**2) * sa**2 - 2 * th * sa**2), 0.0736)


def kalman_llevel(y, s2e, s2n, m1, P1):
    """Local-level Kalman filter, Tsay eq. (11.14). Returns filtered means, gains, log-likelihood (11.25)."""
    mu, P, out, Ks, ll = m1, P1, [], [], 0.0
    for yt in y:
        v, V = yt - mu, P + s2e
        K = P / V
        ll += -0.5 * (np.log(2 * np.pi) + np.log(V) + v * v / V)
        mu_f, P_f = mu + K * v, P * (1 - K)
        out.append(mu_f)
        Ks.append(K)
```

```
        mu, P = mu_f, P_f + s2n
    return np.array(out), np.array(Ks), ll


rng = np.random.default_rng(11)
state = np.cumsum(sn * rng.standard_normal(340))
yk = state + se * rng.standard_normal(340)
filt, Ks, ll = kalman_llevel(yk, se**2, sn**2, 0.0, 10.0)
mod = sm.tsa.UnobservedComponents(yk, level="llevel")
mod.ssm.initialize_known(np.array([0.0]), np.array(10.0))
sres_k = mod.smooth([se**2, sn**2])
assert np.allclose(filt, sres_k.filtered_state[0], atol=1e-8) and near(ll,
 ↪  sres_k.llf_obs.sum(), 1e-6)  # llf drops t = 1 (loglikelihood_burn)
assert abs(Ks[14] - Kstar) < 0.005 and near(Ks[0], 10 / (10 + se**2), 1e-12)     #
 ↪  "within about 15 days"
# (18): discounting makes P_{t|t-1} = P_{t-1|t-1}/delta
delta, Pf = 0.99, 0.37
assert near(Pf + (1 / delta - 1) * Pf, Pf / delta, 1e-15)
# (3): Uhlig scalar case, E theta = a/(a+1/2) for theta ~ Beta(a, 1/2)
a = (5 + 2) / 2
assert near(stats.beta(a, 0.5).mean(), a / (a + 0.5), 1e-12) and near(a / (a + 0.5),
 ↪  7 / 8, 1e-12)
ok("§6 K* = 0.1418 = 1 - 0.858; filter = statsmodels UnobservedComponents; discount
 ↪  and Uhlig identities")
```

---

§7 · EWMA, DCC and GARCH persistence, FSV implied correlation (L09-07 (1)-(2), (6), (11)-(12))

The EWMA closed form (L09-07 (1)) equals its recursion (2); the DCC and GARCH persistence numbers of the W&T Offshore / S&P 500 page (half-lives 7.22, 2.43, 13.33 days; $\bar{\sigma}^2 = 25.6$; S&P persistence above one; t kurtosis 7.1); the FSV correlation (11) equals (12), is increasing in h, and gives 0.9387 and 0.4893.

```
rng = np.random.default_rng(7)
A = rng.standard_normal((400, 2))
lam_e = 0.94
t0 = 400
w = (1 - lam_e) * lam_e ** np.arange(t0 - 1) / (1 - lam_e ** (t0 - 1))
S_direct = sum(wj * np.outer(A[t0 - 1 - j], A[t0 - 1 - j]) for j, wj in enumerate(w))
S_rec = np.outer(A[0], A[0])
for t in range(1, t0):
    S_rec = (1 - lam_e) * np.outer(A[t], A[t]) + lam_e * S_rec
assert np.allclose(S_direct, S_rec, atol=1e-9) and near(w.sum(), 1.0, 1e-12)     #
 ↪  (1) = (2) for large t
s_engle, s_tt = 0.8703852 + 0.03813264, 0.7017527 + 0.049999
assert near(np.log(0.5) / np.log(s_engle), 7.22, 5e-3) and near(np.log(0.5) /
 ↪  np.log(s_tt), 2.43, 5e-3)
assert near(s_engle**10, 0.383) and near(s_tt**10, 0.058)
om, al, be = 1.29849458, 0.1061140, 0.8432337
```

```
assert near(om / (1 - al - be), 25.635, 5e-3) and near(np.log(0.5) / np.log(al + be),
 ↪  13.33, 5e-3)
assert 0.1964253 + 0.8065090 > 1
nu = 5.464443
assert near(3 * (nu - 2) / (nu - 4), 7.1, 0.05)
rho11 = lambda b1, b2, p1, p2, h: b1 * b2 * h / np.sqrt((b1**2 * h + p1) * (b2**2 * h
 ↪  + p2))
rho12 = lambda b1, b2, p1, p2, h: np.sign(b1 * b2) / np.sqrt((1 + p1 / b1**2 / h) *
 ↪  (1 + p2 / b2**2 / h))
hh = np.exp(np.linspace(-6, 3, 200))
assert np.allclose(rho11(0.99, 0.9, 0.01, 0.1, hh), rho12(0.99, 0.9, 0.01, 0.1, hh))
assert np.all(np.diff(rho11(0.99, 0.9, 0.01, 0.1, hh)) > 0)                          #
 ↪  increasing in h
assert near(rho11(0.99, 0.9, 0.01, 0.1, 1.0), 0.9387, 5e-5) and near(rho11(0.99, 0.9,
 ↪  0.01, 0.1, 0.05), 0.4893, 5e-5)
ok("§7 EWMA recursion, DCC half-lives 7.22 / 2.43, GARCH 25.6 and 13.3 days, FSV
 ↪  correlation (11) = (12)")

print("ALL L09 CHECKS PASSED")
```

Links

- Unit: L09-00-map · L09-08-formulas · L09-09-pitfalls.
- Section notes: L09-01-var-p-estimation-forecasting · L09-02-vma-irf-fevd · L09-03-structural-var · L09-04-bvar-minnesota-ssvs · L09-05-factor-models-favar · L09-06-tvp-var-sv-state-space · L09-07-fsv-dcc-covariances.
- Class 1 code: L01 python (VAR(1) moments).
- Python tools named in the class plan: `statsmodels.tsa.api.VAR` (IRF, FEVD), `SVAR`, `statsmodels` state space; `arch` and `PyMC` are not installed — see above.